

Scott–Vogelius elements on polygonal approximations of curved walls: the zero-gradient lock, Nitsche’s method and the penalty threshold

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Abstract

Scott–Vogelius elements give pointwise divergence-free velocities, but when a curved no-slip wall is replaced by an inscribed polygon, strongly imposed no-slip can force a zero velocity gradient at polygon vertices, and the $\mathbf{H}^1$ error can degrade to order $\mathbf{h}^{1/2}$. Gjerde and Scott therefore imposed the wall condition by Nitsche’s method, and Scott asked whether the error is $\mathcal{O}(\mathbf{h}_\Gamma^{3/2} + \mathbf{h}_\Omega^k)$ and, if not, why not. A letter written jointly with C. de S. Cavalcante announces the core answer with proof sketches: the printed method leaks through the wall at the rate $(\mathbf{h}/\mu)(\mathbf{p} - \bar{\mathbf{p}})$ unless the wall pressure is constant, and a mean-free pressure traction restores Scott’s rate. This companion paper contains what the letter does not. First, we determine when the zero-gradient lock occurs: under the Guzmán–Scott angle condition strongly imposed no-slip already has $\mathbf{H}^1$ error $\asymp \mathbf{h}_\Gamma^{3/2} + \mathbf{h}^k$, and the lock comes from wall vertices lying in only two triangles, each costing $\mathbf{h}_\Gamma |\partial_n \mathbf{u}(z)|$. Second, we give the penalty theory: the penalty must scale with the ratio of bulk to boundary mesh size, as shown by exact certificates on shape families and a proof on every shape-regular mesh, while exact counterexamples show that necessity is not local. Third, we identify the $\mathbf{H}^1$ constant of the leak as the energy of a Stokes leak flow. Finally, we give complete proofs of the letter’s results, including the sharpness of $\mathbf{h}_\Gamma^{3/2}$ for every $k \geq 4$. Computations confirm the rates and constants.

Keywords: Stokes equations, Scott–Vogelius elements, Nitsche’s method, curved boundaries, divergence-free finite elements, penalty parameter

MSC Classification: 65N30 (primary) , 65N12 , 65N15 , 76D07

1 Introduction

1.1 Exactly divergence-free elements

Most mixed finite element methods for incompressible flow enforce the constraint $\operatorname{div} u = 0$ only weakly: the discrete divergence is orthogonal to the discrete pressure space, but not zero. In 1985 Scott and Vogelius [1] observed that for continuous piecewise- P_k velocities and discontinuous piecewise- P_{k-1} pressures the divergence maps the velocity space *onto* the pressure space when $k \geq 4$ and the mesh has no singular vertices. For such a pair the weak constraint is a strong one: the discrete velocity is divergence-free pointwise. The inf-sup constant of the pair was later made explicit in terms of the local geometry of the mesh by Guzmán and Scott [2], through an angle quantity $\Theta(z)$ at each vertex z that measures how far the vertex is from being singular. Lower-degree pairs with the same property exist on split meshes, for instance on barycentric (Clough–Tocher) refinements [3, 4]; see Neilan’s review [5].

Pointwise incompressibility is more than a cosmetic property. It gives exact mass conservation, and, on meshes fitted to the domain with strongly imposed Dirichlet data, it makes the velocity error independent of the pressure. This *pressure robustness* matters when pressure gradients are large relative to the viscosity, for example in nearly hydrostatic or rotating flows; the survey of John, Linke, Merdon, Neilan and Rebholz [6] explains why the classical inf-sup stable pairs lack it and why divergence-free pairs have it.

1.2 Curved walls and polygonal approximation

Most flows of interest are bounded by curved walls. The simplest way to handle a curved boundary Γ in a finite element code is to replace it by an inscribed polygon Γ_h whose vertices lie on Γ , and to impose the boundary condition on Γ_h . For scalar second-order problems the consequences are classical. Berger, Scott and Strang [7] and Scott [8] showed that transferring a homogeneous Dirichlet condition from Γ to Γ_h costs an H^1 error of order $h_\Gamma^{3/2}$, where h_Γ is the length of the chords: the solution is of size h_Γ^2 on the chords and oscillates on the scale h_Γ , so its $H^{1/2}(\Gamma_h)$ norm is of order $h_\Gamma^{3/2}$. Higher order requires either a better geometry or a correction of the boundary data. The first route is the isoparametric approach, in which the boundary elements are curved by a polynomial map [9]; the second goes back to Bramble, Dupont and Thomée [10], who corrected the boundary condition on the polygon by a Taylor expansion in the normal direction. Both ideas are alive today: boundary-value correction has been combined with Nitsche’s method for cut elements [11, 12] and for polygonal approximations [13], and the shifted boundary method [14] is a related approach.

For exactly divergence-free elements the polygonal approximation has an additional, less obvious cost. A field that is smooth at a polygon vertex and vanishes on both adjacent edges has zero gradient there, since its gradient annihilates two independent directions. Continuous piecewise polynomials can escape this constraint, because their gradients may jump across the interior edges at the vertex; for *divergence-free* piecewise polynomials, however, the jumps are themselves constrained, and such fields inherit gradient constraints at the vertices of a polygonal wall [15, 16]. On the true

curved wall, however, the gradient of the velocity is $\partial_n u \otimes n$, which is not zero where the wall shear stress is not zero. If no-slip is imposed strongly on an inscribed polygon, the computed gradient can therefore be wrong by $O(1)$ at the polygon vertices. Scott and Gjerde–Scott report that the H^1 error of Scott–Vogelius elements then decays only like $h^{1/2}$ [15, 16], compared with $h^{3/2}$ for scalar problems. We call this the *zero-gradient lock*. The standard cure is to fit the geometry, with isoparametric or Piola-mapped Scott–Vogelius elements [17, 18]; these attain the optimal order h^k , at the price of curved elements.

1.3 Nitsche’s method and the prize question

In 1971 Nitsche [19] proposed to impose Dirichlet conditions weakly, by adding to the variational form a consistency term (the normal flux of the solution tested against the test function), its symmetric counterpart, and a penalty term $(\mu/h)\langle u, v \rangle$. Unlike a pure penalty method [20], whose penalty must grow as the mesh is refined because the method is inconsistent, Nitsche’s method is consistent, so the penalty only has to be large enough for coercivity. Stenberg [21] later showed that Nitsche’s method can be read as a stabilised Lagrange multiplier method, which explains both its consistency and the form of its penalty; the multiplier is the boundary flux of the solution. For the Stokes equations the consistency term is the full traction $\sigma(u, p)n = (\nabla u)n - pn$ (or its symmetric-gradient version), and therefore contains the pressure term $\langle p, v \cdot n \rangle$ [21–24]; Burman, Hansbo and Larson note that this term is essential for consistency [12].

Gjerde and Scott [16] combined these ideas: they keep the inscribed polygon, use Scott–Vogelius elements with $k = 4$, and impose no-slip on the polygon by Nitsche’s method, with penalty μ/h and *no pressure term*. They report errors orders of magnitude below those of strong imposition. Scott observed H^1 errors of order $h^{3/2}$ for this Scott–Vogelius–Nitsche method on a benchmark with constant pressure, and posed the question as a prize problem, listed as PPL 115 in the Prize Problem Ledger [15]. The prize asks for “a published proof that Scott–Vogelius–Nitsche converges”, which “should include a computational study of the influence of the Nitsche penalty parameter”. It states the expected error

$$\|u - u_h\|_{H^1} \lesssim h_\Gamma^{3/2} + h_\Omega^k,$$

where h_Γ is the boundary mesh size and h_Ω the bulk mesh size, and it adds: “*If this does not hold, why not?*” Together with the letter [25], of which it is the companion (Section 1.6), this paper answers the question, for general Stokes data, in two dimensions.

1.4 What goes wrong, and why the method still converges

The key observation is simple. Because the Gjerde–Scott form has no pressure term on the wall, the discrete momentum equation does not see the wall pressure through a traction. Testing the exact solution in the discrete equations leaves the residual $-\langle p - \bar{p}, v \cdot n \rangle$ on the polygon, for any constant $\bar{p}$. Something must balance it, and the

only candidate is the penalty. To leading order the discrete velocity therefore satisfies the *leak law*

$$\frac{\mu}{h} u_h \cdot n \approx p - \bar{p} \quad \text{on } \Gamma_h,$$

where $\bar{p}$ is the mean of p over the wall: the fluid leaks through the wall, outward where the pressure is above its mean and inward where it is below. The leak is of size h/μ , so it vanishes as the mesh is refined. This is the classical consistency error of a boundary penalty method [20], acting only on the pressure and with the vanishing parameter h/μ , and it is the reason why the printed method converges at all.

How fast does it converge? The leak is a normal velocity of size h/μ on the wall, so a naive trace argument would lift it to an H^1 error of order $(h/\mu)h_\Gamma^{-1/2}$, that is, rate $\frac{1}{2}$. The true H^1 rate is 1. The reason is incompressibility: for a divergence-free polynomial field on a boundary triangle, the normal derivative of the normal component equals minus the tangential derivative of the tangential component, so the purely *normal* load produced by the missing traction acts like a tangential derivative and can be integrated by parts along the wall. In the mesh-dependent energy norm, which contains $(\mu/h)^{1/2} \|\cdot\|_{L^2(\Gamma_h)}$, the leak is seen directly and the rate is exactly $\frac{1}{2}$. In short: the printed method converges, but not at Scott's rate, unless the pressure is constant along the wall. This explains why Scott's constant-pressure benchmark shows the rate $\frac{3}{2}$, and it is the answer to “why not” for the method as printed.

1.5 The repair

The obvious repair is to add the missing term $\langle p_h, v \cdot n \rangle$ to the momentum equation. With exactly divergence-free elements this does not work as it stands: the square linear system acquires the kernel vector $(u, p) = (0, 1)$, because a constant pressure now drops out of both equations. The standard fix, restricting the pressure to mean-zero functions, destroys pointwise incompressibility: the discrete divergence becomes a nonzero constant. The repair we analyse adds the traction with the *wall mean of the discrete pressure removed*, $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$, to the momentum equation only. The continuity equation is untouched, so the velocity stays exactly divergence-free, and the constant mode is fixed automatically. We call this method CNS* (consistent Nitsche, with the star marking the mean-free correction). The device is not new: Frachon, Nilsson and Zahedi [26] introduced the zero-net-flux formulation for divergence-free cut elements, and the velocity of CNS* coincides with the velocity of their formulation transcribed to our setting. What is new is the analysis on an inscribed polygon.

1.6 This paper and the companion letter

The two observations above, the leak law of the printed method and the rate $h_\Gamma^{3/2} + h_\Omega^k$ of CNS*, are announced with proof sketches, together with the sharpness of the exponent $\frac{3}{2}$ and the penalty threshold, in a short letter written jointly with C. de S. Cavalcante [25]. The present paper is its companion. It is organised around the material that the letter does not contain: when the zero-gradient lock of strong imposition actually occurs, which is the other half of the answer to “why not”; the theory of the penalty threshold beyond a single boundary triangle; and the exact H^1 constant of the leak. It also gives complete proofs of the results announced in [25].

Wherever a result of the letter is stated here, we say so at the statement, and Section 2 lists precisely what is in the letter and what is new here.

1.7 Main results

We state the results informally here; precise statements are in Sections 4–7. We write $h = h_\Omega$ for the bulk mesh size (the h in Scott’s penalty μ/h), h_Γ for the longest chord, $\gamma = \mu h_\Gamma/h$ for the effective boundary penalty, and $G = \|p - \bar{p}\|_{L^2(\Gamma)}$ for the variation of the pressure along the wall. The solution is assumed smooth, the mesh shape regular, and every polygon vertex is assumed to lie in at least three triangles (this is the Guzmán–Scott angle condition at the wall; Section 3).

- (1) *The zero-gradient lock is a property of the wall mesh* (Section 4). Under this angle condition, strongly imposed no-slip on the polygon already has H^1 error $\asymp h_\Gamma^{3/2} + h^k$, two-sided. The lock is caused by polygon vertices lying in only two triangles, which the angle condition excludes: each such vertex costs $h_\Gamma |\partial_n u(z)|$ in H^1 , one lock gives rate 1, and a positive fraction of locks gives rate $\frac{1}{2}$. With three or more triangles the same loss appears continuously as one triangle at the vertex degenerates. The advantages of Nitsche imposition on an inscribed polygon are therefore robustness to such meshes and a smaller constant, not a better rate. None of this is in [25].
- (2) *The penalty must scale with the mesh ratio* (Section 5). With Scott’s normalisation μ/h , the penalty must grow like $\rho = h/\min_e |e|$. Sufficiency holds on every mesh. Necessity is certified in exact rational arithmetic on every mesh whose shortest chord carries a triangle with apex angle at least 35° and base angles at least 5° , and is proved on every shape-regular mesh with explicit but very small constants. A necessity statement based on a single wall star is false in general (the exact counterexamples have small angles). The letter [25] contains only the single-triangle witness and the certified shape families, with a proof sketch.
- (3) *The H^1 constant of the leak* (Sections 6.5 and 8). The normalised H^1 error of the printed method tends to the energy of the Stokes flow that has the leak as Dirichlet data. This constant depends on the whole domain, not only on the wall. The part of the error caused by the missing traction converges to this leak flow uniformly in the penalty, the Dirichlet limit included, at the exact rate $h_\Gamma^{1/2}$ (stated in Section 8, proved in [27]). The letter gives only two-sided bounds for the H^1 error.
- (4) *Complete proofs of the results announced in [25]* (Sections 6 and 7). The printed method converges, but at rate 1 when the wall pressure varies: its H^1 error is at most $C_p h/\mu + C(h_\Gamma^{3/2} + h_\Omega^k)$, and at least $c(h/\mu)G$; in the energy norm it is $\asymp (h/\mu)^{1/2}G$ plus Scott’s terms. To leading order the error is an explicit field whose trace is $-(h/\mu)(p - \bar{p})n$; its slip and energy norms, normalised by $(h/\mu)G$ and $(h/\mu)^{1/2}G$, tend to 1. The mean-free correction attains Scott’s rate: CNS^* is well posed for γ above a threshold, its velocity is divergence-free pointwise, and $\|u - u_h\|_{H^1} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ for general Stokes data and every fixed $k \geq 4$. With a zero-flux correction of the outer boundary data the bound is uniform in the penalty and needs no relation between h_Γ and h_Ω . If the wall shear does not

vanish identically and $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$, with $\theta_{\min}$ the minimum angle of the triangles at the wall, the H^1 error is also bounded below by $ch_\Gamma^{3/2}$, so the term $h_\Gamma^{3/2}$ cannot be improved. The letter gives only the idea of the lower bound; here the argument is carried out in full, with the geometric star condition on which it rests constructed explicitly (its interval certificates are in [27]).

Table 1 summarises the rates.

Table 1 Rates for $k \geq 4$ on meshes satisfying the angle condition (M0)–(M2) of Section 3, fixed penalty γ , smooth solution, $h \asymp h_\Gamma$. “ $G > 0$ ” means the pressure is not constant on the wall; “ $W \not\equiv 0$ ” means the wall shear does not vanish identically.

method	H^1 error	energy error	where
printed (GS), $G = 0$	$\asymp h_\Gamma^{3/2}$ (if $W \not\equiv 0$)	$\lesssim h_\Gamma^{3/2}$	Thms. 6.1, 7.14
printed (GS), $G > 0$	$\asymp h/\mu$	$\asymp (h/\mu)^{1/2}$	Thms. 6.1–6.5
printed, growing penalty $\mu \gtrsim hh_\Gamma^{-3/2}$	$\asymp h_\Gamma^{3/2}$	rate ≤ 1 for any scaling	Cor. 7.16, Sec. 7.4
corrected (CNS*), any G	$\asymp h_\Gamma^{3/2}$ (if $W \not\equiv 0$)	$\lesssim h_\Gamma^{3/2}$	Thms. 7.2, 7.14
strong no-slip, ≥ 3 triangles at every wall vertex	$\asymp h_\Gamma^{3/2}$	–	Thm. 4.4
strong no-slip, locked wall vertices $\mathcal{L}$	$\asymp h_\Gamma^{3/2} + h_\Gamma (\sum_{\mathcal{L}} \partial_n u(z) ^2)^{1/2}$	–	Thm. 4.7

Beyond these four results, the extended version [27] of this paper, of which the present article is a condensed account, contains a theory of the pressure and of drag and lift, L^2 estimates, Clough–Tocher refinements with $k \geq 2$, general curved domains, steady Navier–Stokes flow, nearly singular wall stars, and pressure robustness. We state these results without proof in Section 8. The three-dimensional analysis is the subject of a companion paper [28].

1.8 How to read this paper

Section 2 compares our results with the recent literature and lists precisely what is in the letter [25] and what is new here. Section 3 fixes the setting and collects the geometric and discrete tools. The material that is new relative to the letter comes first: Section 4 treats strong imposition and the zero-gradient lock, and Section 5 the penalty threshold. Section 6 then gives the complete analysis of the printed method, including the H^1 constant of the leak, and Section 7 that of the corrected method and the sharpness of its rate; the main theorems of these two sections are announced in [25]. Section 8 surveys further results, Section 9 reports computations, and Section 10

concludes with open problems. Proofs that are routine, or that rely on long computer-assisted certificates, are sketched and referred to [27]; the code and all raw data are publicly available [29].

2 Related work and novelty

The problem sits at the meeting point of three lines of work: exactly divergence-free elements on curved domains, weak imposition of boundary conditions for Stokes flow, and the stability theory of Scott–Vogelius elements near singular and nearly singular vertices. We discuss each and then state what is new.

2.1 The prize problem and the constant-pressure case

The question we answer was posed by Scott [15], on the basis of the computations of Gjerde and Scott [16]. Cavalcante [30] has analysed the constant-pressure (zero-pressure-gradient) benchmark of the prize: the abstract of that preprint states the bound $\|u - u_h\|_{H^1} \leq C(h_\Omega^k + h_\Gamma^{3/2})$ for this benchmark with quartic velocities on an inscribed polygon, and it reports a coercivity transition of the penalty near $\mu \approx 60.338$ for its test mesh; our code gives $\mu^* = 60.343$ at the same resolution ($N = 20$ chords). In our notation that is the case $G = 0$ of the printed method, which we recover in Theorem 6.1; our proofs do not use [30]. The present paper treats general Stokes data. The difference is essential: once the wall pressure varies, the printed method no longer has Scott’s rate (Theorems 6.1–6.5), and a corrected method is needed. The analysis of the constant-pressure case and ours are thus complementary answers to the two halves of Scott’s question: the “if” (the estimate holds when the pressure is constant on the wall) and the “why not” (it fails otherwise, for an identifiable reason).

The core of our answer for general data, the leak law and the mean-free correction, is announced with proof sketches in a short letter by Cavalcante and the author [25], of which the present paper is the companion; Section 2.6 below separates the content of the letter from what is new here. The constant-pressure analysis is Cavalcante’s [30] and is not reproduced in the present paper.

2.2 Divergence-free elements on curved domains

Neilan and Otus [17] constructed Scott–Vogelius elements on curved two-dimensional domains, with boundary elements mapped to fit the geometry, and proved convergence estimates; they also showed how to recover pressure robustness by reconstructing the load with commuting interpolants [17, Theorem 6.1]. Durst and Neilan [18] extended the approach to general degree on smooth planar domains. In three dimensions, Chen and Liu [31] obtain pressure-robust, optimal-order h^k estimates with isoparametric Piola-mapped elements on curved domains. Methods that fit the boundary can reach rates above $h_\Gamma^{3/2}$ (Chen and Liu attain h^k); the inscribed-polygon Nitsche route analysed here is the low-cost, straight-sided alternative, whose best possible rate is $h_\Gamma^{3/2} + h^k$ (Theorem 7.14). We therefore do not regard it as a competitor on rate, but as the analysis of a method that is easy to implement in any code that already has Scott–Vogelius elements.

The work closest to ours is that of Liu, Neilan and Otus [32]. They use Scott–Vogelius elements on Clough–Tocher splits of a mesh lying inside a curved domain, impose the tangential condition by a Nitsche-type form with a Taylor boundary correction, and enforce the normal component by a separate mean-free boundary multiplier which, as they note, approximates the wall pressure up to a constant. Together with a zero-flux constraint on the velocity space the method is exactly divergence-free and converges at the optimal order h^k ; without the constraint, they remark, it is ill posed [32, Remark 3.2]. Our Proposition 7.1 is the analogue of this remark for the square Scott–Vogelius system. Our subject is different: the method as Gjerde and Scott printed it, and its minimal repair, with no extra unknowns, no boundary correction and the pressure itself on the wall.

For cut (unfitted) meshes, Liu, Neilan and Olshanskii [33] place the boundary pressure term in both the momentum and the continuity equation; the discrete divergence then vanishes only away from an $O(h)$ strip at the boundary. Frachon, Nilsson and Zahedi [26] place it in the momentum equation only, so as not to perturb the divergence condition, and show that fixing the pressure mean by a Lagrange multiplier makes the discrete divergence equal to that multiplier [26, (5.7)–(5.9)]. They avoid this by testing the momentum equation only with velocities of zero net boundary flux. Our CNS* is this device in square form (Remark 7.6); we claim no novelty for it. Burman, Hansbo and Larson give related multiplier formulations for divergence-free cut elements [34]. Exactly divergence-free methods also require discrete Dirichlet data with zero net flux, as Eickmann, Scott and Tscherpel point out for polygonal domains [35]; our flux-corrected outer data (Definition 3.2) are a simple instance.

2.3 Nitsche’s method for Stokes flow

Weak imposition of Dirichlet data in Nitsche’s sense was developed further by Freund and Stenberg [22] and Stenberg [21]; for Stokes flow the consistent form uses the full traction and contains the pressure term [21–24]. Penalty-free variants [36, 37] and Robin-type generalisations [38] exist. That the pressure term is needed for consistency is known [12], and that the Nitsche penalty must scale with the local boundary mesh size follows from the trace-inverse inequality [21, 38]. What has not been done, to our knowledge, is an analysis of a Nitsche–Stokes form *without* the pressure term: whether it converges, at what rate, and what its error looks like. The mechanism, a consistency error of penalty type acting only on the pressure, is that of Babuška’s penalty method [20].

On the geometric side, the polygonal approximation of curved boundaries for scalar problems is classical [7, 8], and higher-order boundary corrections on polygons are due to Bramble, Dupont and Thomée [10] and, more recently, Dupont, Guzmán and Scott [13] (for the Poisson problem). Bernardi, Costabel, Dauge and Girault [39] show that the continuous inf-sup constant of an inscribed polygonal approximation converges to that of the curved domain; we use a simple gluing argument for the same purpose. None of the cut-element or boundary-correction works above uses exactly divergence-free spaces on an inscribed polygon.

2.4 Singular and nearly singular vertices

The stability of Scott–Vogelius elements degenerates near singular vertices, and the Guzmán–Scott quantity $\Theta(z)$ [2] measures this. Gräßle, Bohne and Sauter [40] introduced pressure constraints (the “pressure-wired” element) that restore a shape-robust inf-sup constant at nearly singular vertices, and spurious pressures with robust velocities, together with postprocessing filters, were studied by Park [41] and by Bohne, Gräßle and Sauter [42]. Our constrained inf-sup result for strong imposition (Theorem 4.7(a)) is essentially the wall-vertex case of theirs, and our pressure filter is in the same spirit. Kean, Neilan and Schneier [43] analyse Scott–Vogelius elements on anisotropic meshes obtained by barycentric and incenter refinement. What is different in our setting is that the loss at a locked wall vertex is an *approximation* loss, caused by the zero trace on the polygon, and not a stability loss; the velocity error can therefore be characterised two-sidedly.

The zero-gradient lock at a wall vertex with two triangles, and the statement that no constraint arises with more than two triangles, are due to Scott [15, Theorem 1]; Gjerde and Scott treat the two-triangle case in [16, §3.4, App. B].

2.5 Broader context

Several further strands of the literature enter the analysis or its extensions. The divergence-free pairs used in three dimensions, relevant to the companion paper [28], are stable on Alfeld (barycentric) splits [4, 44], on Worsey–Farin splits [45], and on Freudenthal meshes for large degree [46]; smooth discrete de Rham complexes [47, 48] provide the corresponding smooth stream-function spaces, and Farrell, Mitchell and Scott [49] survey what is known and conjectured. A local Fortin projection for Scott–Vogelius elements on general meshes has recently been constructed by Eickmann, Guzmán, Neilan, Scott and Tscherpel [50]. Our L^2 estimates on graded meshes use local energy estimates in the spirit of Arnold and Liu [51] and of Demlow, Guzmán and Schatz [52], and the regularity result of Kellogg and Osborn [53] on convex polygons. The Navier–Stokes extension follows the framework of nonsingular solution branches of Brezzi, Rappaz and Raviart [54, 55], with a duality argument of Schatz type [56] for explicit thresholds. Sharp trace-inverse constants for polynomials [57] enter the analysis of the pressure in the first row of boundary triangles. For nearly singular wall stars we use anisotropic interpolation [58, 59] and right inverses of the divergence on John domains [60].

2.6 What is known, what is in the companion letter, and what is new

Known, and used or recovered here.

- The pressure term is necessary for consistency of Nitsche’s method for Stokes flow [12]; penalty consistency errors of the leak type are classical [20].
- The mean-free (zero-net-flux) placement of the pressure term that keeps the discrete velocity divergence-free [26]; the need for a flux constraint or correction [32, 35].

- Sufficiency of a penalty proportional to the ratio of bulk to boundary mesh size (the trace-inverse inequality [21, 38]).
- The two-triangle lock [15, 16]; constrained and filtered pressures at nearly singular vertices [40–42].
- The H^1 bound $h_\Omega^k + h_\Gamma^{3/2}$ for the constant-pressure benchmark, for every fixed $k \geq 4$ and for every $k \geq 2$ on Clough–Tocher refinements, due to Cavalcante [30, Thm. 5.2, Cor. 5.3].

Announced in the joint letter [25] (C. de S. Cavalcante and the author), with proof sketches; complete proofs are given here.

- The leak law of the printed Gjerde–Scott method for general data: convergence at the sharp rate h/μ in H^1 and $(h/\mu)^{1/2}$ in energy, and the explicit leading error term with constant 1 in the slip and energy norms (Theorems 6.1–6.5, Corollary 6.6).
- The failure of the naive and mean-zero consistent forms (Proposition 7.1), and the proof that the mean-free correction CNS* attains Scott’s rate $h_\Gamma^{3/2} + h_\Omega^k$ for general Stokes data (Theorem 7.2).
- Announced there with a pointer or an idea of proof only: the penalty-uniform bound under flux-corrected data (Theorem 7.3), the sharpness of $h_\Gamma^{3/2}$ for every $k \geq 4$ (Theorem 7.14, Corollary 7.15), and the effect of a growing penalty (Corollary 7.16, Theorem 7.17).
- The penalty threshold at the level of a single boundary triangle: sufficiency, the single-triangle witness and the certified shape families (Theorem 5.1, Proposition 5.2, Corollary 5.3).
- Computations with $k = 4$ confirming the rates, the leak constants and the measured threshold.

New in this paper, as far as we know, and not contained in [25].

- A two-sided theory of strongly imposed no-slip on the inscribed polygon, which says when the zero-gradient lock occurs: the two-sided rate $h_\Gamma^{3/2} + h^k$ under (M2), the counting lower bound at locked vertices, the sharp two-sided rate with locks, and the needle-angle law (Section 4). The upper bound $h_\Gamma^{3/2} + h^k$ under the Guzmán–Scott condition is close to a corollary of [2] and the scalar theory [7, 8]; what is new there is only the observation that it holds, and that the lock therefore needs two-triangle or nearly singular wall stars.
- The penalty theory beyond a single triangle (Section 5): exact counterexamples to purely local (single-star) necessity, necessity from a locally resolved region and on every shape-regular mesh, and exact witnesses on layered meshes with tall boundary triangles.
- The identification of the H^1 constant of the leak with the energy of a Stokes leak flow, with a two-sided bracket (Section 6.5), and the penalty-uniform leak limit at the exact rate $h_\Gamma^{1/2}$ (Section 8).
- Full proofs of the auxiliary results that the letter only sketches or cites, among them the normal-load cancellation (Lemma 6.2), the divergence-free lift and discrete Dirichlet problem behind the penalty-uniform bound (Lemmas 7.4, 7.5), and the duality argument and star construction behind the lower bound (Lemmas 7.7–7.13).

- The results surveyed in Section 8 (pressure and drag, L^2 , Navier–Stokes, general domains, nearly singular stars, pressure robustness), proved in [27], some of them only under the additional hypotheses listed in Table 6.

The letter is joint work with Cavalcante, and its author contributions are stated there; the present paper is by the author alone.

3 Setting, methods and tools

3.1 The model problem

To keep the geometry explicit we work, as Gjerde and Scott do, with flow past a disk in a box. Let $R = (-L, L)^2$ with $L > 1$, let D be the open unit disk, $\Gamma = \partial D$ the wall, and $\Omega = R \setminus \overline{D}$. We seek (u, p) with

$$-\Delta u + \nabla p = f, \quad \operatorname{div} u = 0 \quad \text{in } \Omega, \quad u = 0 \text{ on } \Gamma, \quad u = g \text{ on } \partial R.$$

Curved domains bounded by several smooth curves, convex or not, are treated in [27] (Section 8). The disk has one feature that the general case lacks and that makes the exposition cleaner: every point of the polygon has an explicit nearest point on Γ .

The polygon. Let P_h be a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h$. Since $P_h \subset D$, the computational domain Ω_h contains Ω ; the difference $S_h = \Omega_h \setminus \Omega$ is a union of thin circular segments, one between each chord and its arc. The outward normal of Ω_h on Γ_h is denoted n ; it points into P_h . On a chord e we write n_e and τ_e for its normal and tangent, and $\langle \cdot, \cdot \rangle$ for the $L^2(\Gamma_h)$ pairing.

The mesh. $\mathcal{T}_h$ is a conforming triangulation of Ω_h and $\mathcal{E}_h^\Gamma$ the set of its edges on Γ_h (the chords). We use four parameters:

$$h = h_\Omega := \max_{T \in \mathcal{T}_h} \operatorname{diam} T, \quad h_\Gamma := \max_{e \in \mathcal{E}_h^\Gamma} |e|, \quad \rho := \frac{h}{\min_{e \in \mathcal{E}_h^\Gamma} |e|}, \quad \gamma := \frac{\mu h_\Gamma}{h}.$$

Here h is the h in Scott's penalty μ/h , and γ is the *effective boundary penalty*: $\mu/h = \gamma/h_\Gamma$. Keeping h_Γ and h_Ω separate matters, because Scott's expected rate $h_\Gamma^{3/2} + h_\Omega^k$ distinguishes them and because the penalty threshold depends on their ratio. We assume:

Assumption 3.1 (Mesh)

- (M0) *Shape regularity and comparable chords*: the triangles are shape regular, and $c_0 h_\Gamma \leq |e| \leq h_\Gamma$ for all $e \in \mathcal{E}_h^\Gamma$. No quasi-uniformity is assumed; $h_\Gamma \ll h$ is allowed.
- (M1) *No singular vertices*: no interior vertex has all its edges on two lines, and no boundary vertex lies in only one triangle or has all its edges on two lines.
- (M2) *The Guzmán–Scott angle condition*: $\Theta(z) := \max_j |\sin(\theta_j + \theta_{j+1})| \geq \Theta_0 > 0$ at every vertex z , where $\theta_1, \theta_2, \dots$ are the consecutive angles at z [2] (at boundary vertices the wrap-around term is omitted).

At a vertex of Γ_h the angle of Ω_h is π plus the two small angles between the chords and the tangent, so it is $\pi + O(h_\Gamma)$. With only two triangles at such a vertex, the only pair of consecutive angles sums to $\pi + O(h_\Gamma)$, and $\Theta(z) = O(h_\Gamma)$. So, for small h_Γ , (M2) at the vertices of Γ_h says precisely that *every polygon vertex lies in at least three*

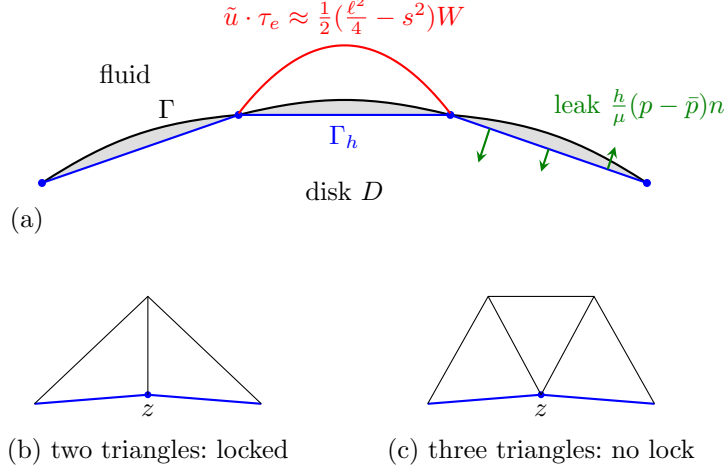


Fig. 1 (a) The wall Γ , the inscribed polygon Γ_h and the thin strip between them (grey, exaggerated). On each chord the extended solution is a tangential velocity whose profile along the chord is a parabola of height $\ell^2 W/8$ (the chordal bump, red), which produces Scott's $h_\Gamma^{3/2}$. The printed method adds a normal leak $(h/\mu)(p - \bar{p})$ (green). (b) A polygon vertex in two triangles: divergence-free fields vanishing on both chords have zero gradient at z . (c) With three triangles the gradient can jump across the interior edges, and the lock disappears (Section 4).

triangles (Lemma 4.1(b) below). This innocent-looking consequence will turn out to be the whole story of the zero-gradient lock (Figure 1 and Section 4).

Spaces. Fix $k \geq 4$. V_h is the space of continuous piecewise- P_k vector fields on $\mathcal{T}_h$, $V_h^R := \{v \in V_h : v|_{\partial R} = 0\}$, $V_h^0 := \{v \in V_h : v|_{\partial \Omega_h} = 0\}$, Π_h is the space of discontinuous piecewise- P_{k-1} functions, $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$ and $Y_h := Z_h \cap V_h^0$. The discrete outer data g_I are the P_k Lagrange interpolant of g , and $W_h := \{z \in V_h : z|_{\partial R} = g_I, \operatorname{div} z = 0\}$.

An exactly divergence-free method is sensitive to the net flux of its Dirichlet data [35]: every $z \in W_h$ has $\int_{\Gamma_h} z \cdot n = -\int_{\partial R} g_I \cdot \nu$ (ν the outward normal of R), while the exact solution has zero flux through Γ . The interpolant g_I has a small nonzero flux $m_R := \int_{\partial R} g_I \cdot \nu$, of order h^{σ_k} with $\sigma_k = k + 2$ for even k and $k + 1$ for odd k (Newton–Cotes). It is removed by a one-line correction.

Definition 3.2 (Flux-corrected outer data) $\tilde{g}_I := g_I - \frac{m_R}{8L^2} x$ on ∂R , where x is the position field. Since $\int_{\partial R} x \cdot \nu = \int_R \operatorname{div} x = 8L^2$, $\tilde{g}_I$ has zero net flux, and it is again continuous and piecewise P_k .

3.2 The two Nitsche methods

With $Du = \nabla u + (\nabla u)^\top$, $a_h(u, v) = \frac{1}{2}(Du, Dv)$ and $\partial_n u := (\nabla u)n$, the symmetric Nitsche form of Gjerde and Scott is

$$N_h(u, v) = a_h(u, v) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + \frac{\mu}{h} \langle u, v \rangle.$$

In words: the viscous form, minus the normal flux of the solution tested against the test function (the consistency term), minus its symmetric counterpart, plus the penalty. The only piece of the Stokes traction that is represented is the viscous flux $(\nabla u)n$; the pressure traction $-pn$ is absent.

Definition 3.3 (The printed method GS [16, (27)–(28)]) Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$ and $p_h \in \Pi_h$ such that

$$N_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^R, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$$

Here $\tilde{f}$ is a smooth extension of f to the strip S_h . Two details of the transcription: Gjerde and Scott write the pressure form with the opposite sign (our p_h is their $-p_h$), and their text defines the test space as vanishing on Γ_h , which would remove the Nitsche terms altogether; the evident intended test space is V_h^R . The outer boundary carries strongly imposed data; the Nitsche terms act only on the polygon.

Definition 3.4 (The mean-free correction CNS*) As GS, with the momentum equation

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle = (\tilde{f}, v) \quad \forall v \in V_h^R,$$

$$\bar{p}_\Gamma(q) := \frac{1}{|\Gamma_h|} \int_{\Gamma_h} q.$$

The added term is the missing pressure traction, with the wall mean of the discrete pressure removed. It sits in the momentum row only, so the continuity row, and with it the pointwise incompressibility of u_h , is unchanged. Removing the mean is what makes the square system nonsingular (Proposition 7.1). Methods “with flux-corrected data” replace g_I by $\tilde{g}_I$.

3.3 The exact solution on the polygonal domain

The exact solution lives on Ω , the discrete one on the larger domain Ω_h . To compare them we extend u across Γ in a divergence-free way. Since $u \cdot n = 0$ on Γ , the stream function ψ with $u = \operatorname{curl} \psi = (\partial_y \psi, -\partial_x \psi)$ is single-valued; let $\tilde{\psi}$ be a smooth extension of ψ to a neighbourhood of $\bar{\Omega}$ and put $\tilde{u} := \operatorname{curl} \tilde{\psi}$, so $\operatorname{div} \tilde{u} = 0$ everywhere. Let $\tilde{p}$ be a smooth extension of p . The extended fields satisfy the momentum equation up to a defect $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$, which vanishes on Ω and is supported in the thin strip S_h .

Assumption 3.5 (Smooth solution) (u, p) and $\tilde{f}$ are smooth and bounded on a fixed neighbourhood of $\bar{\Omega}$, and $h \leq h_*$ is small.

Near Γ this follows from smooth data. At the four corners of the box it is a genuine hypothesis on the *solution*: for generic smooth, compatible data the Stokes velocity at a right-angled corner is only in H^s for $s < 3.7396$ [61]. It holds for the manufactured tests of Section 9. For data-driven problems on meshes that are quasi-uniform at the corners the terms h^k below become $h^{\min(k, 2.7396) - \epsilon}$, and the terms h^{k+1} of the L^2

bounds degrade likewise, unless the mesh is graded there [27]. The terms involving h_Γ are unaffected.

Norms and constants. For fields with traces on Γ_h we use the mesh-dependent energy norm

$$\|v\|^2 := \|\nabla v\|_{\Omega_h}^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2.$$

The middle term is the *slip* part. We write $G := \|p - \bar{p}\|_{L^2(\Gamma)}$, where $\bar{p}$ is the mean of p over Γ ; $G = 0$ exactly when the pressure is constant along the wall. Constants C, c depend on (M0)–(M2), k, L and norms of (u, p, ψ) , never on $h, h_\Gamma, \mu, \gamma, \rho$ unless displayed; C_p may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

3.4 Geometry of chords

Everything that distinguishes the polygon from the circle is contained in the following elementary facts.

Lemma 3.6 (Chords) *Let e be a chord of length ℓ , $x \in e$, s the signed arclength from the midpoint m_e of e , and $x^* = x/|x|$ the nearest point of Γ . Then $d := \text{dist}(x, \Gamma) = \frac{1}{2}(\frac{\ell^2}{4} - s^2) + O(\ell^4)$, in particular $d \leq \ell^2/4$, and, up to a fixed orientation sign,*

$$n(x^*) - n_e = -s\tau_e + O(\ell^2).$$

The arclength Jacobian of the radial projection $\Gamma_h \rightarrow \Gamma$ is $1 + O(\ell^2)$.

Proof $|x|^2 = 1 - \ell^2/4 + s^2$. The normal at x^* is radial, and its angle differs from that of n_e by the central angle to x^* , which is $s + O(s^3)$. $\square$

Lemma 3.7 (Wall structure and extension) *On Γ we have $\nabla u = \partial_n u \otimes n$ and $\partial_n u \cdot n = 0$; hence $\partial_n u = W\tau$ with the wall shear $W := \partial_n u \cdot \tau$. On Γ_h , $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$, and for $x \in e$,*

$$\tilde{u}(x) = dW(x^*)\tau(x^*) + O(d^2) = \frac{1}{2}\left(\frac{\ell^2}{4} - s^2\right)W(m_e^*)\tau_e + O(h_\Gamma^3). \quad (1)$$

Moreover $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ for $v \in V_h^R$.

Proof Tangential derivatives of u vanish on Γ , and $\text{div } u = n \cdot \partial_n u$. Taylor-expand $\tilde{u}$ at x^* and use Lemma 3.6. For the last bound, F is bounded and supported in the strip S_h of width $O(h_\Gamma^2)$; apply a one-dimensional Poincaré inequality across the strip and the uniform Friedrichs inequality of Lemma 3.8. $\square$

Formula (1) is the *chordal bump*: on each chord the extended solution is a parabola of height $\ell^2 W/8$ in the tangential direction. It carries the whole of Scott's $h_\Gamma^{3/2}$ (Section 7).

Lemma 3.8 (Uniform trace, Friedrichs and Korn inequalities) *There are $C_{\text{tr}}, C_F, \kappa$, independent of h , such that every $v \in H^1(\Omega_h)^2$ with $v|_{\partial R} = 0$ satisfies $\|v\|_{L^2(\Gamma_h)} \leq C_{\text{tr}} \|\nabla v\|$, $\|v\| \leq C_F \|\nabla v\|$ and $\|\nabla v\|^2 \leq \kappa a_h(v, v)$.*

Proof Write Γ_h in polar form as $r = \rho_h(\theta)$. On a chord with endpoint angles $\theta_m \pm \alpha$, $\rho_h = \cos \alpha / \cos(\theta - \theta_m)$, so $\|\rho_h - 1\|_\infty \leq Ch_\Gamma^2$ and $\|\rho_h'\|_\infty \leq Ch_\Gamma$. Let χ be smooth with $\chi(1) = 1$ and $\chi \equiv 0$ outside a fixed collar in Ω , and $\Phi_h(r, \theta) = (r + \chi(r)(\rho_h(\theta) - 1), \theta)$. Then Φ_h maps Ω onto Ω_h and Γ onto Γ_h , is the identity near ∂R , and $\|D\Phi_h - I\|_\infty \leq Ch_\Gamma$. For $w = v \circ \Phi_h$ we have $D(w) = D(v) \circ \Phi_h + E$ with $|E| \leq Ch_\Gamma |\nabla v \circ \Phi_h|$. On the fixed domain Ω the function w vanishes on ∂R , so Korn's first inequality, the trace inequality and Friedrichs' inequality hold there; changing variables back gives $\|\nabla v\| \leq C(\|D(v)\| + h_\Gamma \|\nabla v\|)$, and the last term is absorbed for $h_\Gamma \leq h_*$. $\square$

The next lemma is the first place where the geometry produces the exponent $\frac{3}{2}$. Its content is a cancellation: the transposed-gradient term that the symmetric-gradient form leaves on the chords is only $O(h_\Gamma)$ pointwise, but its leading part is odd in s on each chord.

Lemma 3.9 (Transposed-gradient term) *For $v \in V_h^R$, $|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq Ch_\Gamma^{3/2} \|\nabla v\|$.*

Proof By Taylor expansion at x^* and Lemmas 3.6–3.7, $(\nabla \tilde{u})^\top n_e = n(x^*)(\partial_n u(x^*) \cdot (n_e - n(x^*))) + O(h_\Gamma^2) = s a_e + O(h_\Gamma^2)$ on e , with $a_e := n(m_e^*)W(m_e^*)$ constant on e . Since $\int_e s ds = 0$,

$$\left| \int_e s a_e \cdot v \right| = \left| \int_e s a_e \cdot (v - v(m_e)) \right| \leq \frac{|e|^2}{4} \|\partial_\tau v\|_{L^1(e)} \leq C|e|^{5/2} \|\partial_\tau v\|_{L^2(e)}.$$

With the inverse trace inequality $|e| \|\partial_\tau v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|_{T_e}^2$ and $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$, the sum over the chords is at most $Ch_\Gamma^{3/2} \|\nabla v\|$. The remainder $O(h_\Gamma^2)$ is handled by the trace inequality of Lemma 3.8. $\square$

3.5 Discrete tools

Lemma 3.10 (Inverse traces) *For $v \in V_h$, $\sum_{e \in \mathcal{E}_h^\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2 \leq C_I^2 \|\nabla v\|^2$, hence $\|\partial_n v\|_{\Gamma_h} \leq C_I(\rho/h)^{1/2} \|\nabla v\|$. For $q \in \Pi_h$, $\|q\|_{\Gamma_h}^2 \leq C_I^2(\rho/h) \|q\|^2$.*

The following lemma is the standard coercivity argument for Nitsche's method, written so that the role of ρ is visible.

Lemma 3.11 (Coercivity and continuity) *Let $\mu_0 := 4\kappa\rho C_I^2$. If $\mu \geq \mu_0$ (which holds when $\gamma \geq \gamma_0 := 4\kappa C_I^2/c_0$), then*

$$N_h(v, v) \geq c_1 \|v\|^2 \quad \forall v \in V_h^R, \quad |N_h(w, v)| \leq M \|w\| \|v\|,$$

with c_1, M independent of h, μ and ρ .

Proof By Lemma 3.10 and Young's inequality, $2|\langle \partial_n v, v \rangle| \leq \frac{1}{2} a_h(v, v) + 2\kappa\rho C_I^2 h^{-1} \|v\|_{\Gamma_h}^2$. Hence $N_h(v, v) \geq \frac{1}{2} a_h(v, v) + (\mu - 2\kappa\rho C_I^2) h^{-1} \|v\|_{\Gamma_h}^2$, and for $\mu \geq \mu_0$ this is at least $(2\kappa)^{-1} \|\nabla v\|^2 + (\mu/2h) \|v\|_{\Gamma_h}^2$. Adding the inverse trace bound for the last part of $\|\cdot\|$ gives coercivity. Continuity: each ∂_n -term is at most $(\rho/\mu)^{1/2} \|w\| \|v\| \leq (4\kappa C_I^2)^{-1/2} \|w\| \|v\|$. $\square$

Scott–Vogelius theory enters through the next two results.

Lemma 3.12 (Uniform inf-sup constant) *There is $\beta > 0$, depending only on k , Θ_0 , shape regularity and L , such that for every $q \in \Pi_h \cap L_0^2(\Omega_h)$ there is $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq \beta^{-1}\|q\|$.*

Sketch Guzmán and Scott [2, Theorem 1] prove this for $k \geq 4$ under (M1)–(M2), with a local Fortin construction that uses neither quasi-uniformity nor small h . The domain enters only through the continuous inf-sup constant and a Poincaré constant. The latter is uniform by Lemma 3.8. For the former, cover Ω_h by the pieces $W_j \cap \Omega_h$, where the W_j are fixed angular wedges about the centre of the disk; each piece is star-shaped with respect to a ball of fixed radius (every chord line meeting W_j leaves the ball on its outer side, because $P_h \subset D$), and the pieces have diameters, ball radii and overlaps bounded independently of h . Gluing Bogovskiĭ operators over such a finite cover gives a uniform continuous inf-sup constant [27]. Alternatively, Bernardi, Costabel, Dauge and Girault prove directly that the continuous inf-sup constant of an inscribed polygonal approximation differs from that of the curved domain by $O(h)$ [39, Theorem 4.6]. $\square$

Lemma 3.13 (Divergence range and approximation) *Under (M1), $\operatorname{div} V_h^R = \Pi_h$, $W_h \neq \emptyset$, and the velocities of GS and CNS* are divergence-free pointwise. Moreover*

$$E_h := \inf_{z \in W_h} \|\tilde{u} - z\| \leq C\varepsilon_h, \quad \varepsilon_h := (1 + (\gamma h_\Gamma)^{1/2}) h^k + h^{\sigma_k} (h_\Gamma^{-1/2} + (\mu/h)^{1/2}).$$

With flux-corrected data the flux part $h^{\sigma_k}(\dots)$ is replaced by $(1 + \gamma^{1/2})h_\Gamma^{\sigma_k-1/2} + |m_R|$.

Proof Divergence range. Lemma 3.12 gives $\operatorname{div} V_h^0 = \Pi_h \cap L_0^2$. The quadratic edge bubble $b = \beta_e n_e$ on a single chord lies in V_h^R and has $\int \operatorname{div} b = \int_{\Gamma_h} b \cdot n = |e|/6 \neq 0$; this adds the constants, so $\operatorname{div} V_h^R = \Pi_h$. For any $\mathcal{G} \in V_h$ with trace g_I on ∂R we have $\operatorname{div} \mathcal{G} \in \Pi_h = \operatorname{div} V_h^R$, so $\mathcal{G} - v \in W_h$ for a suitable $v \in V_h^R$. For the discrete velocity, $u_h - \mathcal{G} \in V_h^R$ and the continuity equation forces $\operatorname{div} u_h = 0$.

Interpolation. Let $w_h := I_h \tilde{u}$ be the Lagrange interpolant; $w_h = g_I$ on ∂R . Boundary triangles have diameter at most Ch_Γ , so $\|\nabla(\tilde{u} - w_h)\| \leq Ch^k$, $\|\tilde{u} - w_h\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^{k+1}$ and $\|\partial_n(\tilde{u} - w_h)\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^k$, whence $\|\tilde{u} - w_h\| \leq C(1 + (\gamma h_\Gamma)^{1/2})h^k$. Moreover $\operatorname{div} w_h = \operatorname{div}(w_h - \tilde{u}) \in \Pi_h$ has norm at most Ch^k .

Net flux. Since $\tilde{u} = g$ on ∂R and $\operatorname{div} \tilde{u} = 0$, $\int_{\Gamma_h} \tilde{u} \cdot n = -\int_{\partial R} g \cdot \nu = 0$, and therefore

$$m := \int_{\Omega_h} \operatorname{div} w_h = \int_{\partial R} (g_I - g) \cdot \nu + \int_{\Gamma_h} (w_h - \tilde{u}) \cdot n, \quad |m| \leq Ch^{\sigma_k}.$$

Equispaced P_k interpolation on an edge integrates like the closed $(k+1)$ -point Newton–Cotes rule, which is exact for polynomials of degree $\sigma_k - 1$; so each edge contributes $O(|e|^{\sigma_k+1})$.

Removing the mean. Let $b := \sum_{e \in \mathcal{E}_h} \beta_e n_e$, so $\int \operatorname{div} b = |\Gamma_h|/6$, and $a := 6m/|\Gamma_h|$. Since $\#\mathcal{E}_h^\Gamma \leq C/h_\Gamma$, $\|ab\| \leq C|a|(h_\Gamma^{-1/2} + (\mu/h)^{1/2}) \leq C\varepsilon_h$, and $\|\operatorname{div}(ab)\| \leq C|a|h_\Gamma^{-1/2} \leq C\varepsilon_h$.

Divergence correction. The field $q := \operatorname{div}(w_h - ab) \in \Pi_h$ has mean zero and $\|q\| \leq C\varepsilon_h$. Lemma 3.12 gives $v \in V_h^0$ with $\operatorname{div} v = q$ and $\|\nabla v\| \leq C\varepsilon_h$; since $v = 0$ on Γ_h , $\|v\| \leq (1 + C_I^2)^{1/2} \|\nabla v\|$. Then $z := w_h - ab - v \in W_h$ and $\|\tilde{u} - z\| \leq C\varepsilon_h$.

Flux-corrected data. Replace w_h by $I_h \tilde{u} - c I_h(\zeta x)$ with $c = m_R/(8L^2)$ and ζ a cut-off equal to 1 near ∂R and 0 near Γ . Its trace on ∂R is $\tilde{g}_I$, it costs $C|m_R|$ in energy (no boundary terms on Γ_h), and its net flux is now $\int_{\Gamma_h} (I_h \tilde{u} - \tilde{u}) \cdot n = O(h_\Gamma^{\sigma_k})$, because only the chords contribute. The bubble step then costs $C(1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$. $\square$

The flux term is genuine for the plain interpolant: every $z \in W_h$, including the discrete velocity, has $\int_{\Gamma_h} z \cdot n = -m_R$, so $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$. For bounded γ it is below Ch^k whenever $h_\Gamma \geq h^{2(\sigma_k - k)}$ ($h_\Gamma \geq h^4$ for even k , $h_\Gamma \geq h^2$ for odd k).

3.6 The error equation

Testing the extended solution in the Nitsche form produces the pressure traction, which is where the two methods part ways.

Lemma 3.14 (Error equation) *For $v \in V_h^R$,*

$$\begin{aligned} N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle &= (\tilde{f}, v) + \mathcal{G}(v), \\ \mathcal{G}(v) &:= \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v), \end{aligned}$$

and $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2} \|v\|$; for $v \in V_h^0$, $|\mathcal{G}(v)| \leq Ch_\Gamma^{3/2} \|\nabla v\|$.

Proof Since $D(\tilde{u})$ is symmetric and $v = 0$ on ∂R , $a_h(\tilde{u}, v) = -(\operatorname{div} D(\tilde{u}), v) + \langle D(\tilde{u})n, v \rangle$, and $\operatorname{div} D(\tilde{u}) = \Delta \tilde{u}$ because $\operatorname{div} \tilde{u} = 0$. Insert $-\Delta \tilde{u} = \tilde{f} - F - \nabla \tilde{p}$, integrate $-(\nabla \tilde{p}, v) = (\tilde{p}, \operatorname{div} v) - \langle \tilde{p}, v \cdot n \rangle$ by parts, and use $D(\tilde{u})n - \partial_n \tilde{u} = (\nabla \tilde{u})^\top n$. For the bound: the transposed term is Lemma 3.9; $|\langle \tilde{u}, \partial_n v \rangle| \leq Ch_\Gamma^2 \cdot C_I(\rho/h)^{1/2} \|\nabla v\| \leq Ch_\Gamma^{3/2} \|\nabla v\|$; the penalty term is at most $(\mu/h)^{1/2} Ch_\Gamma^2 \|v\| = C\gamma^{1/2} h_\Gamma^{3/2} \|v\|$; and the F -term is Lemma 3.7. For $v \in V_h^0$ the transposed and penalty terms vanish. $\square$

The functional $\mathcal{G}$ is the geometric consistency error, common to both methods and of Scott's order $h_\Gamma^{3/2}$. For GS the error equation has one more term. Testing with divergence-free $v \in Z_h$, which have zero net flux through Γ_h , we can subtract any constant from the pressure:

Corollary 3.15 (The printed method on divergence-free test functions) *For $v \in Z_h$ and every constant c ,*

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - c, v \cdot n \rangle.$$

We will take $c = \bar{p}$. The functional $\mathcal{R}$ is the *missing traction*. It is of size G , not small, and it is the subject of Section 6.

3.7 A roadmap of the error analysis

All the upper bounds below are built from the same three ingredients, and it may help to see them side by side before the details.

- (i) *Approximation.* How well can divergence-free discrete fields with the right outer data approximate the extended solution? Lemma 3.13 answers: to within ε_h , which is h^k plus a flux term that the flux-corrected data remove. This is where Scott–Vogelius theory (Lemma 3.12) and the angle condition (M2) enter.
- (ii) *Geometric consistency.* The extended solution does not vanish on the chords; it has the chordal bump (1) of height $O(h_\Gamma^2)$. Every Nitsche term that sees this trace produces a consistency error, collected in $\mathcal{G}$, of order $h_\Gamma^{3/2}$ (Lemma 3.14); the exponent $\frac{3}{2}$ comes either from odd symmetry on each chord (Lemma 3.9) or from the $H^{1/2}$ size of the bump. This error is common to all methods that impose the wall condition on the polygon, weakly or strongly, and it is sharp (Theorems 7.14 and 4.3).
- (iii) *Traction consistency.* For GS the missing traction $\mathcal{R}$ is an additional consistency error of order one, which the penalty converts into the leak of size h/μ (Section 6). For CNS* it is absent, but the pressure now enters the velocity estimate through the wall term, and it has to be controlled by the inf-sup condition and absorbed, which is why CNS* needs $\gamma \geq \gamma_2$ (Theorem 7.2).

The lower bounds use test fields tailored to each error: a divergence-free field with normal trace $-(p - \bar{p})n$ for the leak (Section 6), divergence-free fields vanishing on the wall and feeling the second moment of the bump for the geometric error (Section 7.3), and local vertex-gradient arguments for strong imposition (Section 4).

3.8 Use of generative AI

The research reported here was carried out by the author with extensive use of a generative AI model, Claude (Anthropic; model Claude Opus 5.5), and we describe that use in detail.

- *Mathematics.* The model was used to develop proof strategies and to draft and check proofs of the lemmas and theorems, including the leak analysis, the lower bounds, the penalty witnesses and the strong-imposition theory.
- *Code.* The model wrote and ran the finite element code (the Scott–Vogelius–Nitsche solver, the penalty-threshold and dual-norm computations), the exact rational-arithmetic and interval certificates, and the scripts that produce the tables and figures.
- *Literature.* The model was used for literature search and for checking bibliographic data and the content of cited results.
- *Writing.* The model drafted and edited the text of this article and of its extended version.

We did not treat the model’s output as verified. The proofs were subjected to independent checks by separate AI instances (fresh sessions of the model instructed to act as critical referees of individual arguments), and every error found in this way was corrected or the corresponding claim weakened; the author reviewed these referee reports and the resulting corrections, which are recorded in the public repository. Claims that could not be fully proved are labelled as numerical, conditional or open, here and in [27]. The computer-assisted steps (the penalty certificates and the star-condition constants) use exact rational or rigorous interval arithmetic, with negative controls. All numbers in the tables were checked against the logs of reproducible runs, and the

code, raw outputs and logs are public [29]. The author directed the project, made the decisions on what to prove and how to present it, and takes full responsibility for the content of this article, including any remaining errors. The AI model is not an author.

4 Strong imposition: when does the zero-gradient lock occur?

The motivation for weak imposition in [15, 16] is the zero-gradient lock of strongly imposed no-slip. Before analysing the Nitsche methods, we take up the original difficulty and ask when, and why, strong imposition on the same polygon, with the same Scott–Vogelius pair, actually loses accuracy. The answer is sharper than one might expect, and it explains the prize question’s “why” from the other side:

- under the mesh condition (M2) used throughout this paper, strong imposition already has H^1 error $\asymp h_\Gamma^{3/2} + h^k$, the rate that CNS* attains (Section 7);
- the lock is caused by polygon vertices lying in only two triangles, which (M2) excludes; each such vertex costs $h_\Gamma |\partial_n u(z)|$ in H^1 , two-sided;
- with three or more triangles the same loss reappears continuously as one triangle at the wall vertex degenerates, and the relevant parameter is its angle, not the Guzmán–Scott quantity $\Theta(z)$.

The results of this section are not in the joint letter [25], which only mentions, with a pointer to [27], that the lock cannot occur under (M2).

The method. Find $u_h \in V_h$ with $u_h|_{\partial R} = g_I$, $u_h|_{\Gamma_h} = 0$, and $p_h \in \Pi_h \cap L_0^2(\Omega_h)$ with $a_h(u_h, v) - (p_h, \operatorname{div} v) = (\tilde{f}, v) \quad \forall v \in V_h^0, \quad (q, \operatorname{div} u_h) = 0 \quad \forall q \in \Pi_h.$

Exact incompressibility requires the outer flux m_R to vanish; this holds for our test data and for flux-corrected data. At a vertex z of Γ_h let $T_1, \dots, T_m$ be its triangles (counterclockwise), $e_1 \subset T_1$ and $e_2 \subset T_m$ its two chords with directions t_1, t_2 , and $s_i = T_i \cap T_{i+1}$ the interior edges. The polygon turns at z by the small angle $\phi_z := \arcsin(|e_1|/2) + \arcsin(|e_2|/2)$, so that the angle of Ω_h at z is $\pi + \phi_z$ and $ch_\Gamma \leq \phi_z \leq 1.1h_\Gamma$. We write $a_z := |\partial_n u(z)|$ for the wall shear at z ; recall $\nabla \tilde{u}(z) = \partial_n u(z) \otimes n$.

4.1 The local picture at a wall vertex

Let v be a continuous, piecewise- P_k , divergence-free field vanishing on $e_1 \cup e_2$, and $G_i := \nabla v|_{T_i}(z)$ its vertex gradients. Divergence-freeness, the boundary condition and continuity across the interior edges give

$$\operatorname{tr} G_i = 0, \quad G_1 t_1 = 0, \quad G_m t_2 = 0, \quad (G_{i+1} - G_i) s_i = 0. \quad (2)$$

Let A_z be the space of tuples (G_i) satisfying (2), the *attainable vertex gradients*.

Lemma 4.1 (Local structure) *(a) If the edges at z lie on at least three distinct lines, $\dim A_z = \max(m - 2, 0)$. In particular $A_z = \{0\}$ (the lock) if and only if $m \leq 2$.*

(b) Let $\phi_z \leq \theta_{\min}/2$, $\theta_{\min}$ the minimum angle. Then $\Theta(z) = \sin \phi_z$ if $m = 2$, and $\Theta(z) \geq \min(\sin 2\theta_{\min}, \sin(\theta_{\min}/2))$ if $m \geq 3$. So for small h_Γ , (M2) at the polygon vertices says exactly that $m \geq 3$.

(c) If $m \geq 3$ there is $(G_i) \in A_z$ with $\max_i |G_i - \nabla \tilde{u}(z)| \leq C(\theta_{\min}) \phi_z a_z$.

(d) Whatever v is, divergence-free or not, if $v = 0$ on a chord e at z then $|(\nabla v|_{T_e} - \nabla \tilde{u})(z) t_e| = a_z |e|/2$ exactly.

Proof (a) Put $H_i = J^{-1}G_i$ with J the rotation by $\pi/2$; trace-free G corresponds to symmetric H (indeed $\nabla \text{curl} \psi = JD^2\psi$). A symmetric matrix annihilating a vector s is a multiple of $s^\perp \otimes s^\perp$, so $H_1 = \alpha_1 t_1^\perp \otimes t_1^\perp$, $H_m = \alpha_m t_2^\perp \otimes t_2^\perp$ and $H_{i+1} - H_i = \beta_i s_i^\perp \otimes s_i^\perp$, subject to one closure condition $\alpha_m t_2^\perp \otimes t_2^\perp - \alpha_1 t_1^\perp \otimes t_1^\perp - \sum_i \beta_i s_i^\perp \otimes s_i^\perp = 0$. The rank-one matrices $u \otimes u$ for unit vectors on three distinct lines are linearly independent (the determinant is a product of sines of the angle differences), so the closure condition has rank three and $\dim A_z = (m+1) - 3$. For $m = 2$ the lines of t_1, s_1, t_2 are distinct and the only solution is zero. (b) The angles at z sum to $\pi + \phi_z$; for $m = 2$ the only pair gives $|\sin(\pi + \phi_z)|$. (c) In Hessian form $\nabla \tilde{u}(z)$ corresponds to $a n \otimes n$ and the chord normals are within ϕ_z of n ; solve for the correction with a 3×3 system whose inverse is bounded by $C(\theta_{\min})$. (d) $\nabla v|_{T_e}(z) t_e = 0$ while $\nabla \tilde{u}(z) t_e = \partial_n u(z) (n \cdot t_e)$, and $|n \cdot t_e| = |e|/2$. $\square$

Part (d) explains an observation that is otherwise puzzling: on meshes with three triangles per wall vertex the vertex-gradient error of strong imposition decays only like h (Table 20). This is harmless for the H^1 error, as we now show. Part (a) says that everything depends on the number of triangles at the wall vertex.

4.2 Under (M2): the rate $h_\Gamma^{3/2}$, two-sided

Theorem 4.2 (Upper bound under (M2)) *Assume (M0)–(M2), Assumption 3.5, $k \geq 4$ and $m_R = 0$. Then strong imposition is well posed and, with $p^\circ := \tilde{p} - |\Omega_h|^{-1} \int_{\Omega_h} \tilde{p}$,*

$$\|\nabla(\tilde{u} - u_h)\| + \|p^\circ - p_h\| \leq C(h^k + h_\Gamma^{3/2}).$$

No relation between h and h_Γ is needed.

Proof An admissible approximation. Let $w_h := I_h \tilde{u} - \sum_x \tilde{u}(x) \varphi_x$, the Lagrange interpolant with its nodal values on Γ_h set to zero (φ_x the nodal basis functions of the nodes x on Γ_h). By Lemma 3.7 $|\tilde{u}(x)| \leq Ch_\Gamma^2$, and $\|\nabla \varphi_x\| \leq C$ by scale invariance; there are $O(h_\Gamma^{-1})$ such nodes, so $\|\nabla(\tilde{u} - w_h)\| \leq C(h^k + h_\Gamma^{3/2})$. Its divergence is in Π_h , has mean $|\Omega_h|^{-1} m_R = 0$ and norm $\leq C(h^k + h_\Gamma^{3/2})$; Lemma 3.12 (which is where (M2) enters) gives $y \in V_h^0$ with $\text{div } y = \text{div } w_h$ and the same bound. So $z := w_h - y$ is admissible, divergence-free, and $\|\nabla(\tilde{u} - z)\| \leq C(h^k + h_\Gamma^{3/2})$.

Consistency and energy. For $v \in V_h^0$, $a_h(\tilde{u}, v) = (\tilde{f} - F - \nabla \tilde{p}, v)$, so $a_h(u_h - \tilde{u}, v) - (p_h - \tilde{p}, \text{div } v) = (F, v)$, with $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$. The field $u_h - z \in V_h^0$ is divergence-free, and Korn's inequality gives $\|\nabla(u_h - z)\| \leq C(\|\nabla(\tilde{u} - z)\| + h_\Gamma^2)$. The pressure bound follows from Lemma 3.12 applied to $p_h - \Pi p^\circ$, Π the L^2 projection. $\square$

Without (M2), at a two-triangle vertex the inf-sup constant is $O(h_\Gamma)$ (Theorem 4.7), and the divergence correction step gives only $h_\Gamma^{1/2}$, which matches the lock. The matching lower bound holds for every field that vanishes on the polygon, discrete or not; it is the Nitsche-free version of the chordal-bump argument, and is the vector analogue of the classical scalar result [7, 8].

Theorem 4.3 (Lower bound) *Under (M0) and (M1), every $v \in H^1(\Omega_h)^2$ with $v|_{\Gamma_h} = 0$ satisfies*

$$\|\nabla(\tilde{u} - v)\| \geq ch_\Gamma^{3/2} \|\partial_n u\|_{L^2(\Gamma)} - Ch_\Gamma^{5/2}.$$

Proof On each boundary triangle T_e , $|w|_{H^{1/2}(e)} \leq C|w|_{H^1(T_e)}$, since the one-dimensional Gagliardo seminorm is dilation invariant. Apply this to $w = \tilde{u} - v$, whose trace on e is $\tilde{u}|_e = \delta(s)a_e + r$ with $a_e := \partial_n u(m_e^*)$, $\delta(s) = \text{dist}(x, \Gamma)$ and $|r|_{H^{1/2}(e)} \leq C\ell^3$. Since $\delta(s) - \delta(t) = (t^2 - s^2)/(|x(t)| + |x(s)|)$ and $|x| \leq 1$, $|\delta|_{H^{1/2}(e)}^2 \geq \frac{1}{4} \iint (s+t)^2 ds dt = \ell^4/24$. Summing over the chords with a Riemann sum for $\sum_e \ell_e |a_e|^2$ gives $\|\nabla(\tilde{u} - v)\|^2 \geq ch_\Gamma^3 (\|\partial_n u\|_{L^2(\Gamma)}^2 - Ch_\Gamma) - Ch_\Gamma^5$. $\square$

Theorem 4.4 (Two-sided rate under (M2)) *Under (M0)–(M2), Assumption 3.5, $m_R = 0$, $h^k \leq h_\Gamma^{3/2}$ and $\partial_n u \not\equiv 0$ on Γ , strongly imposed no-slip satisfies $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq Ch_\Gamma^{3/2}$.*

So, under the hypothesis used throughout this paper, strong imposition attains Scott’s expected rate, with the same exponent as CNS*. Theorem 4.2 is close to a corollary of the Guzmán–Scott inf-sup theory and standard boundary approximation. The remark of [16] that the effect “will occur at the vertices of any polygonal domain” therefore does not hold under (M2). By the results below, the strong-imposition meshes of [16] must have lost uniform (M2), either through a positive fraction of two-triangle wall vertices or through nearly singular wall stars. Their star counts are not reported, so this last statement is an inference, not a check.

4.3 Locked vertices

Let $\mathcal{L}$ be the set of polygon vertices lying in exactly two triangles. At such a vertex the attainable gradients vanish (Lemma 4.1(a)), while the true gradient does not. This costs a fixed amount per vertex.

Theorem 4.5 (Counting lower bound) *Under (M0) and (M1), every divergence-free $v \in V_h$ vanishing on Γ_h satisfies*

$$\|\nabla(\tilde{u} - v)\| \geq ch_\Gamma \left(\sum_{z \in \mathcal{L}} a_z^2 \right)^{1/2} - Ch_\Gamma^{3/2}.$$

Proof Let $\gamma_k^2 := \min\{\|1 - q\|_T^2 / |T| : q \in P_{k-1}(T), q(\text{a vertex}) = 0\}$, which is affine invariant and positive (exactly $4/(k^2(k+1)^2)$ for $k = 4, \dots, 8$). On each triangle T at z , $\nabla v|_T \in P_{k-1}$, so $\|\nabla \tilde{u} - \nabla v\|_T \geq \gamma_k |T|^{1/2} |\nabla \tilde{u}(z) - \nabla v|_T(z)| - Ch_T |T|^{1/2}$. At a locked vertex $\nabla v|_T(z) = 0$, and $|\nabla \tilde{u}(z)| = a_z$; the star has area $\geq ch_\Gamma^2$. Sum over z (a triangle has at most two vertices on Γ_h); the remainders add up to $Ch_\Gamma^{3/2}$. $\square$

A single lock where $a_z \geq \delta > 0$ already reduces the rate to 1, and a positive fraction of locks along an arc where $\partial_n u \neq 0$ reduces it to $\frac{1}{2}$: this is the $h^{1/2}$ phenomenon of [15,

16]. The bound is sharp, but proving the matching upper bound requires controlling the stability of the pair at the locked vertices, where the inf-sup constant degenerates. The structure of a locked star is completely explicit. At $z \in \mathcal{L}$ let $T_1 = [z_-, z, w]$ and $T_2 = [z, z_+, w]$ be its two triangles, $s = [z, w]$ the interior edge, λ_x the hat function of the vertex x , and $g_i := \nabla \lambda_w|_{T_i} = \hat{n}_i/H_i$, where $\hat{n}_i$ is the unit normal of the chord e_i into T_i and $H_i = \text{dist}(w, e_i) \asymp h_z := \text{diam}(T_1 \cup T_2)$. Since λ_w is linear on s from 0 to 1,

$$g_1 \cdot (w - z) = g_2 \cdot (w - z) = 1. \quad (3)$$

Let D_z be the 2×2 matrix with rows $g_1^\top, g_2^\top$.

Lemma 4.6 (Locked star) (a) For every $v \in V_h^0$ there is a unique $c_v \in \mathbb{R}^2$ with $\nabla v|_{T_i}(z) = c_v \otimes g_i$, $i = 1, 2$. Hence the two vertex divergences are $(\text{div } v|_{T_1}(z), \text{div } v|_{T_2}(z))^\top = D_z c_v$.
(b) $|\det D_z| = \sin \phi_z / (H_1 H_2)$, so D_z degenerates as $\phi_z \rightarrow 0$; and $|D_z^{-1}(1, 0)^\top| = H_1 / \sin \phi_z$.
(c) $D_z(w - z) = (1, 1)^\top$. Consequently the radial field $\sigma(w - z)\lambda_w\lambda_z^2 \in V_h^0$ has vertex divergences (σ, σ) at z , zero vertex gradients at all other vertices, zero flux through s , and H^1 norm at most $C|\sigma|h_z$, independent of ϕ_z .

Proof (a) $v = 0$ on e_1 , so $\nabla v|_{T_1}(z)$ annihilates the direction of e_1 and has the form $c \otimes g_1$ (g_1 is normal to e_1); likewise on T_2 . Continuity across s and (3) force the same c . Then $\text{div } v|_{T_i}(z) = \text{tr}(c \otimes g_i) = g_i \cdot c$. (b) $g_1 \times g_2 = \sin \angle(\hat{n}_1, \hat{n}_2) / (H_1 H_2)$, and consecutive chord normals make the angle ϕ_z ; $D_z^{-1}(1, 0)^\top = (\det D_z)^{-1} g_2^\perp$. (c) is (3); the gradient of $(w - z)\lambda_w\lambda_z^2$ at z is $(w - z) \otimes \nabla \lambda_w$, and $w - z$ is parallel to s , so the flux through s vanishes. $\square$

Parts (b) and (c) say that the degenerate direction of D_z is the *difference* of the two vertex divergences: equal divergences are cheap, at a cost independent of the turning angle, while a difference ℓ costs $h_z|\ell|/\sin \phi_z$. This is the whole stability theory of locked vertices.

Theorem 4.7 (Stability and the two-sided rate with locks [27]) Assume (M0), (M1), $k \geq 4$, and (M2) at every vertex except the locked ones, with no condition on their number or arrangement. Let $T_1(z), T_2(z)$ be the two triangles at $z \in \mathcal{L}$.

- (a) Constrained inf-sup. For every $q \in \Pi_h \cap L_0^2$ with $q|_{T_1(z)}(z) = q|_{T_2(z)}(z)$ at all $z \in \mathcal{L}$ there is $y \in V_h^0$ with $\text{div } y = q$ and $\|\nabla y\| \leq C\|q\|$, with C independent of $h, h_\Gamma, (\phi_z)$ and $\#\mathcal{L}$.
- (b) Sharp stability. The divergence maps V_h^0 onto $\Pi_h \cap L_0^2$ always, and the inf-sup constant satisfies $c \min(1, \min_{\mathcal{L}} \phi_z) \leq \beta_h \leq C \min_{\mathcal{L}} \phi_z$; so $\beta_h \asymp h_\Gamma$ as soon as $\mathcal{L} \neq \emptyset$.
- (c) Velocity. If also Assumption 3.5 holds and $m_R = 0$, then

$$\|\nabla(\tilde{u} - u_h)\| \leq C \left(h^k + h_\Gamma^{3/2} + h_\Gamma \left(\sum_{z \in \mathcal{L}} a_z^2 \right)^{1/2} \right),$$

and, when $h^k \leq h_\Gamma^{3/2}$ and $\partial_n u \not\equiv 0$, the error is $\asymp h_\Gamma^{3/2} + h_\Gamma(\sum_{\mathcal{L}} a_z^2)^{1/2}$.

- (d) Pressure. The raw pressure error is only bounded by C/h_Γ times the velocity error, and numerically this is attained; the L^2 -orthogonal projection of p_h onto the constrained space of (a), computable by a banded Gram solve, converges at the velocity rate.

Proof sketch (a) The construction has four steps, each local; the constants involve only k , the angles, c_0 , Θ_0 and L .

- A. *Element means.* Take $w \in H_0^1(\Omega_h)^2$ with $\operatorname{div} w = q$ and $\|\nabla w\| \leq C\|q\|$ (continuous level, as in Lemma 3.12) and its P_2 Fortin interpolant $v_A \in V_h^0$, which preserves edge integrals and hence $\int_T \operatorname{div} v_A = \int_T q$ for every triangle [55]. At each locked vertex subtract the field $c\lambda_w\lambda_z^2 + \delta n_s\lambda_z^2\lambda_w^2$ (n_s the normal of s) with $c = c_{v_A}$ and δ chosen so that the field is flux-neutral on both triangles; this kills the vertex gradient of v_A at z without changing any element mean or any other vertex gradient. The remainder q_1 of q has zero element means, and at a locked vertex its two vertex values are equal (to σ_z), because $q \in$ the constrained space.
 - B. *Locked vertices.* The radial fields $\sigma_z(w_z - z)\lambda_{w_z}\lambda_z^2$ of Lemma 4.6(c) match (σ_z, σ_z) , with zero element means, at a cost independent of ϕ_z .
 - C. *Other vertices.* At a vertex y with $\Theta(y) \geq \Theta_0$ and m triangles, fields of the form $\lambda_y^2\ell + \sum_E \delta_E n_E \lambda_y^2 \lambda_{x_E}^2$ (ℓ linear on each triangle, $E = [y, x_E]$ the interior edges) realise every vector of vertex values with zero element means [2, 27]; the map is onto because at least three edge lines meet at y , and uniformly so by compactness.
 - D. *Element bubbles.* What remains has zero element means and vanishes at all vertices, and the element bubble lemma of Scott and Vogelius [1] inverts the divergence triangle by triangle.
- (b) The upper bound uses the same construction with the difference of the vertex values paid at cost $h_z/\sin\phi_z$; the lower bound tests with the L^2 representer of $q \mapsto q|_{T_1}(z) - q|_{T_2}(z)$. (c) Modify the interpolant of Theorem 4.2 so that its vertex gradient vanishes at every locked vertex, at cost $Ch_\Gamma a_z$ each; its divergence then satisfies the constraint of (a). (d) follows from (a), (b) and the error equation. $\square$

At a locked vertex the loss is thus an *approximation* loss caused by the zero trace, not a stability loss: the velocity bound in (c) does not see the degenerate inf-sup constant of (b).

4.4 Nearly singular wall stars

With a lower bound on the angles, a wall star with $m \geq 3$ triangles is uniformly nonsingular, and there is nothing to add. Near-singularity requires a small angle ε_z at z , and the Guzmán–Scott quantity $\Theta(z)$ does not detect it: there are three-triangle stars with $\Theta(z) \geq 0.84$ whose attainable-gradient cost is that of a lock. The two relevant shapes are a *needle at the wall* (the triangle next to a chord has angle ε_z at z) and a *split edge* (the middle triangle has angle ε_z , i.e. a lock with its interior edge doubled). For a wall vertex z define the area-weighted distance of the exact gradient from the attainable ones,

$$d_z^2 := \min_{(G_T) \in A_z} \sum_{T \ni z} |T| |\nabla \tilde{u}(z) - G_T|^2.$$

It is what the counting argument of Theorem 4.5 actually bounds from below, and at a locked vertex $d_z^2 = |\operatorname{star}(z)| a_z^2$.

Table 2 Strong imposition on mesh (s) (every wall vertex in three triangles), test A:
 H^1 error, observed rate, rate predicted by the fit $E = ah_\Gamma^{3/2}(1 + \beta h_\Gamma)$, $E/h_\Gamma^{3/2}$, and
the largest vertex-gradient error, which decays like h_Γ (Lemma 4.1(d)).

N	h_Γ	H^1 error	rate	predicted	$E/h_\Gamma^{3/2}$	vertex gradient
64	0.1243	$4.557 \cdot 10^{-2}$	1.657	1.653	1.040	0.307
96	0.0831	$2.384 \cdot 10^{-2}$	1.610	1.614	0.995	0.212
128	0.0624	$1.518 \cdot 10^{-2}$	1.576	1.584	0.974	0.162
160	0.0500	$1.073 \cdot 10^{-2}$	1.558	1.566	0.961	0.130
192	0.0416	$8.097 \cdot 10^{-3}$	1.547	1.555	0.953	0.109

Theorem 4.8 (Needle-angle law) *Let z have $m = 3$ triangles with exactly one angle $\varepsilon_z \leq \varepsilon_0$ at z (the needle, in either position), the other two angles at z at least θ_0 , ray lengths in $[c_r h_z, h_z]$, and (M0). Then, with constants depending only on θ_0, c_r, c_0 ,*

$$ch_z a_z \min\left(1, \frac{\phi_z}{\sqrt{\varepsilon_z}}\right) \leq d_z \leq C h_z a_z \min\left(1, \frac{\phi_z}{\sqrt{\varepsilon_z}}\right).$$

Proof By Lemma 4.1(a), $\dim A_z = 1$. In Hessian form the attainable tuples are t times a fixed tuple whose entries are ratios of 3×3 determinants of the rank-one matrices of the four edge lines; the sine formula gives them exactly. With the regular triangle next to a chord normalised to carry the exact gradient ($t = a_z$), the other regular triangle carries a defect $O(a_z \phi_z)$ and the needle a defect $a_z \frac{\phi_z}{\varepsilon_z} (1 + O(\phi_z + \varepsilon_z))$. *Upper bound:* with $t = a_z$, $d_z^2 \leq Ch_z^2 a_z^2 (\phi_z^2 + \varepsilon_z \phi_z^2 / \varepsilon_z^2)$, since the needle has area $\leq \varepsilon_z h_z^2$; with $t = 0$, $d_z^2 \leq |\text{star}| a_z^2$. *Lower bound:* either t differs from a_z by a fixed fraction of $\min(1, \phi_z / \varepsilon_z) a_z$, which costs that much on a regular triangle of area $\asymp h_z^2$, or the needle carries a defect $\gtrsim a_z \phi_z / \varepsilon_z$ on its area $\asymp \varepsilon_z h_z^2$. In both cases $d_z^2 \geq ch_z^2 a_z^2 \min(1, \phi_z^2 / \varepsilon_z)$. $\square$

The square root comes from the needle's area: the cheapest admissible tuple puts a gradient defect $a_z \phi_z / \varepsilon_z$ on an area $\varepsilon_z h_z^2$. The maximum-norm defect, $a_z \min(1, \phi_z / \varepsilon_z)$, overstates the H^1 cost. Numerically the ratio $d_z / (h_z a_z \min(1, \phi_z / \sqrt{\varepsilon_z}))$ lies in $[0.56, 0.92]$ over $\phi_z \in [0.005, 0.08]$, $\varepsilon_z \in [\phi_z^2 / 4, 0.5]$. The counting argument of Theorem 4.5 turns this into an H^1 lower bound, and under a mesh hypothesis that isolates each needle in its own pair of wall needles the matching upper bound is proved in [27]:

$$\|\nabla(\tilde{u} - u_h)\| \asymp h_\Gamma^{3/2} + h_\Gamma \left(\sum_z w_z^2 a_z^2 \right)^{1/2},$$

$$w_z = \min\left(1, \frac{\phi_z}{\sqrt{\varepsilon_z}}\right) \text{ at needles, } w_z = 1 \text{ at locks.}$$

One needle at the wall gives rate $\frac{3}{2}$ for $\varepsilon \gtrsim h_\Gamma$ and rate 1 for $\varepsilon \lesssim h_\Gamma^2$, and a positive fraction of needles with $\varepsilon \asymp h_\Gamma^p$ gives rates $\frac{3}{2}, 1, \frac{1}{2}$ for $p = 0, 1, 2$. Needles come in pairs across their short edge, and for a straight pair the inf-sup constant is at most $C\varepsilon$; the velocity estimate does not see this, because its proof never inverts the divergence on a needle.

Table 3 Strong imposition with locked vertices, test A: H^1 velocity error, raw pressure error $\|\pi_h\|$ and filtered pressure error $\|\pi_h^f\|$ (Theorem 4.7(d)), with $\pi_h = p_h - \Pi p^\circ$.

mesh	$\#\mathcal{L}$	N	H^1 error	rate	$\ \pi_h\ $	rate	$\ \pi_h^f\ $	rate
one lock	1	32	$2.123 \cdot 10^{-1}$	1.36	7.291	-0.45	0.900	1.71
	1	64	$9.376 \cdot 10^{-2}$	1.16	9.160	-0.29	0.328	1.39
	1	128	$4.464 \cdot 10^{-2}$	1.06	10.447	-0.16	0.145	1.12
mesh (a)	16	32	$5.724 \cdot 10^{-1}$	0.70	19.747	-0.49	1.378	0.79
	32	64	$3.746 \cdot 10^{-1}$	0.60	26.384	-0.42	0.868	0.65
	64	128	$2.543 \cdot 10^{-1}$	0.55	35.653	-0.45	0.580	0.57

4.5 What this says about Nitsche’s method

Strong imposition and the Nitsche methods have the same rate under (M2) (Theorem 4.4 and, for CNS*, Theorem 7.2); the constant of strong imposition is about 2.5 times larger in our tests (Table 20). On the alternating mesh, where every other wall vertex is locked, strong imposition degrades to rate $\frac{1}{2}$ while GS and CNS* are unaffected to within 6% (an observation; that mesh lies outside our hypotheses for the Nitsche methods). The advantages of Nitsche imposition on an inscribed polygon are therefore robustness to locked and nearly singular wall stars, and a smaller error constant, not a better rate.

5 How large must the penalty be?

Scott’s prize asks for a study of the influence of the Nitsche penalty. The analysis of the Nitsche methods in Sections 6–7 below needs $\gamma \geq \gamma_0$ (or γ_2), and the penalty enters the error bounds; but they say nothing about how large the smallest admissible μ actually is, or how it depends on the mesh. This section answers that question. Its first part, the sufficiency and the single-triangle witness with its certified shape families (Theorem 5.1, Proposition 5.2, Corollary 5.3), was announced in the joint letter [25] with a proof sketch; the rest of the section, from Section 5.3 on, is not in the letter.

5.1 Sufficiency, and why ρ appears

Lemma 3.11 shows that $\mu \geq \mu_0 = 4\kappa C_I^2 \rho$ with $\rho = h / \min_e |e|$ suffices for coercivity of N_h on V_h^R . The proof is local at the wall: what is really needed is $\mu/h \gtrsim 1/|e|$ for each chord e , a lower bound on the *local* penalty in units of the chord. The appearance of the global ratio ρ is the price of Scott’s normalisation of the penalty by the global mesh size h . It has a practical consequence: with a fixed μ , refining the boundary while keeping the far field coarse will eventually violate coercivity. The question is whether this is a real effect or an artefact of the proof. We show that it is real: the threshold μ^* (the infimum of the μ for which N_h is coercive on Z_h) scales like ρ on broad classes of meshes. Note that the relevant space is Z_h , the divergence-free fields: coercivity on Z_h is what the GS analysis uses, and it is easier to have than coercivity on V_h^R , so necessity statements on Z_h are the strong ones.

5.2 A witness in a single boundary triangle

Coercivity fails as soon as there is a single field $v \in Z_h$ with $N_h(v, v) < 0$. For v supported near the wall,

$$N_h(v, v) = A(v) - 2B(v) + \frac{\mu}{h}C(v),$$

$$A = \sum_T \int_T \frac{1}{2} Dv : Dv, \quad B = \int_{\Gamma_h} (\partial_n v) \cdot v, \quad C = \int_{\Gamma_h} |v|^2.$$

In two dimensions A and B are invariant under dilation and C scales like length. So a witness on a patch of size ℓ gives a lower bound $\mu^* \geq \lambda h/\ell$, where $\lambda = (2B - A)/C$ is computed on the patch scaled to unit size. The smallest possible patch is one boundary triangle.

Let T have an edge $e \subset \Gamma_h$ and not meet ∂R . Scale and rotate so that $e = [(0, 0), (1, 0)]$, the fluid lies above, and the apex is (a, b) with $b > 0$. With barycentric coordinates λ_0 (apex), λ_1, λ_2 (base vertices), and $c \in \mathbb{R}^3$, put

$$\psi_c := \lambda_1^2 \lambda_2^2 (c_0 \lambda_0 + c_1 \lambda_1 + c_2 \lambda_2) \in P_5, \quad v_c := \text{curl } \psi_c \in P_4^2.$$

ψ_c vanishes to second order on the two interior edges of T , so v_c , extended by zero, is continuous and divergence-free: $v_c \in Z_h$ for every $k \geq 4$ and every mesh containing T , with no condition on the other triangles. Let $A_T = \int_T \frac{1}{2} Dv : Dv$, $B_T = \int_e (\partial_n v) \cdot v$ and $C_T = \int_e |v|^2$, with $n = (0, -1)$, as quadratic forms in c . They are computed exactly; with $\varpi := 4a^2 + 4b^2$,

$$C_T = \frac{1}{1260 b^2} \begin{pmatrix} 2 & a-6 & -(a+5) \\ a-6 & 2(\varpi-3a+9) & \varpi-4a+15 \\ -(a+5) & \varpi-4a+15 & 2(\varpi-5a+10) \end{pmatrix},$$

$$B_T = \frac{1}{840 b^3} \begin{pmatrix} 12 & -(\varpi-6a+27) & -(\varpi-2a+25) \\ -(\varpi-6a+27) & 6(\varpi-3a+9) & 3(\varpi-4a+15) \\ -(\varpi-2a+25) & 3(\varpi-4a+15) & 6(\varpi-5a+10) \end{pmatrix},$$

and $A_T = (\alpha_{ij})/b^3$ with

$$\begin{aligned} 630 \alpha_{00} &= 3a^4 - 6a^3 + 6a^2b^2 + 12a^2 - 6ab^2 - 9a + 3b^4 + 4b^2 + 9, \\ 420 \alpha_{01} &= 3a^4 + 6a^2b^2 - 5a^2 + 6a + 3b^4 - 7b^2 - 9, \\ 420 \alpha_{02} &= 3a^4 - 12a^3 + 6a^2b^2 + 13a^2 - 12ab^2 - 8a + 3b^4 - b^2 - 5, \\ 70 \alpha_{11} &= 3a^4 - 3a^3 + 6a^2b^2 + 5a^2 - 3ab^2 - 3a + 3b^4 + 7b^2 + 3, \\ 140 \alpha_{12} &= a^4 - 2a^3 + 2a^2b^2 + 5a^2 - 2ab^2 - 4a + b^4 + 7b^2 + 5, \\ 70 \alpha_{22} &= 3a^4 - 9a^3 + 6a^2b^2 + 14a^2 - 9ab^2 - 10a + 3b^4 + 10b^2 + 5. \end{aligned}$$

The trace form is positive definite: $\det C_T = (3a^4 - 6a^3 + 7a^2b^2 + 3a^2 - 7ab^2 + 4b^4 + 5b^2)/(83\,349\,000\,b^6) > 0$. The forms were checked against an independent quadrature, and the resulting λ_T against our threshold code on single-triangle patches.

Theorem 5.1 (Single-triangle necessity; announced in [25]) *Let λ_T be the largest eigenvalue of the pencil $(2B_T - A_T, C_T)$; it depends only on the shape of T . If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. Consequently every mesh satisfies the necessary condition*

$$\mu \geq h \max_{e \in \mathcal{E}_h^i} \frac{\lambda_{T_e}}{|e|}.$$

Proof For the eigenvector c , $N_h(v_c, v_c) = A_T - 2B_T + (\mu|e|/h)C_T = (\mu|e|/h - \lambda_T) C_T(c) < 0$, by the scaling of the forms. $\square$

The theorem is useful only where $\lambda_T > 0$, and λ_T is not given in closed form. We therefore certify positive lower bounds on families of shapes, in exact arithmetic.

Proposition 5.2 (Exact certificates over shape families; announced in [25]) *Let Q_φ be the set of apex positions for which the apex angle is at least φ and $b \geq s \max(a, 1 - a)$ with $s = 0.087 < \tan 5^\circ$; it contains every triangle with apex angle at least φ and base angles at least 5° . Then*

$$\lambda_T > 9.3 \text{ on } Q_{60^\circ}, \quad \lambda_T > 4 \text{ on } Q_{45^\circ}, \quad \lambda_T > 1 \text{ on } Q_{35^\circ}.$$

Method Cover Q_φ by dyadic boxes in the (a, b) plane. On each box, take a rational vector q (the eigenvector at the box centre, rounded) and verify that the polynomial $2520b^3q^\top(2B_T - A_T - cC_T)q$, whose coefficients are rational, is positive on the whole box, by exact rational interval arithmetic; split boxes until this succeeds. The certificates use 259–716 boxes, and the same procedure fails, as it must, for $c = 9.4$ on Q_{60° and $c = 4.5$ on Q_{45° . The code is in [29]. $\square$

A scan (not certified) gives the minimum 9.31381 over Q_{60° , at the equilateral triangle, so the first bound is nearly sharp.

Corollary 5.3 (ρ -scaling is necessary on a broad class of meshes) *If a shortest chord carries a triangle with apex angle at least 35° and base angles at least 5° , then coercivity of N_h on Z_h requires $\mu > \rho$; with apex angle at least 45° (resp. 60°) it requires $\mu > 4\rho$ (resp. $\mu > 9.3\rho$). Together with Lemma 3.11, $\mu^* \asymp \rho$ on every such mesh.*

5.3 Tall triangles, and why necessity is not local

Single triangles cannot do everything: $\lambda_T < 0$ for tall boundary triangles, for example every right triangle with the right angle at a base vertex and apex angle at most 30° . Such triangles occur in practice, for instance near the diagonal directions of our test meshes. One might hope that the whole star of a boundary vertex always carries a witness when the mesh satisfies (M2). This is false.

Proposition 5.4 (Exact counterexamples [27]) *Let $\mathcal{S}^{(H)}$ be the three-triangle star at $(0, 0)$ with rim points $(1, 0), (1, H), (0, H), (-1, 0)$ and wall $\{y = 0\}$, the star of a boundary vertex in a mesh of columns of width 1 and height H split by the forward diagonal. It satisfies (M1),*

(M2) and has three triangles at the wall vertex. In exact rational arithmetic, on the divergence-free continuous piecewise- P_4 fields vanishing on the rim, $A - 2B - C$ is positive definite for $H = 5$ (minimum angle 11.3°), and $A - 2B - \frac{1}{10}C$ is positive definite on the union of the two stars of one chord for $H = 6$. So no field supported in these patches is a witness for any $\mu \geq 0$. Similar failures occur for $k = 5, 6$ and on Clough–Tocher split stars.

These stars have small angles, and the counterexamples do not exclude a star witness under a minimum-angle bound (floating-point searches found star failures at minimum angle 15° but none at 20°). On the same meshes wider patches do work (three consecutive stars at $H = 5$ give a witness), and on a global (M0)–(M2) mesh built from these stars the measured threshold still scales like ρ (numerically, $\mu^*/\rho \approx 3.1$). Necessity therefore has to come from patches that extend along the wall and into the domain. Two results of this kind are proved in [27]. We state them without their technical proofs.

Theorem 5.5 (Necessity from a locally resolved region [27]) *Let $e \in \mathcal{E}_h^\Gamma$ have midpoint x_0 , let $0 < Y \leq 10^{-6}$ and suppose every triangle meeting $B(x_0, 3Y)$ has diameter at most $\kappa_0(\theta)Y$, where θ is the minimum angle and $\kappa_0(\theta)$ is explicit (0.0117 at 30° , 0.0161 at 60°). Then the curl of the Argyris interpolant of an explicit profile of height Y is a witness with $(2B - A)Y/C \geq 15$, so N_h is indefinite on Z_h whenever $\mu < 15h/Y$.*

The constant 15 is close to the half-plane value 15.487 of the profile. The smallness conditions come from worst-case interpolation bounds and are far from sharp: numerically the witness works with $\kappa \approx 1$, and on dyadically graded boundary layers. A statement that needs no resolution at all is:

Theorem 5.6 (Necessity on every shape-regular mesh [27]) *Assume (M0) with minimum angle θ . There are $c_*(\theta, c_0) > 0$ and $h_*(\theta, L) > 0$ such that for $h \leq h_*$ and every $k \geq 4$, coercivity on Z_h requires $\mu \geq c_* \rho$. Neither (M1) nor (M2) is used.*

The idea of the proof is worth recording, because it shows how to beat the lack of resolution that defeats Theorem 5.5. In coordinates in which a chord lies on $\{y = 0\}$ and the fluid is above, take the stream function $F(x, y) = \Phi(x/X)\chi(y/X)\Psi(y)$, with Φ and χ fixed cut-offs, X a fixed multiple of h_Γ , and a one-dimensional profile Ψ with $\Psi(0) = 0$, $\Psi'(0) = 1$. The witness is $v = \text{curl } I_A F$, with I_A the Argyris interpolant. The profile is chosen *exactly quadratic* on the height $y_1 = 2d_1 h_\Gamma$ of every triangle touching the wall, so that the Argyris interpolant reproduces it there and the boundary forms B and C carry no interpolation error of order one. Beyond y_1 the profile’s slope decays in two steps, through two thin transition bands, to zero; the bands are placed where shape regularity alone bounds the triangle diameters by a multiple of the distance to the wall, and the decay rates are chosen so that the interpolation error in the bands costs only a small fraction of the energy A . The price is that the bands must be very wide, which is where the astronomically small constants come from. The constants are explicit but astronomically small ($\log_{10}(c_*/c_0) \approx -39$ at $\theta = 30^\circ$): the theorem

settles the *scaling* $\mu^* \gtrsim \rho$ on every shape-regular mesh, not a usable threshold. The loss comes from the worst-case growth factor between neighbouring triangles and from Taylor-based interpolation constants, both entering squared.

For layered meshes with tall boundary triangles, the case in which single triangles and single stars fail, there are exact witnesses at every mesh size, with no interpolation error at all.

Proposition 5.7 (Stacked columns) *Let a half-plane mesh consist of the cells $[i, i+1] \times [jH, (j+1)H]$ split by the forward diagonal (every boundary star is the star $\mathcal{S}^{(H)}$ of Proposition 5.4). Let Φ be the uniform cubic B-spline with knot spacing $d = DH \geq 4H$ on vertical mesh lines, and G the C^1 piecewise quadratic with $G = y - y^2/(2H)$ on $[0, H]$, $G'' = -1/(8H)$ on $[H, 3H]$, $G'' = 1/(8H)$ on $[3H, 5H]$ and $G = 0$ beyond. Then $F = \Phi(x)G(y)$ is C^1 and piecewise P_5 on the mesh, so $v = \text{curl } F \in Z_h$ exactly for every $k \geq 4$, and*

$$H \frac{h_\Gamma (2B - A)}{C} = \frac{2265D^4 - 2800D^2 - 6944}{2416D^4} \geq 0.8538.$$

Hence $\mu^* \geq 0.85 h/(Hh_\Gamma)$ on every mesh containing this configuration over the support of the witness.

Proof For $F = \Phi G$ with $G(0) = 0$ and compact support, $B = H^{-1} \int \Phi^2$, $C = \int \Phi^2$ and $A = 2 \int \Phi'^2 \int G'^2 + \int \Phi^2 \int G''^2 + \int \Phi''^2 \int G^2$ (the cross term integrates to zero by parts). The integrals are rational: $\int \Phi^2 = \frac{151d}{315}$, $\int \Phi'^2 = \frac{2}{3d}$, $\int \Phi''^2 = \frac{8}{3d^3}$, $\int G^2 = \frac{31H^3}{60}$, $\int G'^2 = \frac{5H}{12}$, $\int G''^2 = \frac{17}{16H}$. The resulting expression is increasing in D . $\square$

So on layered meshes the threshold decreases as the first layer gets taller: numerically $\mu^* \approx (15\text{--}16) h/\delta_1$ with δ_1 the first-layer height, close to the half-plane constant of Theorem 5.5, and the decay with the apex angle of any constant $c(\theta)$ in a bound $\mu^* \geq c(\theta)\rho$ is genuine.

5.4 The measured threshold

On the meshes of Section 9 the measured threshold $\gamma^* = \mu^* h_\Gamma/h$ is about 18–21 for $k = 4$, and it is linear in h/h_Γ over more than a decade when the far field is coarsened (Table 15, Figure 4): the threshold is set locally at the wall, and the scaling with h/h_Γ is the cost of normalising by the global h . Table 4 shows how the threshold is built up from patches of increasing size. The gap between the certified local constants (9–10) and the measured 18–21 is accounted for [27] by three effects, in order of size: the first layer of our meshes is flatter than the reference shapes (height $\frac{3}{4}$ of the chord at the longest chords, which is where γ^* is decided), the optimal witness spreads over 4–8 chords, and the second layer adds a little. Numerically $\gamma^*(N)$ approaches the threshold of a periodic half-plane cell, $\gamma_\infty^* \approx 21.41$, roughly like $\gamma_\infty^* - 22/N$.

Practical advice. For $k = 4$, $\gamma = \mu h_\Gamma/h \geq 25$ is safe on our meshes, so $\mu \geq 25\rho$ is safe too; a fixed μ fails once the far field is coarsened beyond $h/h_\Gamma \approx \mu/20$. The threshold grows quickly with k ($\gamma^* \approx 33$ for $k = 5$ and 48 for $k = 6$) and is larger on Clough–Tocher refinements (Table 16), so these values do not transfer between elements. For CNS* the relevant quantity is the largest μ at which the square system is singular; it is somewhat above the GS threshold (Section 9).

Table 4 Patch thresholds γ on the meshes of Section 9 ($k = 4$): the largest $h_\Gamma(2B - A)/C$ over fields supported in a single boundary triangle, a single wall star, windows of W consecutive first-layer columns, the first J layers, and globally.

N	triangle	star	win2	win4	win8	$J = 2$	global
8	6.13	13.71	13.91	14.65	–	14.74	14.74
16	8.93	16.58	16.89	17.91	17.94	18.09	18.09
32	10.08	17.48	17.82	19.54	19.69	19.85	19.85
64	10.49	17.70	18.03	20.29	20.68	20.80	20.80

6 The printed method: the leak law

Relation to [25]. Theorems 6.1, 6.3 and 6.5 and Corollary 6.6 were announced in the joint letter [25], with proof sketches; we give complete proofs here. The identification of the H^1 constant with a Stokes leak flow (Section 6.5) is not in the letter.

Throughout this section u_h is the velocity of GS and $\gamma \geq \gamma_0$, so that N_h is coercive (Lemma 3.11). The error equation of Corollary 3.15 has two right-hand sides. The geometric part $\mathcal{G}$ is of Scott's order. The traction part $\mathcal{R}(v) = -\langle \tilde{p} - \bar{p}, v \cdot n \rangle$ is of order one; it is a normal load on the wall, and the only term of N_h that can balance a normal load of order one on the wall is the penalty. This suggests that u_h carries a normal wall velocity $(h/\mu)(p - \bar{p})$. We now make this precise: first in the energy norm, where the leak is seen directly, then in the H^1 seminorm, where incompressibility improves the rate, and finally at the level of the exact leading term and its H^1 constant.

6.1 A divergence-free field with prescribed normal trace

All lower bounds need a discrete divergence-free field whose trace on the wall looks like the leak. Let

$$\varphi_0(\theta) := \int_0^\theta (p - \bar{p})(\theta') d\theta',$$

which is periodic because $p - \bar{p}$ has zero mean on Γ , and let $\varphi := \chi_1(r)\varphi_0(\theta)$ with a cut-off χ_1 equal to 1 near $r = 1$ and 0 near ∂R . Then $w := \text{curl } \varphi$ satisfies

$$w = (p - \bar{p})e_r = -(p - \bar{p})n \quad \text{on } \Gamma.$$

Let Π_{k+1} be the C^1 piecewise- P_{k+1} interpolant of Morgan and Scott [62] (for $k = 4$, the Argyris interpolant [63]); it exists because $k + 1 \geq 5$. The field $v_\varphi := \text{curl } \Pi_{k+1}\varphi$ is continuous, piecewise P_k , divergence-free and vanishes near ∂R , so $v_\varphi \in Z_h$, and $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$.

6.2 The energy norm: rate exactly one half

Theorem 6.1 (Energy error of the printed method; announced in [25]) *The error of GS satisfies*

$$\|\tilde{u} - u_h\| \leq C \left[(h/\mu)^{1/2} G + (1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of $\gamma \geq \gamma_0$, then also

$$\|\tilde{u} - u_h\| \geq (2M)^{-1} (h/\mu)^{1/2} G - C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

In particular, for $G = 0$ the printed method satisfies $\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, which is Scott's expected estimate.

Proof Upper bound. For $z \in W_h$ the difference $e_h := z - u_h$ lies in Z_h . By Corollary 3.15 and Lemma 3.11,

$$c_1 \|e_h\|^2 \leq N_h(z - \tilde{u}, e_h) + \mathcal{R}(e_h) + \mathcal{G}(e_h).$$

With the Jacobian bound of Lemma 3.6, $|\mathcal{R}(e_h)| \leq (G + Ch_\Gamma^2)\|e_h\|_{\Gamma_h} \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2}\|e_h\|$. Lemmas 3.14 and 3.13 bound the other two terms.

Lower bound. On a chord, $v_\varphi \cdot n_e = -(p - \bar{p})(x^*)n(x^*) \cdot n_e + O(h^k)$ and $n(x^*) \cdot n_e = 1 + O(s^2)$, so

$$\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k), \quad \|v_\varphi\|^2 \leq \frac{\mu}{h}(G + Ch_\Gamma^2 + Ch^k)^2 + C.$$

By continuity of N_h , $M\|\tilde{u} - u_h\| \geq (|\mathcal{R}(v_\varphi)| - |\mathcal{G}(v_\varphi)|)/\|v_\varphi\|$, and the quotient $|\mathcal{R}(v_\varphi)|/\|v_\varphi\|$ is at least $\frac{1}{2}(h/\mu)^{1/2}G$ once $h \leq h_0(G)$. $\square$

The lower bound is carried by the slip part $(\mu/h)^{1/2}\|\cdot\|_{\Gamma_h}$ of the energy norm. When $G > 0$, $\gamma h_\Gamma \ll G$ and $\varepsilon_h \ll (h/\mu)^{1/2}G$, the energy error is therefore $\asymp (h/\mu)^{1/2}$: the energy rate is exactly $\frac{1}{2}$. The H^1 seminorm behaves better.

6.3 The H^1 seminorm: incompressibility turns a normal load into a tangential derivative

A normal velocity of size h/μ on the chords, lifted into the domain by a generic discrete field, costs $(h/\mu)h_\Gamma^{-1/2}$ in H^1 , because the lift oscillates on the element scale. The following lemma shows that divergence-free fields do not pay this price for *normal* loads.

Lemma 6.2 (Normal-load cancellation) *Let S be edgewise $W^{1,\infty}$ on Γ_h , continuous at the vertices of Γ_h , with $|S \cdot \tau_e| \leq Ch_\Gamma$ on every $e \in \mathcal{E}_h^\Gamma$ (for instance S smooth with $S \cdot \tau = 0$ on Γ). Then for $v \in Z_h$,*

$$|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2}\|\nabla v\|) \leq C\|\nabla v\|.$$

Proof The tangential part of S is small: $|\langle (S \cdot \tau)\tau, \partial_n v \rangle| \leq Ch_\Gamma\|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2}\|\nabla v\|$ by Lemma 3.10 and (M0). For the normal part, v is a polynomial on the triangle T_e with $\text{div } v = 0$, so on e

$$\partial_n v \cdot n_e = \partial_n(v \cdot n_e) = -\partial_\tau(v \cdot \tau_e).$$

With $\sigma_e := S \cdot n_e$, integration by parts along the chord gives

$$\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}.$$

At a polygon vertex x shared by e and e' the endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e)$, which is of size $C(|e| + |e'|)|v(x)|$ because S is continuous and the chords turn by $O(h_\Gamma)$. Summing, $\sum_e |e| \|v\|_{L^\infty(e)} \leq C|\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$, and Lemma 3.8 finishes. $\square$

Computations confirm that this is exactly what incompressibility buys: the discrete dual norm of $v \mapsto \langle S, \partial_n v \rangle$ stays bounded on Z_h for normal S , but grows like $h_\Gamma^{-1/2}$ on V_h^R and also on Z_h for tangential S (Table 14).

Theorem 6.3 (H^1 error of the printed method; announced in [25]) *With $C_p := C(1 + \|p - \bar{p}\|_{W^{1,\infty}(\Gamma)})$,*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h}{\mu} + C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

Proof Split. For $z \in W_h$ write $z - u_h = \delta_R + \delta_G$ with $\delta_R, \delta_G \in Z_h$ defined by

$$N_h(\delta_R, v) = \mathcal{R}(v), \quad N_h(\delta_G, v) = \mathcal{G}(v) - N_h(\tilde{u} - z, v) \quad \forall v \in Z_h.$$

The geometric part is bounded in energy by coercivity: $\|\nabla \delta_G\| \leq \|\delta_G\| \leq C(1 + \gamma^{1/2}) h_\Gamma^{3/2} + CE_h$.

Ansatz for the traction part. We expect $\delta_R \approx (h/\mu)w$, with $w = -(p - \bar{p})n$ on the wall. Put $r_h := \delta_R - (h/\mu)v_\varphi \in Z_h$. Then $N_h(r_h, v) = m(v) + \ell(v)$ with

$$m(v) := -\langle (\bar{p} - p)n + v_\varphi, v \rangle, \quad \ell(v) := -\frac{h}{\mu} [a_h(v_\varphi, v) - \langle \partial_n v_\varphi, v \rangle - \langle v_\varphi, \partial_n v \rangle].$$

The penalty applied to $(h/\mu)v_\varphi$ has cancelled the traction up to the mismatch m .

The mismatch. On a chord, $(\bar{p} - p)n_e + v_\varphi = (p - \bar{p})(x^*)(n_e - n(x^*)) + O(h_\Gamma^2 + h^k) = (p - \bar{p})(m_e^*) s \tau_e + O(h_\Gamma^2 + h^k)$. The leading term is odd in s with constant coefficient, and the argument of Lemma 3.9 gives $|m(v)| \leq C_p(h_\Gamma^{3/2} + h^k) \|\nabla v\|$.

The remaining terms. $|a_h(v_\varphi, v)| + |\langle \partial_n v_\varphi, v \rangle| \leq C_p \|\nabla v\|$. The dangerous term is $\langle v_\varphi, \partial_n v \rangle$, which a trace-inverse estimate would bound only by $h_\Gamma^{-1/2} \|\nabla v\|$. But v_φ is within Ch^k of the normal field w , so Lemma 6.2 applies:

$$|\langle v_\varphi, \partial_n v \rangle| \leq |\langle w, \partial_n v \rangle| + |\langle v_\varphi - w, \partial_n v \rangle| \leq C_p \|\nabla v\| + Ch^k h_\Gamma^{-1/2} \|\nabla v\|.$$

Hence $|\ell(v)| \leq (h/\mu)(C_p + Ch^k h_\Gamma^{-1/2}) \|\nabla v\|$.

Close. Testing with $v = r_h$ and using $(h/\mu)h^k h_\Gamma^{-1/2} = \gamma^{-1} h^k h_\Gamma^{1/2} \leq Ch^k$,

$$\|\nabla \delta_R\| \leq \frac{h}{\mu} \|\nabla v_\varphi\| + \|\nabla r_h\| \leq C_p \frac{h}{\mu} + C(h_\Gamma^{3/2} + \varepsilon_h).$$

Collect $\tilde{u} - u_h = (\tilde{u} - z) + \delta_R + \delta_G$. $\square$

Remark 6.4 (Why rate one and not one half) Without incompressibility Lemma 6.2 would give only $Ch_\Gamma^{-1/2}$, and the H^1 error would be $(h/\mu)h_\Gamma^{-1/2}$, with an element-scale oscillation in the trace. The inconsistency $-pn$ is purely *normal*, and $\operatorname{div} v = 0$ converts $\partial_n v \cdot n$ into a tangential derivative, which can be moved onto the smooth load. This is why the printed method still converges at rate 1 in H^1 .

The rate h/μ is sharp. The next theorem shows that, as long as the traction term dominates, the slip is of size exactly $(h/\mu)G$, and the H^1 error cannot be smaller.

Theorem 6.5 (Slip and H^1 lower bounds; announced in [25]) *Let $G > 0$ and let $X := C_p(h/\mu) + (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$ be the bound of Theorem 6.3. Suppose that, for a fixed small δ ,*

$$X \leq \delta G \min(G, h_\Gamma^{1/2}), \quad h/\mu \leq \delta G, \quad h_\Gamma \leq h_0(G),$$

which holds for fixed γ and bounded ρ as $h_\Gamma \rightarrow 0$. Then

$$c \frac{h}{\mu} G \leq \|\tilde{u} - u_h\|_{L^2(\Gamma_h)} \leq \left(\frac{h}{\mu}\right)^{1/2} \|\tilde{u} - u_h\|, \quad \|\nabla(\tilde{u} - u_h)\| \geq c \frac{h}{\mu} G - Ch^{k+1/2}.$$

Together with Theorem 6.3, $\|\nabla(\tilde{u} - u_h)\| \asymp h/\mu$ in this regime.

Proof Let $e_u := \tilde{u} - u_h$ and $v := v_\varphi$. Corollary 3.15 rearranges to

$$\frac{\mu}{h} \langle e_u, v \rangle = \mathcal{R}(v) + \mathcal{G}(v) - a_h(e_u, v) + \langle \partial_n e_u, v \rangle + \langle e_u, \partial_n v \rangle.$$

We bound the five terms on the right, using that v is smooth up to $O(h^k)$.

1. $\mathcal{R}(v) = G^2 + O(h_\Gamma^2 + h^k)$, as in the proof of Theorem 6.1.
2. $|\mathcal{G}(v)| \leq C(h_\Gamma^{3/2} + \gamma h_\Gamma^3 + \gamma h_\Gamma h^k)$. The only term that could be large is the penalty part $(\mu/h)\langle \tilde{u}, v \rangle$; but on the chords $\tilde{u}$ is tangential to leading order (the chordal bump (1)) while $v \approx -(p - \bar{p})n$ is normal, so $\tilde{u} \cdot v = O(h_\Gamma^4 + h_\Gamma^2 h^k)$.
3. $|a_h(e_u, v)| \leq C_p \|\nabla e_u\| \leq C_p X \leq C_p \delta G^2$ by Theorem 6.3.
4. By Theorem 6.1 and $h/\mu \leq \delta G$, $|\langle e_u, \partial_n v \rangle| \leq C_p \|e_u\|_{\Gamma_h} \leq C_p (h/\mu)^{1/2} \|e_u\| \leq C_p [(h/\mu)G + X] \leq 2C_p \delta G^2$.
5. Split $e_u = (\tilde{u} - z) + (z - u_h)$ with $z \in W_h$ from Lemma 3.13; by Lemma 3.10, $|\langle \partial_n e_u, v \rangle| \leq \|\partial_n(\tilde{u} - z)\|_{\Gamma_h} \|v\|_{\Gamma_h} + C(\rho/h)^{1/2} (\|\nabla(z - \tilde{u})\| + \|\nabla e_u\|) \|v\|_{\Gamma_h} \leq Ch_\Gamma^{-1/2} XG \leq C\delta G^2$.

For δ small, $|\langle e_u, v \rangle| \geq \frac{1}{2}(h/\mu)G^2$. Dividing by $\|v\|_{\Gamma_h} \leq G + Ch_\Gamma^2$ gives the lower bound for the slip; the upper bound is the definition of $\|\cdot\|$. For the H^1 bound, let $E \in H^1(\Omega_h)$ lift $(g - g_I)|_{\partial R}$ with $E|_{\Gamma_h} = 0$ and $\|E\|_{H^1} \leq Ch^{k+1/2}$; Lemma 3.8 applied to $e_u - E$ gives $\|e_u\|_{\Gamma_h} \leq C_{\text{tr}}(\|\nabla e_u\| + \|\nabla E\|)$. $\square$

6.4 The leading term

We can now identify the error itself, not only its size.

Corollary 6.6 (Exact leading term; announced in [25]) *Let $G > 0$ and let $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$ and $\varepsilon_h/(h/\mu)^{1/2} \rightarrow 0$ (for example γ fixed, ρ bounded). Then*

$$\tilde{u} - u_h = \frac{h}{\mu} v_\varphi + \vartheta_h, \quad \|\vartheta_h\| = o((h/\mu)^{1/2}), \quad \|\vartheta_h\|_{\Gamma_h} = o(h/\mu),$$

and consequently

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1.$$

Proof In the notation of the proof of Theorem 6.3, $\tilde{u} - u_h = (\tilde{u} - z) + (h/\mu)v_\varphi + r_h + \delta_G$. The same energy argument gives $\|r_h\| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$, and $\|\delta_G\| + \|\tilde{u} - z\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$; on Γ_h each of these carries the extra factor $(h/\mu)^{1/2}$. Relative to $(h/\mu)^{1/2}G$ and $(h/\mu)G$ they are $o(1)$ under the hypotheses. Finally $\|v_\varphi\|_{\Gamma_h} = G(1 + O(h_\Gamma^2)) + O(h^k)$ and $\|v_\varphi\|^2 = (\mu/h)\|v_\varphi\|_{\Gamma_h}^2 + O(1)$. $\square$

Since $v_\varphi \approx -(p - \bar{p})n$ on Γ_h , the printed method replaces the no-slip condition by the leak condition

$$\frac{\mu}{h} u_h \cdot n \approx p - \bar{p} \quad \text{on } \Gamma_h,$$

with universal constant 1 in the slip and energy norms (Figure 3). The leak does not change the total mass: since $\int_\Gamma (p - \bar{p}) = 0$, the fluid that leaves through the parts of the wall where the pressure is above its mean re-enters where it is below, and the net flux through Γ_h is exactly that dictated by the outer data. What is violated is the local no-penetration condition, by an amount proportional to the local excess pressure. The picture is that of a slightly porous wall whose permeability is h/μ ; as $h \rightarrow 0$ with μ fixed, the wall becomes impermeable at rate h . This is the sharp answer to Scott's question for the printed method: *the estimate $h_\Gamma^{3/2} + h^k$ holds if and only if p is constant on Γ , and otherwise the error is the explicit leak field.*

6.5 The H^1 constant is a Stokes flow

Corollary 6.6 identifies the slip and energy constants but not the H^1 constant $\|\nabla(\tilde{u} - u_h)\|_{\mu/(hG)}$, because the remainder r_h has the same H^1 order as $(h/\mu)v_\varphi$. The correct H^1 description is global: the leak drives a Stokes flow in the whole domain.

Definition 6.7 (Leak flow) Let (U, P) solve $-\Delta U + \nabla P = 0$, $\operatorname{div} U = 0$ in Ω , with $U = -(p - \bar{p})n$ on Γ and $U = 0$ on ∂R , and put $K := \|\nabla U\|_{L^2(\Omega)}/G$.

The data are compatible ($\int_\Gamma U \cdot n = 0$), and U minimises the Dirichlet energy among divergence-free fields with these boundary values. In particular K depends on $p|_\Gamma$ and on the outer boundary.

Theorem 6.8 (The leak flow is the H^1 limit) *Let $\tilde{U}$ be a smooth divergence-free extension of U across Γ . For GS with $\gamma \geq \gamma_0$,*

$$\left\| \nabla \left(\frac{\mu}{h} (\tilde{u} - u_h) - \tilde{U} \right) \right\|_{\Omega_h} \leq C_U \left[\min(\gamma h_\Gamma^{1/2}, (\gamma h_\Gamma)^{1/2}) + h_\Gamma^{1/2} + (h/\mu)^{1/2} + E_U \right] \\ + C \frac{\mu}{h} \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right],$$

where $E_U := h + (1 + (\gamma h_\Gamma)^{1/2})h^k + (1 + \gamma^{1/2})h_\Gamma^{\sigma_k - 1/2}$ is an approximation error for $\tilde{U}$ and C_U depends on norms of p and U near Γ . Consequently, if $h_\Gamma \rightarrow 0$ with $h/\mu \rightarrow 0$, $\gamma h_\Gamma \rightarrow 0$, $\gamma(1 + \gamma^{1/2})h_\Gamma^{1/2} \rightarrow 0$ and $(\mu/h)\varepsilon_h \rightarrow 0$ (all of which hold for fixed γ and bounded ρ),

$$\frac{\|\nabla(\tilde{u} - u_h)\|}{(h/\mu)G} \rightarrow K.$$

Proof We treat the traction response δ_R of the proof of Theorem 6.3; the rest of the error, $(\tilde{u} - z) + \delta_G$, is bounded in energy by $C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$, which after multiplication by μ/h is the last term of the bound. By a variant of Lemma 3.13 (with $g = 0$ and only H^2 regularity of U away from Γ) there is $z_U \in Z_h$ with $\|\tilde{U} - z_U\| \leq CE_U$. Put $\epsilon := \delta_R - (h/\mu)z_U \in Z_h$. For $v \in Z_h$,

$$N_h(\epsilon, v) = \mathcal{R}(v) - \frac{h}{\mu}N_h(\tilde{U}, v) + \frac{h}{\mu}N_h(\tilde{U} - z_U, v).$$

Green's formula for $\tilde{U}$. Let $\tilde{P}$ extend P and $F_U := -\Delta\tilde{U} + \nabla\tilde{P}$, which vanishes on Ω , is bounded and is supported in the strip. As in Lemma 3.14, since $\operatorname{div} v = 0$, $v|_{\partial R} = 0$ and $\int_{\Gamma_h} v \cdot n = 0$,

$$N_h(\tilde{U}, v) = (F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{U}, v \rangle, \quad \sigma_U := (\nabla \tilde{U})^\top n - (\tilde{P} - c)n.$$

Cancellation of the penalty terms. With $\mathcal{R}(v) = -\langle (\tilde{p} - \bar{p})n, v \rangle$ the error equation becomes

$$N_h(\epsilon, v) = -\langle \zeta, v \rangle - \frac{h}{\mu} \left[(F_U, v) + \langle \sigma_U, v \rangle - \langle \tilde{U}, \partial_n v \rangle \right] + \frac{h}{\mu} N_h(\tilde{U} - z_U, v),$$

$$\zeta := (\tilde{p} - \bar{p})n + \tilde{U}.$$

The mismatch. Since $\tilde{U}(x^*) = -(p - \bar{p})(x^*)n(x^*)$, Lemma 3.6 gives $\zeta = (p - \bar{p})(m_e^*)s\tau_e + O(h_\Gamma^2)$ on each chord, odd in s to leading order. As in Lemma 3.9, $|\langle \zeta, v \rangle| \leq Ch_\Gamma^{3/2} \|\nabla v\|$; alternatively $\|\zeta\|_{\Gamma_h} \leq Ch_\Gamma$ gives $|\langle \zeta, v \rangle| \leq Ch_\Gamma(h/\mu)^{1/2} \|v\|$.

The $O(h/\mu)$ terms. $|(F_U, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ and $|\langle \sigma_U, v \rangle| \leq C(h/\mu)^{1/2} \|v\|$. On the chords $\tilde{U}$ is continuous at the vertices and nearly normal, $\tilde{U} \cdot \tau_e = O(h_\Gamma)$, so Lemma 6.2 applies to $S = \tilde{U}$ and $|\langle \tilde{U}, \partial_n v \rangle| \leq C((h/\mu)^{1/2} + h_\Gamma^{1/2}) \|v\|$. Finally $|N_h(\tilde{U} - z_U, v)| \leq ME_U \|v\|$.

Close. Testing with $v = \epsilon$,

$$c_1 \|\epsilon\| \leq C \left[\min(h_\Gamma^{3/2}, h_\Gamma(h/\mu)^{1/2}) + \frac{h}{\mu} (h_\Gamma^2 + (h/\mu)^{1/2} + h_\Gamma^{1/2} + E_U) \right].$$

Divide by h/μ , using $h_\Gamma^{3/2} \mu/h = \gamma h_\Gamma^{1/2}$ and $h_\Gamma(\mu/h)^{1/2} = (\gamma h_\Gamma)^{1/2}$, and add $\|\tilde{U} - z_U\|$. For the limit, multiply out and use $\|\nabla \tilde{U}\|_{\Omega_h}^2 - \|\nabla U\|_\Omega^2 = O(h_\Gamma^2)$. $\square$

Two by-products of the proof are worth recording. First, on the pure-pressure test ($u = 0$, $f = \nabla p$) we have $\tilde{u} = 0$ and $\mathcal{G} \equiv 0$, so the error of GS is exactly δ_R and the theorem applies with no further condition. Second, the remainder $r_h = \delta_R - (h/\mu)v_\varphi$ of Theorem 6.3 satisfies $(\mu/h)r_h \rightarrow U - w$ in H^1 : it is the Stokes correction of the cut-off field w , which explains why r_h has the same H^1 order as the explicit term.

Remark 6.9 (The Dirichlet-dominated regime) The hypothesis $\gamma h_\Gamma = \mu h_\Gamma^2/h \rightarrow 0$ of Corollary 6.6 fails for very large penalties (for example $\gamma h_\Gamma \approx 240$ at $N = 96$, $\mu = 10^4$). There the discrete problem is close to its $\mu \rightarrow \infty$ limit, a discrete Dirichlet problem whose data $-(p - \bar{p})n_e$ jump in direction at every polygon vertex. Theorem 6.8 does not cover this regime, but the penalty-uniform analysis of [27] does (Section 8): the scaled traction response still converges to U in H^1 , at the exact rate $h_\Gamma^{1/2}$.

Computing K

The constant is a property of the continuous problem, and it can be bracketed rigorously. For $\varrho > 1$ let $\mathcal{A}_\varrho = \{1 < r < \varrho\}$ and let $\mathcal{E}(\varrho)$ be the minimal Dirichlet energy of divergence-free fields on $\mathcal{A}_\varrho$ with the leak data on $r = 1$ and zero data on $r = \varrho$. Extending by zero shows that the minimum decreases as the domain grows.

Proposition 6.10 (Annulus bracket) *If $\mathcal{A}_{\varrho_1} \subset \Omega \subset \mathcal{A}_{\varrho_2}$, then $\mathcal{E}(\varrho_2) \leq \|\nabla U\|^2 \leq \mathcal{E}(\varrho_1)$. In the annulus the Fourier modes decouple: if $\Psi|_\Gamma = \sum_n (a_n \cos n\theta + b_n \sin n\theta)$ is the boundary stream function of the leak, then $\mathcal{E}(\varrho) = \sum_n (a_n^2 + b_n^2) E_n(\varrho)$ with*

$$E_n(\varrho) = \pi \int_1^\varrho \left[f''^2 + 2n^2 \left((f/r)' \right)^2 + \left(f'/r - n^2 f/r^2 \right)^2 \right] r dr,$$

where f is the combination of $r^n, r^{-n}, r^{n+2}, r^{2-n}$ (of $r, r^{-1}, r^3, r \log r$ for $n = 1$) with $f(1) = 1$, $f'(1) = 0$ and $f(\varrho) = f'(\varrho) = 0$.

For the pressure of test P, $p - \bar{p} = -\frac{3}{4} \sin \theta + \frac{1}{4} \sin 3\theta + \frac{1}{2} \cos \theta$ on Γ , so $\Psi|_\Gamma = \frac{3}{4} \cos \theta + \frac{1}{2} \sin \theta - \frac{1}{12} \cos 3\theta$ and $G^2 = 7\pi/8$. With $\varrho_1 = L = 2.5$ and $\varrho_2 = L\sqrt{2}$ the proposition gives the rigorous bracket $2.0746004 \leq K \leq 3.0525769$. The exact value on the box is computed by a series in the odd Fourier modes (imposing the outer conditions by least squares at 16 000 points); it is stable to eight digits under refinement of both the truncation and the quadrature:

number of modes $n_{\max}$	max boundary residual	$\ \nabla U\ $	K
15	$4.4 \cdot 10^{-3}$	4.57012143	2.75644148
31	$6.1 \cdot 10^{-5}$	4.57024239	2.75651444
47	$5.1 \cdot 10^{-6}$	4.57024449	2.75651571
79	$1.8 \cdot 10^{-7}$	4.57024450	2.75651571

The same formulas describe the dependence on the box. As $\varrho \rightarrow \infty$, $E_3(\varrho) \rightarrow 45\pi$ and $E_1(\varrho) \rightarrow \pi$; the $n = 1$ data split into a potential dipole, whose exterior energy is exactly π , and a rigid translation, whose energy vanishes as the box grows (the Stokes paradox). Hence

$$K(L) \longrightarrow \sqrt{\frac{(\frac{9}{16} + \frac{1}{4})\pi + \frac{45\pi}{144}}{7\pi/8}} = \frac{3}{\sqrt{7}} \approx 1.134 \quad (L \rightarrow \infty),$$

but only logarithmically: in annuli the values are 1.420, 1.242, 1.183 at $\varrho = 10, 10^2, 10^4$ (the logarithmic rate is observed, not proved).

So the constant 2.7–2.8 observed in Table 11 belongs to the box, not to the method. The approach to K is pre-asymptotic at moderate penalties: at $\mu = 100$ the measured constants increase towards K , at large μ they decrease towards it, and on test P the relative distance $\|\nabla((\mu/h)(\tilde{u} - u_h) - \tilde{U})\|/\|\nabla U\|$ decays at the exact rate $h_\Gamma^{1/2}$ for every penalty, the Dirichlet limit $\mu \rightarrow \infty$ included (Section 8).

7 The corrected method: Scott's rate and its sharpness

Relation to [25]. Proposition 7.1 and Theorem 7.2 were announced in the joint letter [25] with proof sketches, and Theorems 7.3 and 7.14 and the results of Section 7.4 were announced there with a pointer or an idea of proof only. We give complete proofs here, except for the interval certificates of the star condition (Theorem 7.12), which are in [27].

7.1 Why the plain pressure term fails

Adding the missing traction $\langle p_h, v \cdot n \rangle$ to the momentum equation looks like the natural repair. With an exactly divergence-free pair it makes the square system singular, and the usual way of removing the singularity destroys exact incompressibility.

Proposition 7.1 (Naive consistent form, mean-zero variant; announced in [25]) (a)

Replace $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ in CNS by $\langle p_h, v \cdot n \rangle$. The resulting square system has the kernel vector $(u, p) = (0, 1)$, so it is singular and solvable only under one linear compatibility condition on the data.*

(b) *Restrict the pressure to $\Pi_h \cap L_0^2(\Omega_h)$ and test the continuity equation only with mean-zero q . Any solution has $\text{div } u_h \equiv c$, a constant with*

$$c |\Omega_h| = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} u_h \cdot n.$$

(c) *For $\gamma \geq \gamma_2$ (Theorem 7.2) the naive system of (a) is solvable if and only if the CNS* pressure has zero mean on Γ_h , and then its solutions are the CNS* solution up to a constant pressure.*

Proof (a) For $v \in V_h^R$, $-(1, \text{div } v) + \langle 1, v \cdot n \rangle = -\int_{\partial\Omega_h} v \cdot n + \int_{\Gamma_h} v \cdot n = 0$, and the continuity row vanishes at $u = 0$. (b) Since $\text{div } u_h \in \Pi_h$ is orthogonal to all mean-zero $q \in \Pi_h$, it is constant; integrate over Ω_h . (c) The naive pressure form evaluated at r equals the CNS* pressure form at $r - \bar{p}_\Gamma(r)$; use the uniqueness of Theorem 7.2. $\square$

With weak imposition nothing forces $\int_{\Gamma_h} u_h \cdot n$ to cancel the outer flux, even for flux-corrected data, so the constant c in (b) is nonzero in general. In our computations it is: for test A at $N = 16$, $\mu = 100$, with the nodal interpolant g_I as outer data, $|c| = 9.69 \cdot 10^{-4}$, so the discrete divergence has L^2 norm $|c| |\Omega_h|^{1/2} = 4.54 \cdot 10^{-3}$, while CNS* on the same mesh gives $4 \cdot 10^{-13}$ (script `code/120check.py`, archived output `logs/120/`; compare [26, (5.9)]). The singularity in (a) is the analogue, for our square system, of the ill-posedness that Liu, Neilan and Otus observe when their zero-flux constraint is omitted [32, Remark 3.2].

7.2 The mean-free correction: well posed and of Scott's order

The correction CNS* removes the wall mean of the discrete pressure inside the traction term. Its pressure form $B^*(q, v) := -(q, \text{div } v) + \langle q - \bar{p}_\Gamma(q), v \cdot n \rangle$ satisfies $B^*(q + c, v) =$

$B^*(q, v) - c \int_{\Gamma_h} v \cdot n$, so a constant pressure is no longer invisible: it is tested by the net flux, and the chord bubbles have nonzero flux. At the same time the exact pressure normalised by its own wall mean, $p^* := \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, is consistent:

$$N_h(\tilde{u} - u_h, v) + B^*(p^* - p_h, v) = \mathcal{G}(v) \quad \forall v \in V_h^R, \quad (4)$$

by Lemma 3.14. The only remaining inconsistency is the geometric one, of Scott's order.

Theorem 7.2 (CNS* is well posed and has Scott's rate; announced in [25]) *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$ the method CNS* has a unique solution, its velocity is divergence-free pointwise, and*

$$\|\tilde{u} - u_h\| \leq C \left[(1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h \right].$$

In particular, for every fixed $k \geq 4$, γ in a bounded range $[\gamma_2, \gamma_{\max}]$ and $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k).$$

Proof Write $e_u := \tilde{u} - u_h$ and $e_p := p^* - p_h = \eta + \xi$ with $\eta := p^* - \pi_h p^*$ (π_h the L^2 projection onto Π_h) and $\xi \in \Pi_h$.

Pressure. For $v \in V_h^0$ the boundary terms vanish and $(\eta, \operatorname{div} v) = 0$ because $\operatorname{div} V_h^0 \subset \Pi_h$, so (4) gives $(\xi - \bar{\xi}, \operatorname{div} v) = a_h(e_u, v) - \langle e_u, \partial_n v \rangle - \mathcal{G}(v)$. The right side is at most $C(\|e_u\| + h_\Gamma^{3/2})\|\nabla v\|$, and Lemma 3.12 yields $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$.

Velocity. For $z \in W_h$ put $e_h := z - u_h \in Z_h$ and test (4) with it. Since $\operatorname{div} e_h = 0$ and $\int_{\Gamma_h} e_h \cdot n = 0$, only the mean-free parts of the pressure error survive:

$$c_1 \|e_h\|^2 \leq M E_h \|e_h\| + |\mathcal{G}(e_h)| + |\langle \eta, e_h \cdot n \rangle| + |\langle \xi - \bar{\xi}, e_h \cdot n \rangle|.$$

Here $|\langle \eta, e_h \cdot n \rangle| \leq C h_\Gamma^k (h/\mu)^{1/2} \|e_h\|$, and by Lemma 3.10

$$|\langle \xi - \bar{\xi}, e_h \cdot n \rangle| \leq C_I (\rho/h)^{1/2} \|\xi - \bar{\xi}\| (h/\mu)^{1/2} \|e_h\| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|e_h\|.$$

The pressure error enters the velocity estimate with the small factor $\gamma^{-1/2}$. Insert the pressure bound and $\|e_u\| \leq E_h + \|e_h\|$; for $\gamma \geq \gamma_2$ with $C(c_0 \gamma_2)^{-1/2} \beta^{-1} \leq c_1/2$ the pressure term is absorbed.

Uniqueness. For zero data the estimates give $u_h = 0$, hence p_h constant, and then $-c \int_{\Gamma_h} v \cdot n = 0$ for all v forces $c = 0$. The system is square, so uniqueness gives existence. $\square$

The proviso $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$ only serves to make the flux term of ε_h smaller than h^k . It disappears with flux-corrected outer data, and then the H^1 bound is even uniform in the penalty.

Theorem 7.3 (Uniformity in the penalty; flux-corrected data) *For GS with any $\gamma \geq \gamma_0$ and for CNS* with any $\gamma \geq \gamma_2$ (no upper bound),*

$$\|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h_\Gamma}{\gamma} \mathbf{1}_{\text{GS}} + C \left(h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2} \right),$$

where the first term is present only for GS. With flux-corrected data the term $h^{\sigma_k} h_\Gamma^{-1/2}$ is replaced by $h_\Gamma^{\sigma_k-1/2} + |m_R|$, which is of higher order; in particular CNS* with flux-corrected data satisfies, for every $\gamma \geq \gamma_2$ and every $h_\Gamma \leq h$,

$$\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h^k),$$

with C independent of γ and no relation between h_Γ and h . This is the rate Scott conjectured, for general Stokes data.

The proof compares the Nitsche solution with a strongly imposed discrete Dirichlet solution. It needs two lemmas. Let $b = \sum_e \beta_e n_e$ be the sum of the chord bubbles, so that $\int_{\Gamma_h} b \cdot n = |\Gamma_h|/6$.

Lemma 7.4 (Divergence-free lift of a trace) *Let t be continuous and piecewise P_k on Γ_h with $\int_{\Gamma_h} t \cdot n = 0$. There is $L \in Z_h$ with $L|_{\Gamma_h} = t$ and $\|\nabla L\| \leq C h_\Gamma^{-1/2} \|t\|_{L^2(\Gamma_h)}$.*

Proof Let $L_0 \in V_h^R$ take the nodal values of t at the Lagrange nodes on Γ_h and vanish at all other nodes. The nodal values on a chord are bounded by $C|e|^{-1/2} \|t\|_{L^2(e)}$, and nodal basis functions have $\|\nabla \phi_i\| \leq C$ by scale invariance, so $\|\nabla L_0\| \leq C h_\Gamma^{-1/2} \|t\|_{\Gamma_h}$. Its divergence has mean $\int_{\Gamma_h} t \cdot n = 0$, and Lemma 3.12 removes it by a field in V_h^0 . $\square$

Lemma 7.5 (The discrete polygonal Dirichlet problem) *Put $a := -6m_R/|\Gamma_h|$ and $W_h^D := \{z \in W_h : z|_{\Gamma_h} = a b\}$. There is a unique $u^D \in W_h^D$ with $a_h(u^D, v) = (\tilde{f}, v)$ for all $v \in Y_h$, and*

$$\|\nabla(\tilde{u} - u^D)\| \leq C(h_\Gamma^{3/2} + h^k + |m_R| h_\Gamma^{-1/2}).$$

Proof Subtract from $I_h \tilde{u}$ the nodal extension of $I_h \tilde{u}|_{\Gamma_h} - ab$, whose trace has L^2 norm $C(h_\Gamma^2 + |m_R|)$ by Lemma 3.7; the result has the right traces and total divergence $m_R + a|\Gamma_h|/6 = 0$, and Lemma 3.12 makes it divergence-free. This gives $z \in W_h^D$ with $\|\nabla(\tilde{u} - z)\| \leq C(h^k + h_\Gamma^{3/2} + |m_R| h_\Gamma^{-1/2})$. For $v \in Y_h$, $a_h(\tilde{u}, v) = (\tilde{f} - F, v)$, so $a_h(\tilde{u} - u^D, v) = -(F, v)$, and a Céa argument with Korn's inequality on Y_h finishes. $\square$

Proof of Theorem 7.3 We give the proof for CNS*; for GS one first removes the traction response δ_R of Theorem 6.3, whose H^1 norm is $C_p(h_\Gamma/\gamma + h_\Gamma^{3/2} + h^k)$ uniformly in γ , and applies the same argument to $u_h + \delta_R$.

1. *The identity on Y_h .* By Lemma 7.7 below, $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v) = a_h(u^D, v)$ for all $v \in Y_h$. On the fields that vanish on the polygon, CNS* is the Dirichlet problem, except for the symmetric Nitsche term acting on the trace of u_h .
2. *The trace of u_h is small.* By Lemma 3.7 and Theorem 7.2,

$$\|u_h\|_{\Gamma_h} \leq \|\tilde{u}\|_{\Gamma_h} + (h/\mu)^{1/2} \|\tilde{u} - u_h\| \leq C(h_\Gamma^2 + (h_\Gamma/\gamma)^{1/2} \varepsilon_h),$$

since $(h_\Gamma/\gamma)^{1/2}\gamma^{1/2}h_\Gamma^{3/2} = h_\Gamma^2$. Moreover $h_\Gamma^{-1/2}(h_\Gamma/\gamma)^{1/2}\varepsilon_h = \gamma^{-1/2}\varepsilon_h \leq C(h^k + h^{\sigma_k}h_\Gamma^{-1/2})$: the two γ -growing parts of ε_h are exactly compensated, because $\gamma^{-1/2}(\gamma h_\Gamma)^{1/2} = h_\Gamma^{1/2}$ and $\gamma^{-1/2}(\mu/h)^{1/2} = h_\Gamma^{-1/2}$.

3. *Comparison.* $w := u_h - u^D \in Z_h$ has trace $u_h - ab$ on Γ_h with zero flux. Let L be its lift from Lemma 7.4; then $y := w - L \in Y_h$ and, by step 1, $a_h(y, y) = \langle u_h, \partial_n y \rangle - a_h(L, y)$. With Lemma 3.10 and $(\rho/h)^{1/2} \leq (c_0 h_\Gamma)^{-1/2}$,

$$\|\nabla w\| \leq \|\nabla L\| + \|\nabla y\| \leq C h_\Gamma^{-1/2} (\|u_h\|_{\Gamma_h} + |m_R|) \leq C (h_\Gamma^{3/2} + h^k + h^{\sigma_k} h_\Gamma^{-1/2}).$$

4. *Assemble* $\tilde{u} - u_h = (\tilde{u} - u^D) - w$ and use Lemma 7.5.

With flux-corrected data, $a = 0$ in Lemma 7.5, the term $|m_R| h_\Gamma^{-1/2}$ is absent in step 3, and ε_h is replaced by its flux-corrected version from Lemma 3.13; this gives the second bound. The L^2 part of the H^1 norm follows from the trace estimate as in Theorem 6.5. $\square$

The mechanism is the classical one for Nitsche's method as $\mu \rightarrow \infty$: the H^1 semi-norm never sees the penalty directly, only through the trace of the discrete solution, which the energy estimate controls.

Remark 7.6 (Relation to the zero-flux formulation) Let $V_h^{R,b} := \{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$. Transcribed to our setting, the device of Frachon, Nilsson and Zahedi [26, Section 5.1] tests the momentum equation with the naive form only for $v \in V_h^{R,b}$ and fixes the pressure constant separately. On $V_h^{R,b}$ the naive and the CNS* pressure forms coincide, and the remaining direction only fixes the constant. So, for $\gamma \geq \gamma_2$, the two formulations have the *same velocity* and pressures that differ by a constant. CNS* is a square packaging of their device, with the constant fixed automatically so that $p_h \approx \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

7.3 The rate $h_\Gamma^{3/2}$ is sharp

Theorem 7.2 is an upper bound. Could CNS* (or GS at constant wall pressure) be better than $h_\Gamma^{3/2}$ in H^1 ? The scalar analogy suggests not: the chordal bump (1) has $H^{1/2}(\Gamma_h)$ size $h_\Gamma^{3/2}$, and no discrete velocity that respects the no-slip condition on Γ_h can reproduce it. But the Nitsche velocities do *not* vanish on Γ_h , and the natural route to a lower bound, through the transposed-gradient term of Lemma 3.9, fails for CNS*: a test field with nonzero normal trace brings in the pressure error, which is itself of order $h_\Gamma^{3/2}$ and could cancel the bound. We use instead test fields that vanish on the polygon. On them the pressure and the penalty drop out completely, and what remains is Nitsche's symmetric term acting on the bump.

Test spaces. Recall $Y_h = Z_h \cap V_h^0$ and put

$$Y_h^1 := \left\{ v \in Y_h : \int_e \partial_n v = 0 \quad \forall e \in \mathcal{E}_h^\Gamma \right\}.$$

For $v \in Y_h$ the normal component $\partial_n v \cdot n_e$ vanishes on each chord (v vanishes on e , so $0 = \operatorname{div} v = \partial_\tau(v \cdot \tau_e) + \partial_n(v \cdot n_e)$ there), so only the tangential component $\partial_n v \cdot \tau_e$ matters.

Lemma 7.7 (Error identity on fields vanishing on the wall) *Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$) or of GS ($\gamma \geq \gamma_0$, any G), and $e_u := \tilde{u} - u_h$. Then for all $v \in Y_h$,*

$$a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v).$$

Proof For $v \in Y_h$ every pressure term vanishes ($\operatorname{div} v = 0$ and $v = 0$ on Γ_h), and so do the penalty and transposed terms. What remains of (4), or of Corollary 3.15, is the displayed identity. $\square$

So the discrete velocity satisfies $a_h(u_h, v) - \langle u_h, \partial_n v \rangle = (\tilde{f}, v)$ on Y_h , and $\tilde{u}$ violates this by $-\langle \tilde{u}, \partial_n v \rangle$: Nitsche's symmetric term applied to the nonzero trace of $\tilde{u}$ on the chords.

Lemma 7.8 (Lower bound by duality) *There is Λ_0 , depending only on shape regularity, such that*

$$\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} \frac{|\langle \tilde{u}, \partial_n v \rangle + (F, v)|}{\|\nabla v\|}.$$

Proof On each chord $\int_e \partial_n v = 0$, so $\int_e w \cdot \partial_n v = \int_e (w - \bar{w}_{T_e}) \cdot \partial_n v$ for the mean $\bar{w}_{T_e}$ of w over T_e . A scaled trace–Poincaré inequality and the inverse trace inequality give $|\langle w, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla w\| \|\nabla v\|$ for every $w \in H^1(\Omega_h)^2$. Apply this to $w = e_u$ in Lemma 7.7, together with $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$. $\square$

Lemma 7.9 (The bump seen by Y_h^1) *For $v \in Y_h^1$,*

$$\left| \langle \tilde{u}, \partial_n v \rangle - \ell_W(v) \right| \leq C h_\Gamma^{5/2} \|\nabla v\|, \quad \ell_W(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} W(m_e^*) \int_e s^2 \partial_n v \cdot \tau_e ds.$$

Proof By (1) and $\partial_n v \cdot n_e = 0$, $\tilde{u} \cdot \partial_n v = \frac{1}{2}(\frac{\ell^2}{4} - s^2)W(m_e^*) \partial_n v \cdot \tau_e + O(h_\Gamma^3)|\partial_n v|$ on e . The constant part of the parabola integrates to zero against $\partial_n v \cdot \tau_e$, whose mean vanishes. Summing the remainders with $\sum_e \|\partial_n v\|_{L^1(e)} \leq C h_\Gamma^{-1/2} \|\nabla v\|$ gives the claim. $\square$

Only the oscillating part $-\frac{1}{2}s^2 W$ of the bump is seen, and it has amplitude h_Γ^2 and wavelength h_Γ : its $H^{1/2}$ size is $h_\Gamma^{3/2}$. What is needed now is a supply of fields in Y_h^1 that actually feel the second moment $\int_e s^2 \partial_n v \cdot \tau_e$. For the fields supported near a polygon vertex z we measure this by

$$\mathcal{M}(v) := -\frac{1}{2} \sum_{e \in \mathcal{E}_h^\Gamma} \int_e s^2 \partial_n v \cdot \tau_e ds,$$

and we let e_z be the chord following z counterclockwise.

Definition 7.10 (Star condition) The mesh satisfies the *star condition* with constants $\kappa_0 > 0$, K_{ov} and C_s if every vertex z of Γ_h carries a field $v_z \in Y_h^1$ with $\|\nabla v_z\| = 1$, supported within distance $C_s h_\Gamma$ of z , with every triangle in at most K_{ov} supports, and with $\mathcal{M}(v_z) \geq \kappa_0 |e_z|^2$.

For large degree such fields are easy to write down.

Lemma 7.11 (Single-triangle fields for $k \geq 7$) *If $k \geq 7$, the star condition holds with v_z supported in the triangle T_{e_z} , $K_{\text{ov}} = 1$, and κ_0 depending only on shape regularity.*

Proof Let $\lambda_1, \lambda_2, \lambda_3$ be the barycentric coordinates of $T = T_{e_z}$ with $\lambda_3 = 0$ on $e = e_z$, $b := \lambda_1 \lambda_2 \lambda_3$, and

$$\phi := b^2((\lambda_1 - \lambda_2)^2 - \frac{1}{7}) \in P_8, \quad v := \text{curl } \phi \in P_7^2.$$

ϕ vanishes to second order on ∂T , so v , extended by zero, is continuous, divergence-free and vanishes on ∂T ; it lies in Y_h for $k \geq 7$. With $\sigma = \lambda_1 - \lambda_2$ on e , $\partial_n v \cdot \tau_e = \pm \frac{1}{8} |\nabla \lambda_3|^2 (1 - \sigma^2)^2 (\sigma^2 - \frac{1}{7})$, whose mean vanishes since $\int_{-1}^1 (1 - \sigma^2)^2 = \frac{16}{15}$ and $\int_{-1}^1 \sigma^2 (1 - \sigma^2)^2 = \frac{16}{105}$; so $v \in Y_h^1$. Its second moment is proportional to $\int_{-1}^1 \sigma^2 (1 - \sigma^2)^2 (\sigma^2 - \frac{1}{7}) d\sigma = \frac{64}{2205} \neq 0$. The ratio $|\mathcal{M}(v)|/(|e|^2 \|\nabla v\|)$ is a continuous nonvanishing function of the shape of T on a compact set of shapes; normalise and fix the sign. $\square$

For $4 \leq k \leq 6$ no field of Y_h^1 lives in one triangle, or in the two triangles of a first-layer quadrilateral, and the construction is genuinely harder. The reason is instructive. Writing $v = \text{curl } \phi$ with ϕ a C^1 piecewise quintic, $\partial_n v \cdot \tau_e$ is, up to sign, the second normal derivative of ϕ on the chord, a cubic with zero mean. At the far end of the chord the Hessian of ϕ vanishes, and if it also vanished at z the cubic would be odd about the midpoint and its second moment zero. So $\mathcal{M}(v) \neq 0$ requires the Hessian of ϕ to *jump* across the interior edges at z , which is possible only when z lies in at least three triangles. The star condition for small degree is thus tied to the same condition (M2) that governs the zero-gradient lock (Section 4).

Theorem 7.12 (The star condition for every $k \geq 4$ [27]) *Assume (M0)–(M2), let every triangle with a vertex on Γ_h have all angles at least $\theta_{\min}$, and let $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$. Then for every $k \geq 4$ the star condition holds with $K_{\text{ov}} = 3$, $C_s = (\sin \theta_{\min})^{-3}$ and $\kappa_0 = \kappa_*(\theta_{\min}, \theta_{\min}/2) > 0$, where κ_* is the minimum of an explicit continuous function over a compact family of three-triangle fans. Rigorous interval certificates give $\kappa_*(30^\circ, 10^\circ) \geq 0.0012$ and $\kappa_*(20^\circ, 15^\circ) \geq 0.0003$.*

Idea On a fan of three triangles at z the homogeneous quadratic C^1 splines that vanish to second order on the two wall rays form a one-dimensional space, spanned by a spline whose Hessians are given by a Plücker-type identity and have a sign when consecutive angle sums are below π . This quadratic part is completed in closed form by a cubic part to a C^1 quintic stream function with zero boundary data and zero chord means; its second moment is a sum of two terms of the same sign. The witness is a P_4 field, so it serves every $k \geq 4$. The algebra and the certificates are in [27]. $\square$

The witness in more detail. Put $z = 0$ and let the fan be $T_j = (0, p_{j-1}, p_j)$, $j = 1, 2, 3$, counterclockwise, with $[0, p_0]$ and $[0, p_3]$ the two chords; write $a \times b := a_1 b_2 - a_2 b_1$ and $D_{ij} := p_i \times p_j$. On T_j let L_j be the barycentric coordinate of z , and $N_j(x) := p_j \times x$,

a linear form vanishing on the ray through p_j . A piecewise quintic ϕ with $\phi = |\nabla\phi| = 0$ on the boundary of the fan has the form $\phi|_{T_j} = L_j^2(Q_j + C_j)$ with Q_j, C_j homogeneous of degrees 2 and 3, and the C^1 conditions across the interior rays reduce to four explicit conditions on the values of Q_j, C_j and ∇C_j at p_j [27]. With

$$\begin{aligned} A &:= D_{13}D_{23} = r_1 r_2 r_3^2 \sin(\alpha_2 + \alpha_3) \sin \alpha_3, \\ A_3 &:= D_{01}D_{02} = r_0^2 r_1 r_2 \sin \alpha_1 \sin(\alpha_1 + \alpha_2), \end{aligned}$$

where $p_i = r_i(\cos \vartheta_i, \sin \vartheta_i)$ and α_j are the angles at z , both A and A_3 are positive whenever $\alpha_1 + \alpha_2 < \pi$ and $\alpha_2 + \alpha_3 < \pi$. In barycentric coordinates $(\lambda_0, \lambda_1, \lambda_2)$ of T_j (with $\lambda_0 = L_j$) the witness is

$$\begin{aligned} \phi_1 &= AD_{01}^2 \lambda_0^2 \lambda_2^2 (\lambda_0 - 3\lambda_1), & \phi_3 &= A_3 D_{23}^2 \lambda_0^2 \lambda_1^2 (\lambda_0 - 3\lambda_2), \\ \phi_2 &= \lambda_0^2 \left[AD_{01}^2 \lambda_1^2 (\lambda_0 + \lambda_2) + A_3 D_{23}^2 \lambda_2^2 (\lambda_0 + \lambda_1) + 2AD_{01}D_{02} \lambda_1 \lambda_2 (\lambda_0 + \lambda_1 + \lambda_2) \right. \\ &\quad \left. + AD_{01}(6D_{12} - 5D_{02} + 2D_{01}) \lambda_1^2 \lambda_2 + A_3 D_{23}(6D_{12} - 5D_{13} + 2D_{23}) \lambda_1 \lambda_2^2 \right], \end{aligned}$$

and $v = \text{curl } \phi$. One checks that ϕ is C^1 , vanishes with its gradient on the boundary of the fan, and that on both chords $\int \partial_{nn} \phi = 0$ while

$$\int_{[0, p_0]} s^2 \partial_{nn} \phi ds = \frac{|p_0|^5 A}{30}, \quad \int_{[0, p_3]} s^2 \partial_{nn} \phi ds = \frac{|p_3|^5 A_3}{30}.$$

Since $\partial_n v \cdot \tau_e = \pm \partial_{nn} \phi$ with the same sign on both chords, $v \in Y_h^1$ and $|\mathcal{M}(v)| = (|p_0|^5 A + |p_3|^5 A_3)/60$ when the fan is the whole star, a sum of two terms of the same sign. (When the star has more than three triangles, the three next to e_z are used, and $|\mathcal{M}(v)| = |p_0|^5 A/60$.) The normalised constant $|\mathcal{M}(v)|/(|e_z|^2 \|\nabla v\|)$ is a continuous positive function of the shape of the fan, and the compactness of the admissible shapes gives $\kappa_* > 0$; an elementary closed-form lower bound and the interval certificates are in [27].

The constants are small but not tiny on real meshes. On every mesh of Section 9 ($N = 8, \dots, 256$, and also $N = 512, 1024$) the three-triangle star at every polygon vertex carries a one-dimensional space of admissible fields, and the star constants $\kappa_z := \mathcal{M}(v_z)/|e_z|^2$ (with $\|\nabla v_z\| = 1$) satisfy $\min_z \kappa_z \geq 0.00287$, converging to about 0.0029 as N grows (Table 5); the minimum is attained in the diagonal directions of the square, where the first-layer triangles are most elongated.

With the local fields in hand, the lower bound is a matter of assembling them with weights $W(z)$ and comparing a Riemann sum with $\|W\|_{L^2(\Gamma)}^2$.

Lemma 7.13 (Assembly) *Assume the star condition, $W \not\equiv 0$ and $h_\Gamma \leq h_1$, with h_1 depending on $\|W\|_{W^{1,\infty}(\Gamma)}/\|W\|_{L^2(\Gamma)}$ and the constants. Then $v := \sum_z W(z) v_z \in Y_h^1$ satisfies*

$$\ell_W(v) \geq \frac{1}{2} \kappa_0 c_0^{5/2} K_{\text{ov}}^{-1/2} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} \|\nabla v\|.$$

Table 5 The star condition on the meshes of Section 9: star constants κ_z of the three-triangle wall stars (exact witnesses of Theorem 7.12; a floating-point optimisation over the star gives the same values), and the assembled field of Lemma 7.13: $\langle \tilde{u}, \partial_n v \rangle / (\|\nabla v\| h_\Gamma^{3/2})$, with the leading term $\ell_W(v) / (\|\nabla v\| h_\Gamma^{3/2})$ in parentheses, for test A.

N	$\min_z \kappa_z$	mean	$\max_z \kappa_z$	N	assembled field	rate
8	0.00288	0.0055	0.0080	32	0.0606 (0.0605)	–
16	0.00555	0.0109	0.0179	64	0.0651 (0.0651)	1.39
64	0.00366	0.0145	0.0252	128	0.0665 (0.0665)	1.47
256	0.00303	0.0149	0.0260	256	0.0669 (0.0669)	1.49
1024	0.00288	0.0149	0.0261	512	0.0670 (0.0670)	1.50

Proof By bounded overlap $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_z W(z)^2$. Since W varies by $O(h_\Gamma)$ over a support,

$$\begin{aligned} \ell_W(v) &\geq \sum_z W(z)^2 \mathcal{M}(v_z) - Ch_\Gamma^3 \|W\|_{W^{1,\infty}} \sum_z |W(z)| \\ &\geq \kappa_0 c_0^2 h_\Gamma \sum_z W(z)^2 |e_z| - Ch_\Gamma^2 \|W\|_{W^{1,\infty}}^2. \end{aligned}$$

The sum $\Sigma_W := \sum_z W(z)^2 |e_z|$ is a Riemann sum for $\|W\|_{L^2(\Gamma)}^2$ and $\sum_z W(z)^2 \leq \Sigma_W / (c_0 h_\Gamma)$. For small h_Γ , $\Sigma_W \geq \frac{3}{4} \|W\|_{L^2(\Gamma)}^2$ and the negative term is at most a quarter of the positive one. $\square$

Theorem 7.14 (H^1 lower bound; announced in [25]) *Assume (M0)–(M2), Assumption 3.5 and the star condition (which holds for every $k \geq 4$ when $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$, by Theorem 7.12), and let the wall shear $W = \partial_n u \cdot \tau$ not vanish identically on Γ . Let u_h be the velocity of CNS* with $\gamma \geq \gamma_2$, or of GS with $\gamma \geq \gamma_0$ and any G . Then for $h_\Gamma \leq h_1$,*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c_{\text{LB}} \kappa_0 \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - Ch_\Gamma^2, \quad c_{\text{LB}} := \frac{c_0^{5/2}}{2\Lambda_0 K_{\text{ov}}^{1/2}},$$

with constants independent of γ , μ , ρ , h_Ω , the pressure and ε_h .

Proof Insert the field of Lemma 7.13 in Lemma 7.8 and use Lemma 7.9 and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$. $\square$

Corollary 7.15 (Sharpness of Scott's rate) *Under the assumptions of Theorem 7.14, for CNS* with $\gamma \in [\gamma_2, \gamma_{\max}]$, and for GS when p is constant on Γ , and whenever $\varepsilon_h \leq Ch_\Gamma^{3/2}$ (for instance with flux-corrected data and $h_\Omega^k \leq h_\Gamma^{3/2}$),*

$$Ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq Ch_\Gamma^{3/2}.$$

So the H^1 rate is exactly $\frac{3}{2}$; it is the term $h_\Gamma^{3/2}$ of Scott's estimate, not h_Ω^k , that is shown to be sharp.

Three remarks put the lower bound in context. First, the bound is carried by the part of $\tilde{u}|_{\Gamma_h}$ that oscillates on each chord; the chord mean of the bump contributes

only $O(h_\Gamma^2)$. Second, the lower bound holds for CNS^* even though its pressure error is itself of order $h_\Gamma^{3/2}$, because every pressure term vanishes on Y_h . Third, Lemma 7.8 is method-independent: it applies to every Nitsche variant with this volume form and homogeneous wall data, tested on Z_h .

Why the obvious route fails

It is instructive to see why the natural first attempt at a lower bound does not work for CNS^* . That attempt tests the error equation with a field whose trace is odd on each chord, and uses that the transposed-gradient part of $\mathcal{G}$ is $\langle s a_e, v \rangle$ to leading order (Lemma 3.9).

1. Since $a_e = n(m_e^*)W_e$, the relevant trace is the *normal* one: $\langle s a_e, v \rangle = \sum_e W_e \int_e s v \cdot n_e + O(h_\Gamma^2) \|v\|_{L^1}$. An odd tangential trace pairs to zero.
2. Incompressibility allows an odd normal trace: for $v = \text{curl } \phi$, $v \cdot n_e = -\partial_\tau \phi$, and a zero chord mean only says that ϕ takes equal values at the chord ends.
3. On the normal trace the penalty part of $\mathcal{G}$ reinforces the transposed part, by the factor $1 + (\mu/h)d(s)$ with $0 \leq (\mu/h)d \leq \gamma h_\Gamma/8$.
4. But a test field with $v \cdot n \neq 0$ lies outside Y_h , and for CNS^* the error equation then contains the pressure term $\langle \xi - \bar{\xi}, v \cdot n \rangle$, which by the proof of Theorem 7.2 is only bounded by $C(c_0\gamma)^{-1/2}\beta^{-1}(\|e_u\| + h_\Gamma^{3/2})\|v\|$ – the same order as the term one is trying to bound from below. It can cancel the lower bound unless γ is large compared with constants we do not control.
5. Even where the route works (for GS with $G = 0$, where the traction term is of lower order), it gives only an *energy* lower bound $\gtrsim h_\Gamma^{3/2}/(1 + \gamma^{1/2})$, which may be carried entirely by the slip part of the norm; numerically, for GS on test A at $N = 64$, $\mu = 100$, the slip part is 0.0795 against $\|\nabla e_u\| = 0.0182$.

The fields of Y_h^1 avoid all of this: they see neither the pressure nor the penalty, and the symmetric Nitsche term $-\langle \tilde{u}, \partial_n v \rangle$, which pairs the bump with $\partial_n v \cdot \tau_e$, has the same order $h_\Gamma^{3/2}$ as the transposed term.

7.4 A growing penalty for the printed method

Since the leak is of size $h/\mu = h_\Gamma/\gamma$, one might hope to rescue the printed method by letting the penalty grow. Theorem 7.3 shows that this works in H^1 .

Corollary 7.16 (Rescue in H^1) *If $\gamma \geq \max(\gamma_0, c h_\Gamma^{-1/2})$, that is $\mu \geq c h h_\Gamma^{-3/2}$ ($\mu \gtrsim h^{-1/2}$ when $h \asymp h_\Gamma$), then GS with flux-corrected data satisfies $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C_p(h_\Gamma^{3/2} + h^k)$, and by Theorem 7.14 the H^1 rate is exactly $\frac{3}{2}$ when $W \not\equiv 0$ and $h^k \leq h_\Gamma^{3/2}$.*

The energy norm cannot be rescued. The penalty consistency term $(\mu/h)\langle \tilde{u}, v \rangle$ of $\mathcal{G}$ pairs the tangential chordal bump with the penalty, and its size $\gamma^{1/2} h_\Gamma^{3/2}$ in the energy norm is genuine.

Theorem 7.17 (The penalty consistency error is real in the energy norm) *Let u_h be the velocity of GS ($\gamma \geq \gamma_0$) or of CNS* ($\gamma \geq \gamma_2$), and $W \neq 0$. There are γ_1 and h_1 , depending on $\|W\|_{L^2(\Gamma)}$ and the data, such that for $\gamma \geq \gamma_1$, $h_\Gamma \leq h_1$ and $h^k \leq h_\Gamma$,*

$$\|\tilde{u} - u_h\| \geq \left(\frac{\mu}{h}\right)^{1/2} \|\tilde{u} - u_h\|_{\Gamma_h} \geq \frac{c_0^2}{72} \|W\|_{L^2(\Gamma)} \gamma^{1/2} h_\Gamma^{3/2} - C (h_\Gamma/\gamma)^{1/2}.$$

Proof Let $e := \tilde{u} - u_h$ and let $v_W := \text{curl} \Pi_{k+1} \varphi_W \in Z_h$ with $\varphi_W = \pm \chi_1(r)(r-1)W(\theta)$, so that $v_W \approx W\tau$ on Γ_h and $\|v_W\|_{\Gamma_h} \leq 2\|W\|_{L^2(\Gamma)}$. Testing the discrete momentum equation with v_W (the pressure term $(p_h, \text{div } v_W)$ vanishes, and for CNS* the wall term is $O(h_\Gamma^{1/2})$ because $v_W \cdot n = O(h_\Gamma)$) shows that the discrete trace is small along v_W : $\frac{\mu}{h} |\langle u_h, v_W \rangle| \leq C(1 + h_\Gamma^{-1/2} \|e\|)$. The exact trace is not: by the chordal bump (1), $\int_e d s = |e|^3/12 + O(|e|^5)$ and a Riemann sum,

$$\langle \tilde{u}, v_W \rangle = \sum_e W_e^2 \frac{|e|^3}{12} + O(h_\Gamma^3) \geq \frac{c_0^2 h_\Gamma^2}{12} (\|W\|_{L^2(\Gamma)}^2 - C h_\Gamma).$$

Now $\|e\| \geq (\mu/h)^{1/2} \langle e, v_W \rangle / \|v_W\|_{\Gamma_h}$, and $(\mu/h)^{1/2} h_\Gamma^2 = \gamma^{1/2} h_\Gamma^{3/2}$, $(h/\mu)^{1/2} h_\Gamma^{-1/2} = \gamma^{-1/2}$. For $\gamma \geq \gamma_1$ the term $\gamma^{-1/2} \|e\|$ produced by the discrete trace is at most $\frac{1}{2} \|e\|$ and can be moved to the left. $\square$

Combined with Theorem 6.1, the leak and the penalty consistency error cannot both be below h_Γ in the energy norm: their product is $(h_\Gamma/\gamma)^{1/2} \gamma^{1/2} h_\Gamma^{3/2} = h_\Gamma^2$. For a power-law penalty $\gamma \asymp h_\Gamma^{-\alpha}$ with $G > 0$, $W \neq 0$, bounded h/h_Γ and $h^k \leq h_\Gamma^{3/2}$, Theorems 6.1, 6.5, 7.3, 7.14 and 7.17 give the following rates for GS [27]:

	H^1 seminorm	energy norm
$0 \leq \alpha < \frac{1}{2}$	$\asymp h_\Gamma^{1+\alpha}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\frac{1}{2} \leq \alpha < 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(1+\alpha)/2}$
$\alpha = 1$	$\asymp h_\Gamma^{3/2}$	$\lesssim h_\Gamma$
$\alpha > 1$	$\asymp h_\Gamma^{3/2}$	$\asymp h_\Gamma^{(3-\alpha)/2}$

So the H^1 rate is $\min(1 + \alpha, \frac{3}{2})$, optimal for every $\alpha \geq \frac{1}{2}$, and the energy rate is $\min(\frac{1+\alpha}{2}, \frac{3-\alpha}{2}) \leq 1$. At $\alpha = 1$ the energy error is $\asymp h_\Gamma$ except possibly in a window of values of γh_Γ where the leak (normal) and the geometric slip (tangential) could partly cancel; we have no matching lower bound there. No scaling of the penalty gives the printed method the energy rate $\frac{3}{2}$ of CNS*. The growing penalty also costs in conditioning: the condition number of the velocity block grows by the factor $1 + \gamma$ and that of the pressure Schur complement by $1 + \mu/h$ [27]. For CNS*, and for GS with $G = 0$, bounded γ is optimal.

8 Further results

The extended version [27] contains a number of further results, which we state here without proof. Unless said otherwise they are proved there under the hypotheses of the main text ((M0)–(M2), smooth solution, $k \geq 4$); where a result depends on an additional hypothesis that is not proved, we say so.

8.1 The leak limit, uniformly in the penalty

Theorem 6.8 identifies the H^1 limit of the scaled error $(\mu/h)(\tilde{u} - u_h)$ of GS for bounded γ . The traction response δ_R (the part of the error caused by the missing traction; on the pure-pressure test it is the whole error) converges to the leak flow uniformly in the penalty, at an exact rate:

$$cG h_\Gamma^{1/2} \leq \left\| \nabla \left(\frac{\mu}{h} \delta_R - \tilde{U} \right) \right\| \leq C(h_\Gamma^{1/2} + h^k) \quad \text{for every } \gamma \geq \gamma_0,$$

the Dirichlet limit $\mu \rightarrow \infty$ included (the lower bound excludes, on each mesh, a finite set of penalties). The leak constant $\|\nabla \delta_R\| \mu / (hG)$ converges to K for every penalty; for general data the normalised GS constant diverges when $\gamma h_\Gamma^{1/2} \rightarrow \infty$. As $\gamma \rightarrow \infty$ the discrete error acquires an element-scale oscillation nearly orthogonal to U , and $K_h^2 = K^2(1 + d_h^2) + O(h_\Gamma^2)$, where d_h is the relative distance above; this is why large-penalty constants approach K from above (Table 12).

8.2 Pressure, stress and drag

For CNS* with flux-corrected data and γ in a bounded range above γ_2 , the pressure converges at Scott's rate, $\|p^\circ - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ with $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$, including the constant. For GS with $G > 0$ the pressure is uniquely determined but carries an explicit first-layer field, produced by the symmetric Nitsche term acting on the leak. The leak enters the pressure equation only through the coupling $\langle u_h, \partial_n v \rangle$, which for test fields vanishing on the wall is a boundary source $\langle u_h \cdot n, \operatorname{div} v \rangle_{\Gamma_h}$; the discontinuous pressure resolves it exactly, as a field supported in the first row of triangles, of pointwise size $(h/\mu)/H_e \asymp 1/\gamma$ (H_e the height of the triangle on the chord e) and L^2 norm $\asymp (h/\mu)h_\Gamma^{-1/2}G$; for fixed γ and bounded ρ its L^2 error converges at the exact rate $\frac{1}{2}$, and not at all pointwise in the first row of triangles. The variational drag, lift and torque coincide with the integral of the Nitsche flux. For GS (fixed γ , bounded ρ) their error is $-(h/\mu) \int_\Gamma (p - \bar{p})(R_\psi - \bar{R}_\psi) + O(h^{3/2})$, with R_ψ the pressure of a dual Stokes problem, so the drag converges at rate 1 with an explicit reciprocity constant; for CNS* with $\gamma \geq \max(1, \gamma_2)$ and $\gamma h_\Gamma \leq 1$ the error is $O(h_\Gamma^2 + \varepsilon_h)$, so the drag converges at rate 2. The torque computed with Gjerde and Scott's volume functional is an exception: its error is of first order in h/μ even for CNS* (the leading term is proved as $\gamma \rightarrow \infty$; at fixed γ only the rate-1 upper bound is proved). Lower bounds $ch_\Gamma^{3/2}$ (in L^2) and ch_Γ (in L^∞) for the CNS* pressure error are proved under a hypothesis on a boundary pressure moment. A periodic first-layer cell problem indicates when that hypothesis holds (exactly when an explicit cell density is not constant along the wall; it fails on mirror-symmetric first layers), but this part rests on a cell-decay property that is only verified numerically and on a transplant lemma that is not proved.

8.3 L^2 estimates

The Stokes $H^2 \times H^1$ regularity needed for duality is proved on the model domain (Kellogg–Osborn [53] on the convex box and smooth-domain regularity at Γ). For CNS* with flux-corrected data,

$$\|\tilde{u} - u_h\|_{L^2} \leq C(h_\Gamma^2 + h h_\Gamma^{3/2} + h^{k+1})$$

uniformly in γ , improving to $C(h_\Gamma^2 + h^{k+1})$ on meshes graded towards the wall. For bounded γ the term h_Γ^2 is sharp when the wall shear does not vanish identically; this lower bound needs no regularity assumption. For GS with $G > 0$ the L^2 rate is exactly 1, with leading term $(h/\mu)\|U\|_{L^2}$ given by the leak flow.

8.4 Clough–Tocher refinements

On the Clough–Tocher (barycentric) refinement of any shape-regular mesh with comparable chords, the convergence results for GS (Theorems 6.1–6.5, Corollary 6.6) and for CNS* (Theorem 7.2), and the sufficiency part of the penalty analysis, hold for every $k \geq 2$, with no angle or singular-vertex condition. For example CNS* satisfies $\|\tilde{u} - u_h\|_{H^1} \leq C(h_\Gamma^{3/2} + h^2)$ for $k = 2$ if $h_\Gamma \geq h^4$. The penalty-necessity certificates are redone exactly on split stars. The leak constant, the lower bound of Section 7.3, and the results below have not been checked on split meshes, and we make no claim about them there.

8.5 General curved domains

The convergence theory (Theorems 6.1–7.2 and Corollary 6.6) extends to bounded connected planar domains whose curved boundary consists of finitely many closed smooth curves, convex or not, with a polygonal outer component on which the data are imposed strongly. The geometric lemmas need only C^3 curves; the results assume the regularity of the solution, which for generic f requires $\Gamma \in C^{k+1,\alpha}$ and vanishing corner coefficients at the vertices of the outer polygon. With several curved components the correct pressure constant is the *global* boundary mean: CNS* needs one global correction, and a per-component correction is inconsistent. Stokes H^2 regularity with L^2 data holds on such a domain if and only if the outer polygon is convex.

8.6 Steady Navier–Stokes flow

Gjerde and Scott’s application is cylinder drag, a Navier–Stokes computation. On a nonsingular solution branch the discrete linearisation is uniformly stable for $h \leq h_0$, with h_0 independent of the penalty (explicit, but far too small to be useful). Then GS converges at H^1 rate h/μ with the same leak law (slip and energy constants 1) and an H^1 constant given by an Oseen leak flow, and CNS* attains $h_\Gamma^{3/2} + h_\Omega^k$; the CNS* drag, lift and torque converge at rate 2. Both natural discrete convective forms give the same conclusions, and no inflow/outflow correction is needed.

8.7 Pressure robustness

A Stokes discretisation is pressure-robust (gradient-robust) if a change of the force by a gradient changes only the discrete pressure; equivalently, its velocity error does not depend on the pressure and does not grow like ν^{-1} as the viscosity $\nu \rightarrow 0$ [6]. Exact incompressibility does not by itself give this when the boundary condition is imposed weakly on an approximate boundary. The velocity error splits exactly as $E_M - \nu^{-1}W_M(\tilde{p})$, where E_M is a pressure-free error (up to a strip term of order h_Γ^4) and $W_M(\phi)$ is the discrete velocity produced by the purely hydrostatic data $f = \nabla\phi$;

a method is gradient-robust if and only if $W_M \equiv 0$. GS is not pressure-robust: the pressure part of its velocity error at viscosity ν is the leak flow times $h/(\nu\mu)$. CNS* has exactly one pressure-dependent defect, the projection error $p - \pi_h p$ on Γ_h , whose H^1 effect is of order $\nu^{-1}\gamma^{-1}h_\Gamma^{k+1/2}$ for generic pressures on locally periodic meshes. A local BDM $_k$ correction of the load on the boundary triangles makes CNS* exactly gradient-robust, without changing its matrix or its rates. The load-reconstruction idea is that of [6, 17, 31]; we claim no novelty for the device.

8.8 Three dimensions

The three-dimensional analysis (inscribed polyhedra, Alfeld splits) is the subject of a companion paper [28]. One ingredient of the present paper is not available there: the odd-symmetry cancellation on each chord behind Lemma 3.9. In two dimensions it is needed only for bounds in the plain H^1 dual norm; the upper bounds of Theorems 6.1 and 7.2 also follow from the cruder energy-dual estimate $|\langle (\nabla \tilde{u})^\top n, v \rangle| \leq C\gamma^{-1/2}h_\Gamma^{3/2}\|v\|$, which uses only the penalty.

8.9 Status of the results

Table 6 records the status of the main claims of this paper and of [27]. “Proved” means proved from the stated mesh hypotheses, the smooth-solution hypothesis and standard cited results; “conditional” means proved except for a named hypothesis that we have not proved; “numerical” means observed only.

9 Numerical experiments

The computations have three purposes: to confirm the rates and constants of the theorems, to separate the leak from the geometric error, and to carry out the study of the penalty that Scott’s question asks for. We report a selection; the full study, with all tables, is in [27], and all code, raw outputs and logs are in the public repository [29].

9.1 Setup

Except where stated we use P_4 Scott–Vogelius elements (all ten P_3 divergence moments per triangle), Nitsche’s method on the inscribed polygon, and strong Dirichlet data on the square $R = (-2.5, 2.5)^2$. The vertices of Γ_h are the radial projections of N equispaced points on the square, so the chords are not all equal ($\max |e|/\min |e|$ grows from 1.43 at $N = 16$ to 1.97 at $N = 256$). The ratio h/h_Γ stays in 3.28–3.48, so rates in h and h_Γ coincide; h decreases from 1.505 ($N = 16$) to 0.109 ($N = 256$). Rates in the tables are computed between consecutive runs of the full sequence $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$, some of which are omitted from the tables. Every polygon vertex lies in three triangles (mesh (s)), except on the alternating mesh (a) of Section 9.7. Saddle-point systems are solved by SuperLU [64] with iterative refinement; over the 148 production solves the relative residual is at most $3.8 \cdot 10^{-15}$ and the discrete divergence at most $7 \cdot 10^{-11}$. As a cross-check, for the constant-pressure benchmark of [30] our code gives the coercivity threshold $\mu^* = 60.343$ at $N = 20$, against the transition $60.33821 < \mu^* < 60.33831$ reported there.

Table 6 Status of the results.

claim	status
Leak analysis of GS: energy, H^1 and slip rates, leading term with constants 1, H^1 constant K (Section 6)	proved
Penalty-uniform leak limit at the exact rate $h_\Gamma^{1/2}$	proved
CNS*: well-posedness and $h_\Gamma^{3/2} + h_\Omega^k$, uniform in γ with flux-corrected data (Section 7)	proved
Sharpness of $h_\Gamma^{3/2}$ for every $k \geq 4$, for $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$ (star condition with certified constants)	proved (computer-assisted certificates)
Growing penalty: H^1 rescue, energy rate ≤ 1	proved
Penalty: sufficiency; single-triangle necessity; certificates on shape families; exact counterexamples to local necessity; necessity on every shape-regular mesh	proved (exact arithmetic for certificates)
Limit $\gamma_\infty^* \approx 21.41$ of the threshold on our meshes	lim inf proved; equality conditional (localisation and spline cut-off lemmas); value numerical
Strong imposition: two-sided $h_\Gamma^{3/2}$ under (M2); counting bound; constrained and sharp inf-sup; two-sided rate with locks; pressure filter (Section 4)	proved
Needle-angle law: area-weighted distance; two-sided velocity rate	proved; the upper bound under a hypothesis isolating each wall needle pair
CNS* pressure $h_\Gamma^{3/2} + h_\Omega^k$; GS pressure rate $\frac{1}{2}$; drag, lift and torque expansions	proved
CNS* pressure lower bounds	conditional (boundary moment hypothesis); the cell-problem analysis of that hypothesis rests on a numerically verified decay property and an unproved transplant lemma
L^2 estimates, regularity, graded meshes	proved
Clough–Tocher refinements ($k \geq 2$), general curved domains, steady Navier–Stokes on nonsingular branches	proved (scope as stated in this section)
Pressure robustness of CNS* and its BDM $_k$ correction	proved; some constants explicit only for $k = 4$
H^1 constants close to K for $k = 5, 6$ and Clough–Tocher elements; stability of CNS* at $\mu = 100$	numerical

The test problems are:

- A** Scott’s benchmark, the rotating shear flow $u = (1 - r^{-2})(-y, x)$ with $p \equiv 0$, so $G = 0$;
- B_a** $\psi = (r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$ (1.658 for $a = 1$, 165.8 for $a = 100$);

Table 7 CNS* attains Scott’s rate $h_T^{3/2}$: H^1 error, $k = 4$, $\mu = 100$.

N	h	A	rate	B_1	rate
32	0.813	$6.41 \cdot 10^{-2}$	1.80	$2.34 \cdot 10^{-2}$	2.43
64	0.422	$2.08 \cdot 10^{-2}$	1.69	$6.83 \cdot 10^{-3}$	1.76
96	0.285	$1.09 \cdot 10^{-2}$	1.64	$3.54 \cdot 10^{-3}$	1.68
128	0.215	$6.96 \cdot 10^{-3}$	1.60	$2.24 \cdot 10^{-3}$	1.63
192	0.144	$3.71 \cdot 10^{-3}$	1.58	$1.19 \cdot 10^{-3}$	1.60
256	0.109	$2.39 \cdot 10^{-3}$	1.56	$7.58 \cdot 10^{-4}$	1.57

P pure pressure: $u \equiv 0$, $f = \nabla p$ with p as in B_1 . CNS* reproduces this exactly, so the GS output is *exactly* the response to the missing traction;

C ψ as in B with the non-polynomial pressure $p = e^{x/2} \cos y$.

The manufactured solutions are smooth on the whole box, so the corner caveat of Assumption 3.5 does not arise.

The production penalty. Most runs use $\mu = 100$, i.e. $\gamma \approx 29$ –30. This is only 1.37–1.41 times the measured coercivity threshold of GS on Z_h . For CNS* the relevant quantity is the largest μ at which the square system is singular: $\mu_c = 78.9, 84.4, 88.1$ at $N = 16, 32, 64$, extrapolating to about 93.

N	μ^* (GS coercivity)	μ_c (CNS* singular)	margin of $\mu = 100$ for CNS*
16	59.3	78.9	27%
32	66.0	84.4	19%
64	70.7	88.1	13%
∞ (extrapolated)	≈ 73	≈ 93	$\approx 7\%$

So $\mu = 100$ is stable for CNS* on every production mesh, but the margin shrinks from 27% ($N = 16$) to 13% ($N = 64$), and to about 7% in the limit. The thresholds γ_0, γ_2 of the proofs are not computed, so these runs illustrate the theorems rather than being covered by them. Near the threshold the discrete solution is amplified by a factor of order $(1 - \mu^*/\mu)^{-1}$, which drifts from about 2.5 to 3.4 over $N = 16 \rightarrow 64$ and can lower measured rates by up to about 0.2 per doubling at $N \leq 64$; normalised constants at $\mu = 100$ are partly pre-asymptotic, and we quote larger penalties alongside where we have them.

9.2 Scott’s rate for the corrected method

Table 7 shows the H^1 errors of CNS*. The rates decrease monotonically towards $\frac{3}{2}$, the geometric term, for both tests; at $N = 256$ the energy rate is 1.52 and the slip rate 2.02 (the slip carries an extra factor $(h/\mu)^{1/2}$). The term h^4 is invisible at these levels. The error does not depend on the pressure: for polynomial pressures this is structural ($p \in \Pi_h$), and B_1 and B_{100} agree to all printed digits; with the non-polynomial pressure of test C the H^1 error differs from that of B_1 by a relative $7.9 \cdot 10^{-7}$, $3.9 \cdot 10^{-8}$ and $9.7 \cdot 10^{-9}$ at $N = 16, 64$ and 96, the size of the higher-order projection term. For $k = 5$ and 6 the CNS* rates on B_1 are 1.58–1.61 at the finest levels, non-increasing, with nearly equal errors for $k = 4, 5, 6$: the geometric term dominates.

Table 8 Test B₁₀₀, $k = 4$, $\mu = 100$: H^1 velocity error and L^2 pressure error $\|p^\circ - p_h\|$, $p^\circ = \tilde{p} - \bar{p}_\Gamma(\tilde{p})$.

N	GS, $ \cdot _1$		CNS*, $ \cdot _1$		GS, pressure		CNS*, pressure	
	error	rate	error	rate	error	rate	error	rate
16	6.14	–	$1.29 \cdot 10^{-1}$	–	27.5	–	$4.97 \cdot 10^{-1}$	–
32	3.51	0.93	$2.34 \cdot 10^{-2}$	2.43	18.2	0.68	$7.67 \cdot 10^{-2}$	3.03
64	1.88	0.97	$6.83 \cdot 10^{-3}$	1.76	11.8	0.65	$1.62 \cdot 10^{-2}$	2.08
96	1.28	0.98	$3.54 \cdot 10^{-3}$	1.68	9.23	0.62	$8.01 \cdot 10^{-3}$	1.80
128	$9.71 \cdot 10^{-1}$	0.98	$2.24 \cdot 10^{-3}$	1.63	–	–	–	–
192	$6.54 \cdot 10^{-1}$	0.99	$1.18 \cdot 10^{-3}$	1.60	–	–	–	–

Table 9 H^1 rate between $N = 128$ and $N = 192$ on test B₁₀₀.

μ	GS	CNS*
100	0.99	1.60
1000	1.00	1.56
10000	1.06	1.58

9.3 The printed method loses the rate once the wall pressure varies

Table 8 compares the two methods on B₁₀₀, where the wall pressure varies strongly. GS has H^1 rate 1 from the start, with errors three orders of magnitude above those of CNS*; its pressure error converges at rate about $\frac{1}{2}-\frac{2}{3}$, while that of CNS* decreases towards $\frac{3}{2}$. The crossover from rate $\frac{3}{2}$ to rate 1 happens for every μ (Table 9): the term $(h/\mu)G$ dominates once $(h/\mu)G \gtrsim h_\Gamma^{3/2}$, which always happens eventually because $h_\Gamma^{3/2}$ decays faster. Larger μ or smaller G delays the crossover but does not remove it: for B₁ at $\mu = 1000$, GS still shows rate 1.51 at $N = 192$. Figure 2 (left) shows the two behaviours.

9.4 Pressure errors

The pressure errors follow Section 8 (Tables 8 and 10). For CNS* the pressure error does not depend on the pressure data (for B₁, B₁₀₀ and C it agrees to 10^{-6} at $N = 16$ and $3 \cdot 10^{-11}$ at $N = 96$), and its L^2 rate decreases towards $\frac{3}{2}$. For GS with $G > 0$ the L^2 rate decreases towards $\frac{1}{2}$; the first layer of triangles carries rate 0.48–0.50 and the rest rate 0.9–1.0, and the L^∞ error on the first layer does not converge (0.78 for B₁, 77 for B₁₀₀). On test P the first-layer pressure error of GS is 0.0958 and 0.0793 at $N = 64$ and 96, against 0.0961 and 0.0796 for the explicit first-layer field predicted by the theory. With $G = 0$ (test A), GS behaves like CNS*.

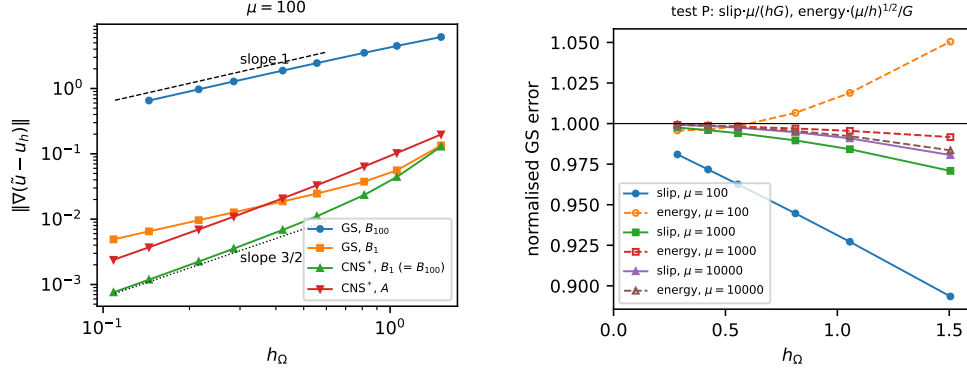


Fig. 2 Left: H^1 errors at $\mu = 100$. Once $G > 0$, GS eventually has slope 1, while CNS* tends to slope $\frac{3}{2}$ independently of the pressure. Right: the normalised slip and energy errors of GS on test P approach 1 (Corollary 6.6).

Table 10 L^2 pressure error $\|p^\circ - p_h\|$, $\mu = 100$. The “off layer” column is the GS error outside the first row of triangles, with the optimal constant removed.

N	GS	rate	test B ₁ ($G > 0$)				test A ($G = 0$)			
			off layer	rate	CNS*	rate	GS	rate	CNS*	rate
16	0.530	—	0.500	—	0.497	—	0.205	—	0.378	—
32	0.180	1.22	0.121	1.84	0.0767	3.03	0.0648	1.85	0.121	1.81
48	0.137	0.72	0.0805	1.08	0.0287	2.59	0.0329	1.79	0.0623	1.75
64	0.115	0.61	0.0632	0.88	0.0162	2.08	0.0204	1.73	0.0392	1.69
96	0.0913	0.60	0.0445	0.89	0.00801	1.80	0.0106	1.68	0.0205	1.65

9.5 The leak law and its constants

Test P isolates the traction response exactly. Table 11 and Figure 2 (right) show that the normalised slip and energy errors approach 1, as Corollary 6.6 predicts: at $N = 96$ the energy ratios are within 0.5% of 1 for every μ , and the defect of the slip ratio shrinks with h/μ (0.019, 0.0025, 0.0007 for $\mu = 10^2, 10^3, 10^4$). Figure 3 shows the leak itself: at $N = 32$, $\mu = 100$, the normal trace of u_h matches $(h/\mu)(p - \bar{p})$ with relative $L^2(\Gamma_h)$ misfit 0.055, and the tangential trace is small and oscillates on the element scale. On test B₁ at $N = 256$, GS itself gives slip and energy ratios 0.995 and 1.001.

The H^1 constant approaches the value $K = 2.7565$ of Section 6.5 from both sides (Table 12): from below at $\mu = 100$, and from above at large μ , where the excess is an element-scale oscillation nearly orthogonal to U ($K_h^2 \approx K^2(1 + d_h^2)$; for example $2.7565\sqrt{1 + 0.175^2} = 2.798$ against 2.796 at $N = 96$, $\mu = 10^8$). The same slip and energy constants, and H^1 constants close to K , are observed for $k = 5, 6$ and on Clough–Tocher refinements with $k = 2, 3$ (Table 13); for these elements the agreement with K is observed, not proved.

Incompressibility is what saves the rate. We computed exact discrete dual norms of the functionals used in the proofs, by saddle-point solves with the pointwise divergence

Table 11 The leading term of Corollary 6.6: normalised GS error on test P ($G = 1.658$), $e = \tilde{u} - u_h$.

N	μ	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
16	100	2.464	0.893	1.050
32	100	2.612	0.945	1.007
64	100	2.684	0.972	0.996
96	100	2.708	0.981	0.996
96	1000	2.770	0.998	0.999
96	10000	2.790	0.9993	0.9994

Table 12 Approach to the H^1 constant $K = 2.7565157$ (Theorem 6.8) on test P: normalised H^1 error K_h and relative H^1 distance d_h between $(\mu/h)(\tilde{u} - u_h)$ and the leak flow, computed pointwise against the series solution.

N	$\mu = 10^2$		$\mu = 10^3$		$\mu = 10^4$		$\mu = 10^8$	
	K_h	d_h	K_h	d_h	K_h	d_h	K_h	d_h
16	2.4642	0.193	2.7844	0.318	3.0562	0.539	3.3254	0.717
32	2.6117	0.124	2.7880	0.222	2.8751	0.323	2.9339	0.385
64	2.6843	0.084	2.7759	0.154	2.8082	0.206	2.8208	0.226
96	2.7084	0.067	2.7701	0.124	2.7899	0.163	2.7959	0.175

Table 13 Normalised GS error on test P for other elements, at the finest level run.

element	μ	N	$\ \nabla e\ \mu / (hG)$	$\ e\ _{\Gamma_h} \mu / (hG)$	$\ e\ (\mu/h)^{1/2} / G$
$k = 5$	1000	128	2.768	0.9983	0.9994
$k = 6$	1000	96	2.769	0.9976	0.9993
Clough–Tocher, $k = 2$	1000	128	2.751	0.9982	0.9992
Clough–Tocher, $k = 3$	1000	128	2.763	0.9983	0.9996

constraint (Table 14). The dual norm of $v \mapsto \langle S, \partial_n v \rangle$ with S normal stays bounded on Z_h , but grows like $h_\Gamma^{-1/2}$ on V_h^R , and also on Z_h for tangential S : exactly the dichotomy of Lemma 6.2. The transposed-gradient functional decays at rate 1.56–1.57, confirming that the rate $\frac{3}{2}$ of Lemma 3.9 is sharp.

9.6 The penalty study

The coercivity threshold μ^* on Z_h is located by bisection on the inertia of the (unpivoted symmetric factorisation of the) velocity block augmented by a large multiple of the divergence penalty. Table 15 and Figure 4 give the results. Under refinement at fixed h/h_Γ , γ^* increases by 3.4, 1.8 and 0.95 per doubling, approaching the periodic-cell value $\gamma_\infty^* \approx 21.4$ of Section 5. When the far field is coarsened with the wall fixed (part (b)), μ^* is linear in h/h_Γ over more than a decade with slope 18–20: the threshold is set locally at the wall, and the global normalisation of the penalty turns it into a ρ -scaling. For higher degree and on Clough–Tocher refinements the threshold is larger

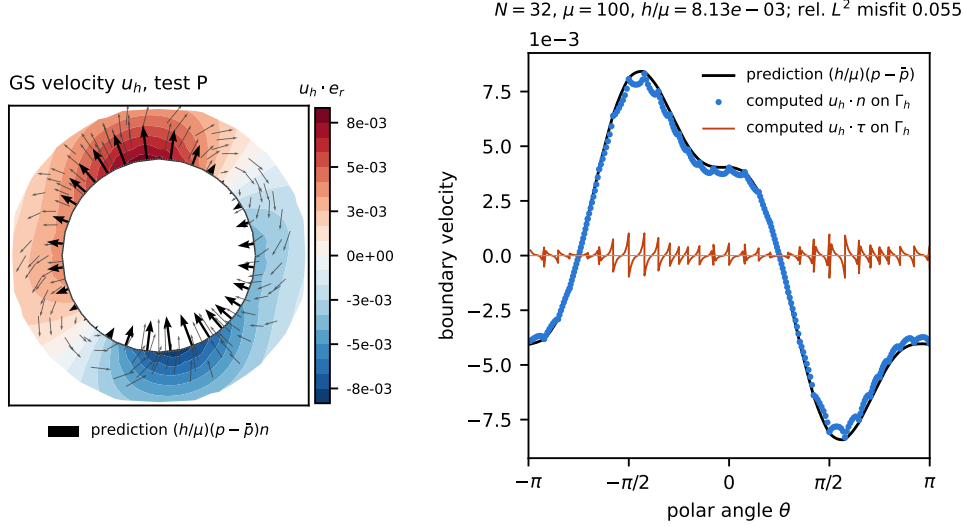


Fig. 3 The leak of the printed method on test P ($u = 0$, so $u_h = -(\tilde{u} - u_h)$), $N = 32, \mu = 100$. Left: radial velocity $u_h \cdot e_r$ near the wall (colour) and u_h (thin arrows); thick arrows show the prediction $(h/\mu)(p - \tilde{p})n$ at the chord midpoints. Right: normal and tangential traces of u_h along Γ_h against $(h/\mu)(p - \tilde{p})$.

Table 14 Exact discrete dual norms; rate in h_Γ between $N = 48$ and $N = 64$ (positive means decay).

functional	space	rate
$\langle S, \partial_n v \rangle, S$ normal	Z_h	+0.05 (bounded, ≈ 1.75)
$\langle S, \partial_n v \rangle, S$ normal	V_h^R	-0.52
$\langle S, \partial_n v \rangle, S$ tangential	Z_h	-0.52
$\langle (\nabla \tilde{u})^\top n, v \rangle$, tests A / B	V_h^R	1.57 / 1.56

(Table 16); in particular $\mu = 100$ is *below* the threshold for $k = 5$ ($N \geq 32$), $k = 6$ and Clough–Tocher $k = 3$, and our runs for those elements use $\mu = 300$ and 1000.

A growing penalty. With $\mu = c(h_{16}/h)^\alpha$, $h_{16} = 1.505$, the rates follow the predictions of Section 7.4 (Table 17). On test P, where only the leak is present, they are exactly $1 + \alpha$ (H^1) and $(1 + \alpha)/2$ (energy). On B_1 the H^1 rate of GS moves from 0.96 to 1.42–1.52 and the error at $N = 96$ halves; on B_1 with $\alpha = \frac{1}{2}$, $c = 1000$ the measured energy rate 1.28 is the geometric branch $(3 - \alpha)/2$, the leak branch taking over only later.

On test A ($G = 0$) a growing penalty only costs: the H^1 constant moves towards that of the Dirichlet limit, 2.5 times larger, and the energy error grows like $\gamma^{1/2} h_\Gamma^{3/2}$.

Flux correction. Test F uses $\psi = (r^2 - 1)^3 \sin(2x + 1) \cos(\frac{3}{2}y + \frac{7}{10})/400$ and $p = x^2y - y + x/2$: the cube makes $W \equiv 0$, so the geometric slip $\|\tilde{u}\|_{\Gamma_h}$ is small, and the phase shifts break the symmetries, so the outer flux m_R is nonzero (it decays like $h^{6.6}$).

(a) Refinement at fixed $h/h_\Gamma \approx 3.3$.

Table 15 Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma/h$ ($k = 4$).

N	8	12	16	20	24	32	48	64
ρ	3.41	4.16	4.70	5.08	5.36	5.73	6.13	6.35
μ^*	50.3	51.8	59.3	60.3	63.3	66.0	69.1	70.7
γ^*	14.7	17.1	18.1	19.1	19.2	19.9	20.5	20.8

(b) Varying h/h_Γ through the outer box $(-L, L)^2$.

	L	2.5	5	10	20	40
$N = 16$	h/h_Γ	3.28	5.90	11.3	22.2	43.9
	μ^*	59.3	112.1	208.0	411.5	809.3
$N = 32$	h/h_Γ	3.33	5.79	10.8	21.1	41.5
	μ^*	66.0	113.9	218.0	422.2	835.7

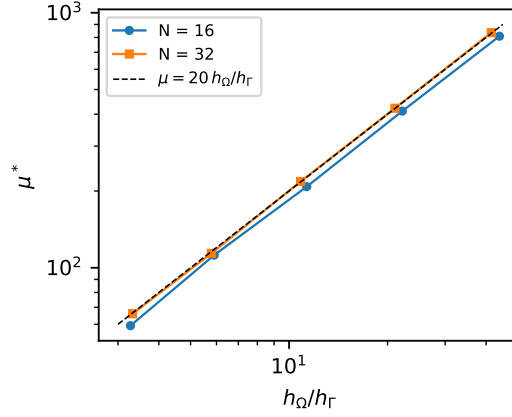


Fig. 4 Coercivity threshold μ^* against h/h_Γ (Table 15(b)).

Every $z \in W_h$ satisfies $\|\tilde{u} - z\| \geq (\mu/h)^{1/2} |m_R|/|\Gamma_h|^{1/2}$, a floor with constant exactly 1. Table 18 shows the floor and its removal by the correction of Definition 3.2. Without correction the slip error of CNS^* is $(|m_R|^2/|\Gamma_h| + \|\tilde{u}\|_{\Gamma_h}^2)^{1/2}$ to three digits; with it, the slip error is $\|\tilde{u}\|_{\Gamma_h}$ to five digits. The H^1 errors with and without correction agree to relative $3 \cdot 10^{-6}$ ($N = 16$) down to 10^{-9} ($N = 64$), as Theorem 7.3 predicts.

Higher degree and Clough–Tocher refinements. Table 19 gives H^1 rates at the finest pair of levels for $k = 5, 6$ and for Clough–Tocher refinements. CNS^* tends to rate $\frac{3}{2}$ for $k = 5, 6$; GS tends to rate 1 once $G > 0$ (for $k = 5$, $\mu = 300$, its rate on B_1 decreases monotonically, 1.63, 1.58, 1.48, 1.38, 1.28, 1.20 for $N = 24, \dots, 128$), and the leak part $D := u_{\text{GS}} - u_{\text{CNS}^*}$ already has rate 1.03–1.15 and normalised slip and energy ratios within 1.5% of 1. At $\mu = 1000$ the crossover of GS comes later, exactly as for $k = 4$.

Table 16 Coercivity threshold γ^* for higher degree and on Clough–Tocher (CT) refinements.

	$N = 8$	16	32	64	128
$k = 4$	14.7	18.1	19.9	20.8	–
$k = 5$	25.7	30.3	32.1	33.0	–
$k = 6$	–	45.2	47.4	48.3	–
CT, $k = 2$	5.7	9.1	10.5	11.2	11.5
CT, $k = 3$	–	31.6	33.9	34.6	–

Table 17 GS (and CNS*) with $\mu = c(h_{16}/h)^\alpha$: rates between $N = 64$ and $N = 96$, and the asymptotic rates predicted in Section 7.4 (on test P, where only the leak is present, $1 + \alpha$ and $(1 + \alpha)/2$).

test	method	(α, c)	$\ \nabla e\ $ at $N = 96$	H^1 rate	energy rate	predicted
P	GS	(0, 100)	$1.28 \cdot 10^{-2}$	0.95	0.49	1, 0.5
P	GS	$(\frac{1}{2}, 100)$	$5.64 \cdot 10^{-3}$	1.45	0.73	1.5, 0.75
P	GS	(1, 100)	$2.47 \cdot 10^{-3}$	1.94	0.97	2, 1
B ₁	GS	(0, 100)	$1.27 \cdot 10^{-2}$	0.96	0.56	1, 0.5
B ₁	GS	$(\frac{1}{2}, 1000)$	$5.81 \cdot 10^{-3}$	1.52	1.28	1.5, 0.75
B ₁	GS	(1, 100)	$5.03 \cdot 10^{-3}$	1.47	1.06	1.5, 1
B ₁	CNS*	(0, 100)	$3.54 \cdot 10^{-3}$	1.64	1.53	1.5, 1.5
B ₁	CNS*	(1, 100)	$5.21 \cdot 10^{-3}$	1.45	1.16	1.5, 1
A	GS	(1, 100)	$1.48 \cdot 10^{-2}$	1.34	1.16	1.5, 1

Table 18 Test F, CNS* with uncorrected (g_I) and flux-corrected ($\tilde{g}_I$) outer data. Slip errors at $\mu = 10^6$; energy errors at $\mu = 10^8$, with the floor $(\mu/h)^{1/2}|m_R|/|\Gamma_h|^{1/2}$.

N	m_R	$\frac{ m_R }{ \Gamma_h ^{1/2}}$	$\ \tilde{u}\ _{\Gamma_h}$	slip, g_I	slip, $\tilde{g}_I$	$\ e\ $, g_I	$\ e\ $, $\tilde{g}_I$	floor
16	$2.04 \cdot 10^{-3}$	$8.19 \cdot 10^{-4}$	$2.40 \cdot 10^{-5}$	$8.204 \cdot 10^{-4}$	$2.396 \cdot 10^{-5}$	6.752	0.955	6.672
32	$2.56 \cdot 10^{-5}$	$1.02 \cdot 10^{-5}$	$1.70 \cdot 10^{-6}$	$1.038 \cdot 10^{-5}$	$1.697 \cdot 10^{-6}$	0.187	0.149	0.114
64	$3.75 \cdot 10^{-7}$	$1.50 \cdot 10^{-7}$	$1.09 \cdot 10^{-7}$	$1.851 \cdot 10^{-7}$	$1.091 \cdot 10^{-7}$	0.0123	0.0121	0.0023

9.7 Strong imposition and the $h^{1/2}$ phenomenon

Table 20 compares strongly imposed no-slip ($u_h = 0$ at every node of Γ_h , no Nitsche terms, same element) with GS and CNS* on test A.

- On mesh (a) the first-layer diagonals alternate, so every other polygon vertex lies in only two triangles. The H^1 rate of strong imposition falls towards $\frac{1}{2}$ (0.56 at $N = 128$); the largest vertex-gradient error tends to 1.99, which is $|\partial_n u \cdot \tau| = 2$ for this flow: the discrete gradient vanishes at the locked vertices, and the pressure grows like $h^{-1/2}$ although $p \equiv 0$. The counting lower bound of Theorem 4.5 is within a factor 3.0–3.15 of the measured error.
- On mesh (s) every polygon vertex lies in three triangles, and strong imposition converges with rates $1.93 \rightarrow 1.61$, like the Nitsche methods, with an error about

Table 19 H^1 rates at the finest pair of levels (energy rate of CNS* in parentheses); $D = u_{\text{GS}} - u_{\text{CNS}^*}$ is the leak part.

element	test	μ	N	GS	CNS*	D
$k = 5$	B ₁	300	96 $\rightarrow$ 128	1.20	1.59 (1.53)	1.03
$k = 5$	B ₁	1000	96 $\rightarrow$ 128	1.53	1.58 (1.54)	1.13
$k = 6$	B ₁	300	64 $\rightarrow$ 96	1.27	1.61 (1.54)	1.04
$k = 6$	B ₁	1000	64 $\rightarrow$ 96	1.57	1.61 (1.54)	1.15
CT, $k = 3$	A	1000	96 $\rightarrow$ 128	1.56	1.56 (1.55)	–
CT, $k = 2$	B ₁	100	192 $\rightarrow$ 256	1.91	1.91 (1.91)	1.01

Table 20 Strong imposition against Nitsche: H^1 error, test A, $k = 4$, $\mu = 100$. Mesh (s): every polygon vertex in three triangles; mesh (a): alternating first-layer diagonals (vertices in two or four triangles).

N	strong, (a)		strong, (s)		GS, (s)		CNS*, (s)	
	error	rate	error	rate	error	rate	error	rate
16	$9.03 \cdot 10^{-1}$	–	$4.62 \cdot 10^{-1}$	–	$1.72 \cdot 10^{-1}$	–	$1.98 \cdot 10^{-1}$	–
32	$5.72 \cdot 10^{-1}$	0.72	$1.43 \cdot 10^{-1}$	1.86	$5.55 \cdot 10^{-2}$	1.79	$6.41 \cdot 10^{-2}$	1.80
64	$3.75 \cdot 10^{-1}$	0.62	$4.56 \cdot 10^{-2}$	1.71	$1.81 \cdot 10^{-2}$	1.68	$2.08 \cdot 10^{-2}$	1.69
96	$2.98 \cdot 10^{-1}$	0.59	$2.38 \cdot 10^{-2}$	1.65	$9.57 \cdot 10^{-3}$	1.63	$1.09 \cdot 10^{-2}$	1.64
128	$2.54 \cdot 10^{-1}$	0.56	$1.52 \cdot 10^{-2}$	1.61	$6.11 \cdot 10^{-3}$	1.59	$6.96 \cdot 10^{-3}$	1.60

2.5 times larger. The slow drift of the rate (to 1.547 at $N = 192$) is a relative $O(h_\Gamma)$ correction: the data fit $E \approx ah_\Gamma^{3/2}(1 + \beta h_\Gamma)$ with $\beta \approx 1.1$ –1.3. The vertex-gradient error decays like h , exactly as Lemma 4.1(d) predicts. Imposing $u_h = \tilde{u}$ instead of 0 at the nodes of Γ_h gives rate about 4: the whole defect is the transfer of the no-slip value from Γ to Γ_h .

- GS and CNS* do not see the change of mesh: at every $N \leq 128$ their errors on meshes (a) and (s) differ by at most 6%.
- With a single locked vertex the strong-imposition rate falls to 1 (1.06 at $N = 128$), the raw pressure error does not converge, and the filtered pressure of Theorem 4.7(d) converges at the velocity rate. With needles at the wall of angle $\varepsilon = 0.3$, h_Γ, h_Γ^2 the rates approach $\frac{3}{2}$, 1, $\frac{1}{2}$ as predicted, although the last two limits are not yet reached at $N \leq 96$.

9.8 Caveats

Three limitations should be kept in mind. First, the production penalty $\mu = 100$ is close to the stability thresholds, as discussed above, so the measured constants are partly pre-asymptotic. Second, the manufactured solutions are smooth up to the corners of the box; for data-driven problems the corner singularities of the box would limit the bulk term (Assumption 3.5), though not the wall term that is the subject of this paper. Third, on Clough–Tocher refinements with $k = 2, 3$ the test pressure is not in the pressure space, and the pressure-independence of CNS* holds only up to a

term of order $h^{k+1/2}$ that scales with the amplitude of p ; on test B₁ the rate $\frac{3}{2}$ is not yet visible there, because the interpolation error of the steep velocity dominates [27].

10 Conclusions and open problems

Scott asked whether the Scott–Vogelius–Nitsche method on an inscribed polygon converges with error $h_\Gamma^{3/2} + h_\Omega^k$ and, if not, why not. The answer has two halves. The first, announced in the joint letter [25] and proved in full here, concerns the method as printed: it converges, but with the expected error only when the pressure is constant along the wall, because without the pressure traction the penalty must balance the wall pressure and the discrete flow leaks through the wall at the rate $(h/\mu)(p - \bar{p})$; restoring the traction with its wall mean removed, a device due to Frachon, Nilsson and Zahedi, gives a well-posed, exactly divergence-free method that attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data, and the term $h_\Gamma^{3/2}$ is sharp. The second half, which is the main new content of this paper, concerns the difficulty that motivated weak imposition in the first place. The zero-gradient lock is a property of the wall mesh: under the Guzmán–Scott angle condition strong imposition has the same two-sided rate $h_\Gamma^{3/2} + h^k$, and the lock comes from wall vertices in only two triangles, each costing $h_\Gamma |\partial_n u(z)|$, or from nearly singular wall stars. With Scott’s normalisation the penalty must grow with the ratio of bulk to boundary mesh size, on every shape-regular mesh, but necessity cannot be read off a single wall star. Finally, the leak is explicit, with constant 1 in the natural norms, and its H^1 size is the energy of a Stokes flow in the whole domain; incompressibility is what keeps the H^1 rate at 1 rather than $\frac{1}{2}$.

For practitioners the conclusions are simple. If exactly divergence-free velocities are to be combined with Nitsche’s method on a polygonal approximation of a curved wall, the pressure traction must be included, with its wall mean removed, and the outer Dirichlet data should be corrected to zero net flux. The penalty should be scaled with the local chord length ($\gamma = \mu h_\Gamma/h \geq 25$ is safe for $k = 4$ on our meshes), and a mesh in which every wall vertex lies in at least three well-shaped triangles should be used, whatever the imposition. The attainable rate is $h_\Gamma^{3/2}$; higher rates require curved elements or boundary corrections [17, 18, 31, 32].

Several questions remain open.

- *Constants in the penalty analysis.* Necessity of $\mu \gtrsim \rho$ is proved on every shape-regular mesh, but with astronomically small constants; numerically $\mu^* \approx (15\text{--}16) h/\delta_1$, with δ_1 the height of the first layer of triangles. An explicit, moderate constant for general meshes is open. The limit $\gamma_\infty^* \approx 21.41$ of the threshold on our meshes is identified with a periodic cell value only modulo a discrete localisation inequality and a spline cut-off lemma that we have not proved.
- *The pressure of the corrected method.* Its lower bound needs a hypothesis on a boundary pressure moment; the cell problem that decides this hypothesis rests on a decay property verified only numerically and on a transplant lemma that is not proved. The sharpness of the rate 2 for the drag of CNS* at fixed penalty is open.
- *Other variants.* We have analysed the symmetric Nitsche form. The leak comes from the missing pressure term, not from the symmetry of the viscous terms, so we expect it in nonsymmetric variants as well, but we have not analysed these,

nor penalty-free variants [36, 37], the slip-condition method of Gjerde and Scott [65], other divergence-free families, or cut meshes.

- *Strong imposition on general anisotropic meshes.* Our two-sided needle law assumes that every needle sits in its own pair at the wall; meshes whose needles do not come in isolated wall pairs are not covered.
- *Three dimensions.* The companion paper [28] treats inscribed polyhedra; the loss of the chordwise odd-symmetry cancellation changes several arguments there.

Finally, although Nitsche’s method does not improve the rate on good meshes, it is robust on bad ones: the Nitsche methods were insensitive to locked wall vertices in all our computations. Proving this robustness, on meshes that violate (M2) at the wall, is perhaps the most natural next question.

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- **Competing interests.** The question studied in this article is the subject of a prize offered by L. R. Scott (Prize Problem Ledger problem PPL 115 [15]), and the author intends to submit this work for it. The author declares no other competing interests.
- **Ethics approval and consent to participate.** Not applicable.
- **Consent for publication.** Not applicable.
- **Data availability.** All code, raw results and logs, and the scripts that regenerate every table and figure, are available at <https://github.com/jaideepsaipadhi/zero-gradient-prize> [29]. The extended version of this article, with all omitted proofs and computations, is archived on Zenodo [27].
- **Materials availability.** Not applicable.
- **Code availability.** The finite element code (the Scott–Vogelius–Nitsche solver), the exact rational-arithmetic and interval certificates, and the scripts that produce every table and figure are available at <https://github.com/jaideepsaipadhi/zero-gradient-prize> [29]; see also [27].
- **Author contributions.** J. S. Padhi is the sole author and is responsible for the conception of the study, the analysis, the software and computations, and the writing of the manuscript (with the use of generative AI described in Section 3.8). Some of the results proved here were announced, with proof sketches, in the letter [25], which is joint work with C. de S. Cavalcante; see Section 2.6.
- **Use of generative AI.** The use of a generative AI model in this work is described in Section 3.8. The AI model is not an author.

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